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Optimal Mean-Variance
Portfolio Selection

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Abstract

Suppose we are given a riskless bond price B with exponential growth and a risky stock price S following a geometric Brownian motion. Let X^u denote the wealth process of an admissible control u corresponding to a self-financing trading strategy. Here admissibility corresponds to an existence and uniqueness assumption of the corresponding wealth process (see Definition 2.10). Let $\mathbb{P}_{t,x}$ denote a probability measure under which $X_t^u = x$ a.s. Fix $0 < c, T < \infty$ and $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Consider the optimization problem

$$\sup_{u \in \mathcal{A}_{t_0, x_0}} (\mathbb{E}_{t_0, x_0}[X_T^u] - c\text{Var}_{t_0, x_0}(X_T^u)),$$

where $\mathcal{A}_{t_0, x_0}$ is a subset of the set admissible controls on $[t_0, T]$ which ensures a certain integrability assumptions (see Lemma 3.2). We find an explicit solution to this problem, hence an optimizing strategy called statically optimal control. It turns out that the resulting family of solutions is not *consistent over time* (see Remark 4.2). Therefore we study a second problem, and we find an explicit expression for the corresponding solution, the so-called dynamically optimal control. This is the unique admissible control on $[0, T[$ such that for every given and fixed $(t_0, x_0) \in [0, T[\times \mathbb{R}$ and every admissible control $v \in \mathcal{A}_{t_0, x_0}$ with $\lim_{x \rightarrow x_0} v(t_0, x)x \neq \lim_{x \rightarrow x_0} u_{1,c}^d(t_0, x)x$, there exists a control $w \in \mathcal{A}_{t_0, x_0}$ satisfying $w(t_0, x_0)x_0 = u_{1,c}^d(t_0, x_0)x_0$ and

$$\mathbb{E}_{t_0, x_0}[X_T^v] - c\text{Var}(X_T^v) < \mathbb{E}_{t_0, x_0}[X_T^w] - c\text{Var}(X_T^w).$$

In particular, this control is in general not admissible on the *closed* interval $[t_0, T]$.

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Chapter 1

Introduction

This thesis relies mainly on the results of J.L. Pedersen and G. Peskir stated in their paper "Optimal Mean-Variance Portfolio Selection" [2]. Suppose we are given a riskless bond price B with exponential growth and a risky stock price S following a geometric Brownian motion. Let X^u denote the wealth process of an admissible control u corresponding to a self-financing trading strategy (see Definition 2.10). Let $\mathbb{P}_{t,x}$ denote a probability measure under which $X_t^u = x$ a.s. Fix $0 < c, T < \infty$ and $(t_0, x_0) \in [0, T[\times \mathbb{R}$. The main goal is to find strategies $u_{1,c}^s$ and $u_{1,c}^d$ maximizing the expression

$$J(t, x; u) := \mathbb{E}_{t,x}[X_T^u] - c\text{Var}_{t,x}(X_T^u)$$

under two different criteria.

The first criterium is the static optimality (see Definition 3.5). An admissible control is statically optimal for fixed $(t_0, x_0) \in [0, T] \times \mathbb{R}$ if it maximizes

$$J(t_0, x_0; u)$$

among all $u \in \mathcal{A}_{t_0, x_0}$, where $\mathcal{A}_{t_0, x_0}$ is a subset of the set of admissible controls satisfying additional integrability assumptions (see Lemma 3.2). We can explicitly write down a closed form expression of an optimizing strategy (later called $u_{1,c}^s$) in $\mathcal{A}_{t_0, x_0}$. However, this solution is *time inconsistent*¹ in the following sense. Denote by u_{t_0, x_0}^s the statically optimal control for $(t_0, x_0) \in [0, T[\times \mathbb{R}$ and its wealth process

¹In the paper of J.L. Pedersen and G. Peskir [2, p.2] it is stated that the time inconsistency does not occur in linear optimal control problems and the reason why it occurs here is the *spatial inconsistency* (dependence of the statically optimal wealth process from the initial wealth x_0) derived from the quadratic nonlinearity of the variance as a function of the mean. However we won't talk about where particularly the time inconsistency comes from.

by X^s . If we follow the strategy u_{t_0, x_0}^s , then at any later time $t_0 < t < T$ the new investment strategy starting at the current time t and wealth X_t^s will be different, i.e.

$$u_{t_0, x_0}^s \big|_{[t, T] \times \mathbb{R}} \neq u_{t, X_t^s}^s.$$

This means that we should actually change our investment strategy at this time. But as we have chosen $t_0 < t < T$ arbitrary, we should change our investment strategy at any time.

In order to give a better approach to this problem, J.L. Pedersen and G. Peskir introduce the notion of dynamic optimality (see Definition 3.5). The dynamically optimal controls do not depend on the initial time and wealth. Therefore they are time consistent in the above sense. Nevertheless they are still optimal in the above problem, but under a different criterium as before: For any other admissible control v that differs from the dynamically optimal control at any initial time and wealth $(t_0, x_0) \in [0, T] \times \mathbb{R}$ there exists a third admissible control w that equals to the dynamically optimal control in (t_0, x_0) and outperforms the second control, i.e.

$$J(t_0, x_0; v) < J(t_0, x_0; w).$$

In a further step we consider the related constrained problems

$$\sup_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \text{Var}(X_T^u) \leq \alpha}} \mathbb{E}_{t_0, x_0}[X_T^u],$$

and

$$\inf_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_T^u] \geq \beta}} \text{Var}_{t_0, x_0}(X_T^u),$$

where $0 < \alpha < \infty$ and $\beta \in \mathbb{R}$ are fixed, under the same two criteria. It will turn out that the supremum resp. infimum in these problems are (mainly) attained in the same control as in the first problem for an intelligent choice of $c = c(\alpha, t_0, x_0)$ resp. $c = c(\beta, t_0, x_0)$.

This thesis is meant to be understood with the knowledge of a Bachelor student. Therefore in Chapter 2 we recall some basic results from stochastic calculus, which we need in order to understand the paper of J.L. Pedersen and G. Peskir [2]. Chapter 3 introduces the problem mathematically. This includes the precise definitions of statically and weakly/strongly dynamically optimal controls for the three optimization problems. In Chapter 4 we solve the problems and discuss the results.

Even though more or less all results of Chapter 4 (and also of the first part of Lemma 3.2) are already stated in the paper of J.L. Pedersen and G. Peskir [2], many computations are omitted there. This thesis provides these computations and additionally explains every step in detail. Moreover we formalize some expressions that are not well defined, complete and correct some proofs and also discuss further properties of the optimal controls.

Chapter 2

Basics of mathematical finance

2.1 Stochastic calculus and stochastic differential equations

We want to recall some basics of stochastic calculus. One of the most important tools we need is Itô's formula. We give the statement without a proof.

Theorem 2.1. *Suppose $X = (X_t)_{t \geq 0}$ is a $\mathbb{R}^d$ -valued semimartingale on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with respect to the filtration $\mathbb{F}$ with $\mathbb{P}$ -almost sure continuous trajectories and $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is in C^2 . Then $f(X) = (f(X_t))_{t \geq 0}$ is a real-valued semimartingale with respect to $\mathbb{F}$ and $\mathbb{P}$ -almost sure $\forall t \geq 0$*

$$f(X_t) = f(X_0) + \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x^i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 f}{\partial x^i \partial x^j}(X_s) d[X^i, X^j]_s$$

or in shorthand notation

$$df(X_s) = \sum_{i=1}^d \frac{\partial f}{\partial x^i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 f}{\partial x^i \partial x^j}(X_s) d[X^i, X^j]_s.$$

Note that by symmetry of the quadratic covariation we can rewrite the right-hand side to

$$\begin{aligned} f(X_t) = f(X_0) &+ \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x^i}(X_s) dX_s^i + \frac{1}{2} \sum_{i=1}^d \int_0^t \frac{\partial^2 f}{(\partial x^i)^2}(X_s) d[X^i, X^i]_s \\ &+ \sum_{\substack{i,j=1 \\ i < j}}^d \int_0^t \frac{\partial^2 f}{\partial x^i \partial x^j}(X_s) d[X^i, X^j]_s. \end{aligned}$$

The next lemma shows that in many cases, the quadratic covariation $[X^i, X^j]_s$ vanishes.

Lemma 2.2. *Let X be a semimartingale. Let Y be a continuous stochastic process with finite variation, i.e*

$$\forall t \geq 0 : \quad \sup_{\Pi \subset [0, t]} \sum_{t_i \in \Pi} |Y_{t_{i+1}} - Y_{t_i}| < \infty \quad \mathbb{P}\text{-a.s.}$$

where the supremum is taken over all partitions of $[0, t]$. Then

$$[Y, Y] \equiv 0 \quad \mathbb{P}\text{-a.s.}$$

and

$$[X, Y] \equiv 0 \quad \mathbb{P}\text{-a.s.}$$

Proof. ¹ Fix $t \geq 0$. We first show that there exists a sequence of partitions Π_n of $[0, t]$, with $\lim_{n \rightarrow \infty} \text{mesh}(\Pi_n) = 0$, such that

$$\begin{aligned} [X, Y]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) \quad \mathbb{P}\text{-a.s.} \\ [X, X]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n} (X_{t_{i+1}} - X_{t_i})^2 \quad \mathbb{P}\text{-a.s.} \\ [Y, Y]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n} (Y_{t_{i+1}} - Y_{t_i})^2 \quad \mathbb{P}\text{-a.s.} \end{aligned}$$

Note that both X and Y are semimartingales. Fix any sequence of partitions $\hat{\Pi}_m$ of $[0, \infty[$, with $\lim_{m \rightarrow \infty} \text{mesh}(\hat{\Pi}_m) = 0$. Then by the good integrator property we have that

$$\begin{aligned} [X, Y]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \hat{\Pi}_n} (X_{t_{i+1} \wedge t} - X_{t_i \wedge t}) (Y_{t_{i+1} \wedge t} - Y_{t_i \wedge t}) \quad \mathbb{P}\text{-a.s.} \\ [X, X]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \hat{\Pi}_n} (X_{t_{i+1} \wedge t} - X_{t_i \wedge t})^2 \quad \mathbb{P}\text{-a.s.} \\ [Y, Y]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \hat{\Pi}_n} (Y_{t_{i+1} \wedge t} - Y_{t_i \wedge t})^2 \quad \mathbb{P}\text{-a.s.} \end{aligned}$$

¹This proof uses the ideas of the proofs of [1, Theorem 3.4, p. 212-213] and [1, Theorem 3.1, p. 209].

converge in ucp^2 , hence also uniformly in probability on $[0, t]$. However, we just need the convergence in probability at time t . Now let $\tilde{\Pi}$ denote the corresponding sequence of partitions on $[0, t]$, i.e. $\tilde{\Pi}_n := \hat{\Pi}_n \setminus]t, \infty[$ for all $n \in \mathbb{N}$. Fix $\epsilon > 0$. Then it holds that

$$\begin{aligned} 0 &= \lim_{m \rightarrow \infty} \mathbb{P} \left[\left| \sum_{t_i \in \tilde{\Pi}_m} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) - [X, Y]_t \right| > \epsilon \right], \\ 0 &= \lim_{m \rightarrow \infty} \mathbb{P} \left[\left| \sum_{t_i \in \tilde{\Pi}_m} (X_{t_{i+1}} - X_{t_i})^2 - [X, X]_t \right| > \epsilon \right], \\ 0 &= \lim_{m \rightarrow \infty} \mathbb{P} \left[\left| \sum_{t_i \in \tilde{\Pi}_m} (Y_{t_{i+1}} - Y_{t_i})^2 - [Y, Y]_t \right| > \epsilon \right]. \end{aligned}$$

Chose a subsequence m_k such that

$$(2.1) \quad \frac{1}{2^k} > \mathbb{P} \left[\left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) - [X, Y]_t \right| > \frac{1}{2^k} \right]$$

$$(2.2) \quad \frac{1}{2^k} > \mathbb{P} \left[\left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i})^2 - [X, X]_t \right| > \frac{1}{2^k} \right]$$

$$(2.3) \quad \frac{1}{2^k} > \mathbb{P} \left[\left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (Y_{t_{i+1}} - Y_{t_i})^2 - [Y, Y]_t \right| > \frac{1}{2^k} \right].$$

This sequence can be found inductively by setting in each step $\epsilon = \frac{1}{2^k}$ and then defining m_k as the smallest $m > m_{k-1}$ such that the above inequalities hold. Now fix $\tilde{\epsilon} > 0$ and define

$$A_k^{\tilde{\epsilon}} := \left\{ \left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) - [X, Y]_t \right| \geq \tilde{\epsilon} \right\}.$$

Then

$$\sum_{k=1}^{\infty} \mathbb{P}[A_k^{\tilde{\epsilon}}] \leq \sum_{k: \frac{1}{2^k} < \tilde{\epsilon}} \underbrace{\mathbb{P}[A_k^{\tilde{\epsilon}}]}_{< \frac{1}{2^k}} + \sum_{k: \frac{1}{2^k} \geq \tilde{\epsilon}} \underbrace{\mathbb{P}[A_k^{\tilde{\epsilon}}]}_{\leq 1} \leq 1 - \log_2(\tilde{\epsilon}) < \infty,$$

²For the complete proof of this statement see [4, Theorem 5.7, p.13].

as $\frac{1}{2^k} \geq \tilde{\epsilon}$ if and only if $k \leq -\log_2(\tilde{\epsilon})$. So by Borel Cantelli

$$0 = \mathbb{P} \left[\bigcap_{k_0 \in \mathbb{N}} \bigcup_{k \geq k_0} A_k^{\tilde{\epsilon}} \right]$$

and therefore

$$\begin{aligned} 1 &= \mathbb{P} \left[\bigcup_{k_0 \in \mathbb{N}} \bigcap_{k \geq k_0} (A_k^{\tilde{\epsilon}})^c \right] \\ &= \mathbb{P}[\exists k_0 \in \mathbb{N} \ \forall k \geq k_0 : \left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) - [X, Y]_t \right| < \tilde{\epsilon}]. \end{aligned}$$

But then this holds for all $\tilde{\epsilon} > 0$ as $\tilde{\epsilon}$ was chosen arbitrary. This leads to

$$\begin{aligned} &\mathbb{P} \left[\lim_{k \rightarrow \infty} \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) = [X, Y]_t \right] \\ &= \mathbb{P} \left[\forall n \in \mathbb{N} \ \exists k_0 \in \mathbb{N} \ \forall k \geq k_0 : \left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) - [X, Y]_t \right| < \frac{1}{n} \right] \\ &= \left(1 - \mathbb{P} \left[\bigcup_{n \in \mathbb{N}} \left\{ \forall k_0 \in \mathbb{N} \ \exists k \geq k_0 : \left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) - [X, Y]_t \right| \geq \frac{1}{n} \right\} \right] \right) \\ &\geq \left(1 - \sum_{n \in \mathbb{N}} \mathbb{P} \left[\forall k_0 \in \mathbb{N} \ \exists k \geq k_0 : \left| \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) - [X, Y]_t \right| \geq \frac{1}{n} \right] \right) \\ &= 1. \end{aligned}$$

The same argumentation shows that

$$\begin{aligned} &\mathbb{P} \left[\lim_{k \rightarrow \infty} \sum_{t_i \in \tilde{\Pi}_{m_k}} (X_{t_{i+1}} - X_{t_i})^2 = [X, X]_t \right] = 1 \\ &\mathbb{P} \left[\lim_{k \rightarrow \infty} \sum_{t_i \in \tilde{\Pi}_{m_k}} (Y_{t_{i+1}} - Y_{t_i})^2 = [Y, Y]_t \right] = 1. \end{aligned}$$

So $\Pi_n = \tilde{\Pi}_{m_n}$ has the desired properties.

By Cauchy-Schwarz it holds that

$$\begin{aligned} [X, Y]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n} (X_{t_{i+1}} - X_{t_i}) (Y_{t_{i+1}} - Y_{t_i}) \\ &\leq \sqrt{\left(\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n} (X_{t_{i+1}} - X_{t_i})^2 \right) \left(\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n} (Y_{t_{i+1}} - Y_{t_i})^2 \right)} \\ &= \sqrt{[X, X]_t [Y, Y]_t}. \end{aligned}$$

Note that again by the good integrator property

$$[X, X]_t < \infty \quad \mathbb{P}\text{-a.s.}$$

and

$$\begin{aligned} [Y, Y]_t &= \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n} (Y_{t_{i+1}} - Y_{t_i})^2 \\ &\leq \lim_{n \rightarrow \infty} \left(\sup_{t_i \in \Pi_n} |Y_{t_{i+1}} - Y_{t_i}| \right) \sum_{t_i \in \Pi_n} |Y_{t_{i+1}} - Y_{t_i}| \\ &= 0 \quad \mathbb{P}\text{-a.s.} \end{aligned}$$

as Y is uniformly continuous on the compact set $[0, t]$ and Y is of finite variation. $\square$

Without repeating in each statement, we consider in this chapter a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ and a $(\mathbb{P}, \mathbb{F})$ -Brownian motion W . Fix $0 \leq t_0 < T$. Let Y be a $\mathcal{F}_0$ -measurable random variable and $a, b : [t_0, T] \times \mathbb{R} \rightarrow \mathbb{R}$. We consider the stochastic differential equation (SDE)

$$(2.4) \quad dX_t = a(t, X_t)dt + b(t, X_t)dW_t, \quad X_{t_0} = Y.$$

Definition 2.3. A *strong solution* of (2.4) on $[t_0, T]$ is a continuous $\mathbb{F}$ -adapted process $X = (X_t)_{t_0 \leq t \leq T}$ satisfying

$$(2.5) \quad \int_{t_0}^T (|a(s, X_s)| + |b(s, X_s)|^2) ds < \infty, \quad X_{t_0} = Y \quad \mathbb{P}\text{-a.s.}$$

and

$$(2.6) \quad X_t = Y + \int_{t_0}^t a(s, X_s)ds + \int_{t_0}^t b(s, X_s)dW_s \quad \mathbb{P}\text{-a.s. for each } t_0 \leq t \leq T.$$

X is called a strong solution of (2.4) on the *half open* interval $[t_0, T[$ if X is a strong solution of (2.4) on $[t_0, \tilde{T}]$ for all $t_0 < \tilde{T} < T$.

We need to be able to solve an easy class of SDEs. Therefore we solve the most general case needed in a separate lemma.

Lemma 2.4. *Let Y be a random variable with finite second moment. Let $a, b \in C([t_0, T]; \mathbb{R})$. The SDE*

$$dX_t = a(t)X_t dt + b(t)X_t dW_t, \quad X_0 = Y$$

has a unique strong solution

$$X_t = Y \exp \left(\int_{t_0}^t \left(a(s) - \frac{b(s)^2}{2} \right) ds + \int_{t_0}^t b(s) dW_s \right).$$

on $[t_0, T]$.

Note that this result can be generalized to continuous functions a, b on the *half open interval* $[t_0, T[$ (existence uniqueness on every $[t_0, T']$ for $t_0 < T' < T$).

Proof. Existence:

Note that the integrals in the exponential are well defined as a and b are continuous, hence bounded on $[t_0, T]$. We first show that X defined as above is a strong solution by checking the properties in Definition 2.3. As Y has finite second moment, it is $\mathbb{P}$ -a.s. finite. Consider now that by the Burkholder-Davis-Gundy inequality it holds that

$$\mathbb{E} \left[\sup_{t_0 \leq t \leq T} \left| \int_{t_0}^t b(s) dW_s \right| \right] \leq \int_{t_0}^T b(s)^2 ds < \infty$$

hence, we have that $\sup_{t_0 \leq t \leq T} \left| \int_{t_0}^t b(s) dW_s \right|$ is $\mathbb{P}$ -a.s. finite and therefore also $|X|_T^* := \sup_{t_0 \leq t \leq T} |X_t| < \infty$ $\mathbb{P}$ -a.s. Therefore

$$\begin{aligned} \int_{t_0}^T (|a(s)X_s| + |b(s)X_s|^2) ds &\leq (T - t_0)(|a(s)|_T^* |X|_T^* + (|b(s)|_T^*)^2 (|X|_T^*)^2) \\ &< \infty \quad \mathbb{P}\text{-a.s.} \end{aligned}$$

Define the processes

$$\begin{aligned} Z_t^1 &:= \int_{t_0}^t \left(a(s) - \frac{b(s)^2}{2} \right) ds, \\ Z_t^2 &:= \int_{t_0}^t b(s) dW_s \end{aligned}$$

and the function

$$f(x_1, x_2) := e^{x_1 + x_2}.$$

Note that $X_t = Yf(Z_t^1, Z_t^2)$ and that Z^1 is continuous and of finite variation as for any partition Π of $[t_0, T]$

$$\begin{aligned} \sum_{t_i \in \Pi} |Z_{t_{i+1}}^1 - Z_{t_i}^1| &= \sum_{t_i \in \Pi} \left| \int_{t_i}^{t_{i+1}} \left(a(s) - \frac{b(s)^2}{2} \right) ds \right| \\ &\leq \sum_{t_i \in \Pi} \int_{t_i}^{t_{i+1}} \left| \left(a(s) - \frac{b(s)^2}{2} \right) \right| ds \\ &= \int_{t_0}^T \left| \left(a(s) - \frac{b(s)^2}{2} \right) \right| ds < \infty. \end{aligned}$$

We used here boundedness of the continuous map $a - \frac{b^2}{2}$ on the compact set $[t_0, T]$. Note that $d[Z^2, Z^2]_s = b(s)^2 ds$. Applying Itô's formula we obtain

$$\begin{aligned} dX_t &= Y df(Z_t) \\ &= Y \left(e^{Z_t^1 + Z_t^2} dZ_t^1 + e^{Z_t^1 + Z_t^2} dZ_t^2 + \frac{1}{2} e^{Z_t^1 + Z_t^2} d[Z^2, Z^2]_t \right) \\ &= X_t \left(a(t) - \frac{b(t)^2}{2} \right) dt + X_t b(t) dW_t + \frac{1}{2} X_t b(t)^2 dt \\ &= X_t a(t) dt + X_t b(t) dW_t \end{aligned}$$

Uniqueness:

Let X be a strong solution of the SDE. Define

$$\Phi_t = X_t \exp \left(- \int_{t_0}^t \left(a(s) - \frac{b(s)^2}{2} \right) ds - \int_{t_0}^t b(s) dW_s \right)$$

Then by applying Itô's formula on $\Phi = f(X, Z)$ for the function $f(x_1, x_2) := x_1 e^{x_2}$ and

$$Z_t = - \int_{t_0}^t \left(a(s) - \frac{b(s)^2}{2} \right) ds - \int_{t_0}^t b(s) dW_s$$

we obtain (using $dX_t = a(t)X_t dt + b(t)X_t dW_t$)

$$d\Phi_t = e^{Z_t} dX_t + X_t e^{Z_t} dZ_t + e^{Z_t} d[X, Z]_t + \frac{1}{2} X_t e^{Z_t} d[Z, Z]_t$$

$$\begin{aligned}
&= e^{Z_t}(a(t)X_t dt + b(t)X_t dW_t) + X_t e^{Z_t} \left(- \left(a(t) - \frac{b(t)^2}{2} \right) dt - b(t) dW_t \right) \\
&\quad + e^{Z_t}(b(t)X_t(-b(t))dt) + \frac{1}{2}X_t e^{Z_t}(b(t)^2 dt) \\
&= 0.
\end{aligned}$$

Therefore for all $t_0 \leq t \leq T$: $\Phi_t = \Phi_{t_0} = Y$, $\mathbb{P}$ -a.s., hence

$$X_t = Y \exp \left(\int_{t_0}^t \left(a(s) - \frac{b(s)^2}{2} \right) ds + \int_{t_0}^t b(s) dW_s \right) \quad \mathbb{P}\text{-a.s.}$$

□

2.2 Geometric Brownian motion and stochastic control processes

We work with a market consisting of a riskless bond price B and a stock price solving a particular stochastic differential equation that we introduce in the following definition.

Definition 2.5. Let $(W_t)_{t \geq 0}$ be a Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let S_0 be an $\mathcal{F}_0$ measurable random variable. Let $\mu \in \mathbb{R}$ and $\sigma > 0$. The strong solution $(S_t)_{t \geq 0}$ to the stochastic differential equation

$$(2.7) \quad \frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

is called geometric Brownian motion.

Note that by Lemma 2.4 the unique strong solution to the above SDE is given by

$$S_t = S_0 \exp \left(\sigma W_t + \left(\mu - \frac{1}{2} \sigma^2 \right) t \right).$$

Now assume we have a riskless bond price B solving

$$(2.8) \quad \frac{dB_t}{B_t} = r dt$$

with $B_0 = b$ for some $b > 0$, where $r \in \mathbb{R}$ is the interest rate. Note that this ordinary differential equation has a unique deterministic solution $B_t = be^{rt}$. Let the risky stock price S follow a geometric Brownian motion.

Definition 2.6. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ be a filtration. The sigma algebra

$$\mathcal{P} := \sigma(Y \mid_{\Omega \times]0, \infty[} \mid Y = (Y_t)_{t \geq 0} \text{ is } \mathbb{F}\text{-adapted and left continuous})$$

is called the predictable σ -field. A stochastic process H is called predictable if it is $\mathcal{P}$ -measurable.

Now consider again the market (B, S) from above. As in discrete time, the filtration given by $\mathcal{F}_t = \sigma(S_s \mid 0 \leq s \leq t)$ models the information about the position of the asset price S up to time t . So it is intuitive, that an agent invests in a predictable manner as in the definition above. We need to have a notion of self-financing investment strategies in continuous time. In order to stay consistent with the definition in discrete time, the following definition will be useful.

Definition 2.7. Let B and S be stochastic processes solving (2.8) resp. (2.7). Let $\varphi = (\eta, \vartheta) = (\eta_t, \vartheta_t)_{t \geq 0}$ be a predictable process, i.e. predictable in each entry. Define the wealth process $X := \vartheta S + \eta B$. We call φ a self-financing investment strategy if the stochastic integrals $\vartheta_t dS_t$ and $\eta_t dB_t$ are well defined and it holds that $dX_t = \vartheta_t dS_t + \eta_t dB_t$.

Note that ϑ_t resp. η_t denote the number of shares held in S resp. B at time t . Consider now the case, in which the corresponding wealth process $X := \vartheta S + \eta B$ satisfies $X > 0$. Let $u_t := \frac{\eta_t S_t}{X_t}$ denote the fraction of wealth held in the risky asset. As the wealth process depends on u_t , we introduce the notation X_t^u . Intuitively $u_t < 0$ means going short on the stock, $u_t > 1$ means to borrow from the bond, and $u_t \in [0, 1]$ means holding a fraction in the stock and one in the bond.

The next lemma shows, that the wealth process of a self-financing trading strategy is a solution of an SDE depending only on u . This allows us to model any self-financing investment strategy with u and thus to find the corresponding wealth process, we can simply solve this SDE.

Lemma 2.8. Let $\varphi = (\eta, \vartheta)$ be a self-financing trading strategy such that the corresponding wealth process $X := \vartheta S + \eta B$ satisfies $X > 0$. Let $u_t = \frac{\eta_t S_t}{X_t}$ be as above. Then the wealth process X_t^u solves the stochastic differential equation

$$(2.9) \quad dX_t^u = (r(1 - u_t) + \mu u_t) X_t^u dt + \sigma u_t X_t^u dW_t$$

Proof.

$$\begin{aligned} dX_t^u &= \vartheta_t dS_t + \eta_t dB_t \\ &= \underbrace{\vartheta_t S_t}_{u_t X_t^u} \underbrace{\frac{dS_t}{S_t}}_{\mu dt + \sigma dW_t} + \underbrace{\eta_t B_t}_{(1-u_t) X_t^u} \underbrace{\frac{dB_t}{B_t}}_{r dt} \\ &= (r(1 - u_t) + \mu u_t) X_t^u dt + \sigma u_t X_t^u dW_t \end{aligned}$$

□

The wealth process of a general self-financing trading strategy won't satisfy $X > 0$. So in general we are not allowed to define u directly via the identity $u_t = \frac{\eta_t S_t}{X_t}$, as this won't be well defined for $X_t = 0$. In order to simplify the problem, we still want to work with the SDE (2.9). Motivated by the computation above, we introduce the notion of stochastic control processes.

Definition 2.9. A stochastic control process $(u_t)_{t \geq 0}$ is an $\mathbb{F}$ -predictable stochastic process such that the stochastic differential equation (2.9) has a unique strong solution for any initial data $X_{t_0} = x_0 \in \mathbb{R}$.

In order to solve optimization problems, we will therefore no longer search for self-financing trading strategies, but for the related stochastic control processes. In order to simplify the problems, we will only consider a certain class of stochastic processes, namely the admissible controls.

Definition 2.10. ³ Let $0 < T < \infty$.

- (i) Fix $0 \leq t_0 < T$. Let $u : [t_0, T] \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$ be a map such that $(t, x) \mapsto u(t, x) \cdot x$ is continuously continuable on $[t_0, T] \times \mathbb{R}$ and the stochastic differential equation

$$dX_t = (r(1 - u(t, X_t)) + \mu u(t, X_t))X_t dt + \sigma u(t, X_t)X_t dW_t, \quad X_{t_0} = x_0$$

has a unique strong solution X_t^u for any $x_0 \in \mathbb{R}$. Then we call u an *admissible control on $[t_0, T]$* and X^u is called *the wealth process of u* .

- (ii) We call a map $u : [0, T[\times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$ an *admissible control on the halfopen interval $[0, T[$* , if its restriction $u|_{[t_0, t_1] \times (\mathbb{R} \setminus \{0\})}$ is an admissible control on $[t_0, t_1]$ for all $0 \leq t_0 < t_1 < T$.

In both cases the stochastic process $u_t := u(t, X_t^u)$ is called *admissible control process on $[t_0, T]$* , resp. $[0, T[$, where X_t^u denotes the strong solution of the above SDE.

Note that an admissible control u on $[t_0, t_1]$ is continuous as a map $u : [t_0, t_1] \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$ since it is a product of two continuous functions $u(t, x) = u(t, x)x \times \frac{1}{x}$. We call both the map u and the process $(u_t)_{t \geq 0}$ admissible controls, if the map u fulfills the above properties. It will be clear from the context which one is meant. Also note that all maps $u(t, x) = \frac{1}{x}v(t, x)$ with v continuous on $[t_0, t_1] \times \mathbb{R}$ fulfill the continuity property. We use the convention $u(t, 0) \cdot 0 = \lim_{x \rightarrow 0} u(t, x) \cdot x$ for admissible controls u . Note that an admissible control process $u_t = u(t, X_t^u)$ is $\mathbb{F}$ -predictable as u is continuous. Also note that admissibility in this thesis does not refer to any integrability assumptions.⁴

³Note that we had to change the definition of admissible controls slightly. As we will come up with controls for which the corresponding wealth process is not defined in T . Moreover it was not clear what continuity property the controls have to fulfill. The one given here is the most intuitive one.

⁴In the paper [2, p.2] admissibility requests $\mathbb{E}_{t,x}[\int_t^T (1 + u_s^2)(X_s^u)^2 ds] < \infty$. However since we will be working with controls that do not fulfill this property, we changed the definition to the one given above. For a more detailed discussion, see Remark 4.4.

Chapter 3

Mathematical formulation of the problems

3.1 Additional preparation

The following is stated in the paper of J.L. Pedersen and G. Peskir [2, p.4]: Given an admissible¹ control u on $[0, T]$ for $0 < T < \infty$ fixed and a Brownian motion W' on a probability space $(\Omega', \mathcal{F}', \mathbb{P}')$. Then there exists a random variable X^u on $(C[0, T], \mathcal{B}(C([0, T])))$ such that for each $(t_0, x_0) \in [0, T] \times \mathbb{R}$ there exists a probability measure $\mathbb{P}_{t_0, x_0}^u$ on $(C[0, T], \mathcal{B}(C([0, T])))$ and a $\mathbb{P}_{t_0, x_0}^u$ -Brownian motion W under which X^u is the solution to the SDE

$$dX_t = (r(1 - u(t, X_t)) + \mu u(t, X_t))X_t dt + \sigma u(t, X_t)X_t dW_t, \quad X_{t_0} = x_0.$$

We suppose that this assumption holds true in the sequence, but we prove this statement only for a slightly easier case and not for this general setting.

Lemma 3.1. *Let W' be a Brownian Motion on a probability space $(\Omega', \mathcal{F}', \mathbb{P}')$. Fix $0 < T < \infty$ and $(t_0, x_0) \in [0, T] \times \mathbb{R}$. Let u be an admissible control on $[t_0, T]$ for W' such that the unique strong solution X^{u, t_0, x_0} on $(\Omega', \mathcal{F}', \mathbb{P}')$ to the SDE*

$$(3.1) \quad dX_t = (r(1 - u(t, X_t)) + \mu u(t, X_t))X_t dt + \sigma u(t, X_t)X_t dW'_t, \quad X_{t_0} = x_0$$

satisfies $u(s, X_s^{u, t_0, x_0})X_s^{u, t_0, x_0} \neq 0$, $\mathbb{P}'$ -a.s. for all $t_0 \leq s \leq T$. Then there exists a measure $\mathbb{P}_{t_0, x_0}^u$ on $(\Omega, \mathcal{F}) := (C[0, T], \mathcal{B}(C([0, T])))$ and a $\mathbb{P}_{t_0, x_0}^u$ -Brownian motion W s.t. the random variable X defined as $X_t(\omega) = \omega(t)$ solves the SDE

$$dX_t = (r(1 - u(t, X_t)) + \mu u(t, X_t))X_t dt + \sigma u(t, X_t)X_t dW_t, \quad X_{t_0} = x_0$$

¹Note that admissibility depends on the choice of the Brownian motion.

on $(\Omega, \mathcal{F}, \mathbb{P})$. In particular u is also admissible for W . Moreover X and $X_{t_0}^{u, t_0, x_0}$ have the same distribution.

Proof. Define the probability measure $\mathbb{P}_{t_0, x_0}^u = (X_{t_0}^{u, t_0, x_0})_* \mathbb{P}'$ on $(\Omega, \mathcal{F})$. Define $X_t(\omega) = \omega(t)$ on $(\Omega, \mathcal{F})$. Then X and $X_{t_0}^{u, t_0, x_0}$ have the same distribution. Now rearrange (3.1) to see that

$$\begin{aligned} W'_t = & \int_0^t \frac{1}{\sigma u(s, X_s^{u, t_0, x_0}) X_s^{u, t_0, x_0}} dX_s^{u, t_0, x_0} \\ & - \int_0^t \frac{(r(1 - u(s, X_s^{u, t_0, x_0})) + \mu u(s, X_s^{u, t_0, x_0})) X_s^{u, t_0, x_0}}{\sigma u(s, X_s^{u, t_0, x_0}) X_s^{u, t_0, x_0}} ds \quad \mathbb{P}'\text{-a.s.} \end{aligned}$$

We define

$$\begin{aligned} W_t = & \int_0^t \frac{1}{\sigma u(s, X_s) X_s} dX_s \\ & - \int_0^t \frac{(r(1 - u(s, X_s)) + \mu u(s, X_s)) X_s}{\sigma u(s, X_s) X_s} ds. \end{aligned}$$

Then W and W' have the same distribution, hence W is a brownian motion defined on the new probability space $(\Omega, \mathcal{F}, \mathbb{P}_{t_0, x_0}^u)$ and by construction of W , the process X solves

$$(3.2) \quad dX_t = (r(1 - u(t, X_t)) + \mu u(t, X_t)) X_t dt + \sigma u(t, X_t) X_t dW_t,$$

with $X_{t_0} = x_0$, $\mathbb{P}_{t_0, x_0}^u$ -almost sure. $\square$

We now turn our attention to the main problem. Given $0 < c, T < \infty$ and $(t_0, x_0) \in [0, T[\times \mathbb{R}$. We want to find an optimal admissible control u on $[t_0, T]$ that maximizes

$$J(t_0, x_0; u) := \mathbb{E}_{t_0, x_0}[X_T^u] - c \text{Var}_{t_0, x_0}(X_T^u).$$

Note that we have to make sure that this expression is well defined, i.e. that $\mathbb{E}_{t_0, x_0}[(X_T^u)^2] < \infty$. We want to simplify the problem by maximizing over a smaller set of controls. In particular, these controls have to satisfy $\mathbb{E}_{t_0, x_0}[(X_T^u)^2] < \infty$.

Lemma 3.2. *Let $0 < T < \infty$ and $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Let u be an admissible control. Then*

$$\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T u_t^2 (X_t^u)^2 dt \right] < \infty$$

gives a sufficient condition for

$$\mathbb{E}_{t_0, x_0}[(X_T^u)^2] < \infty.$$

Moreover, define

$$\mathcal{A}_{t_0, x_0} := \left\{ u \text{ admissible control on } [t_0, T] \mid \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T (u_t^2 + 1)(X_t^u)^2 dt \right] < \infty \right\}$$

and let $a, b : [t_0, T] \rightarrow \mathbb{R}$ be continuous and $u \in \mathcal{A}_{t_0, x_0}$. Then

$$M_t := \int_{t_0}^t (a(s)X_s^u + b(s))u_s X_s^u dW_s$$

is well defined and a $(\mathbb{P}_{t_0, x_0}, \mathbb{F})$ -martingale. Moreover it holds that

$$(3.3) \quad \mathbb{E}_{t_0, x_0} \left[\sqrt{\int_{t_0}^T (a(s)X_s^u + b(s))^2 u_s^2 (X_s^u)^2 ds} \right] < \infty.$$

Proof. We first claim that the inequalities

$$(3.4) \quad \forall x_1, \dots, x_n \geq 0 : \quad \sqrt{x_1 + \dots + x_n} \leq \sqrt{x_1} + \dots + \sqrt{x_n}$$

$$(3.5) \quad \forall x_1, \dots, x_n \in \mathbb{R} : \quad (x_1 + \dots + x_n)^2 \leq n(x_1^2 + \dots + x_n^2)$$

hold.

For the first inequality we notice that $\forall x_1, \dots, x_n > 0 : \quad x_1^2 + \dots + x_n^2 \leq (x_1 + \dots + x_n)^2$. Thus $\sqrt{x_1^2 + \dots + x_n^2} \leq x_1 + \dots + x_n$. Replacing x_i by $\sqrt{x_i}$ for $i = 1, \dots, n$ gives the result.

The second inequality follows from the fact that $\forall x, y \in \mathbb{R} : \quad 0 \leq (x - y)^2 = x^2 - 2xy + y^2$. Hence $xy \leq \frac{x^2 + y^2}{2}$. Therefore we have

$$\begin{aligned} (x_1 + \dots + x_n)^2 &\leq \sum_{i=1}^n x_i^2 + 2 \sum_{\substack{i, j=1, \dots, n \\ i < j}}^n x_i x_j \\ &\leq \sum_{i=1}^n x_i^2 + \sum_{\substack{i, j=1, \dots, n \\ i < j}}^n (x_i^2 + x_j^2) \\ &= \sum_{i=1}^n x_i^2 + (n-1) \sum_{i=1}^n x_i^2 \end{aligned}$$

$$= n \sum_{i=1}^n x_i^2$$

Now let u be an admissible control on $[t_0, T]$ satisfying

$$\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T u_t^2 (X_t^u)^2 dt \right] < \infty.$$

We show that $\mathbb{E}_{t_0, x_0} [(X_T^u)^2] < \infty$. Consider the process

$$Y_t^u := e^{-rt} X_t^u.$$

Clearly it suffices to show that $\mathbb{E}_{t_0, x_0} [(Y_T^u)^2] < \infty$ as then

$$\mathbb{E}_{t_0, x_0} [(X_T^u)^2] = e^{2rT} \mathbb{E}_{t_0, x_0} [(Y_T^u)^2] < \infty.$$

Note that by assumption it holds that

$$\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T u_t^2 (Y_t^u)^2 dt \right] \leq \max\{e^{-2rt_0}, e^{-2rT}\} \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T u_t^2 (X_t^u)^2 dt \right] < \infty.$$

Define $f \in C^2(\mathbb{R} \times [0, T])$ by $f(x, t) := e^{-rt}x$. We then have $Y_s^u = f(X_t^u, t)$. By Itô's formula on Y_t^u and (2.9) we obtain

$$\begin{aligned} dY_t^u &= f_x(X_t^u, t) dX_t^u + f_t(X_t^u, t) dt + \frac{1}{2} f_{xx}(X_t^u, t) d[X^u, X^u]_t \\ &= e^{-rt} dX_t^u - r e^{-rt} X_t^u dt + 0 d[X^u, X^u]_t \\ &= e^{-rt} ((r(1 - u_t) + \mu u_t) X_t^u dt + \sigma u_t X_t^u dW_t) - r e^{-rt} X_t^u dt \\ (3.6) \quad &= (\mu - r) u_t Y_t^u dt + \sigma u_t Y_t^u dW_t \end{aligned}$$

Writing this out we obtain

$$(3.7) \quad Y_T^u = e^{-rt_0} x_0 + \int_{t_0}^T (\mu - r) u_t Y_t^u dt + \int_{t_0}^T \sigma u_t Y_t^u dW_t \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

By Cauchy-Schwarz we obtain

$$\begin{aligned} (3.8) \quad \mathbb{E}_{t_0, x_0} \left[\left(\int_{t_0}^T (\mu - r) u_t Y_t^u \cdot 1 dt \right)^2 \right] &\leq \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T ((\mu - r) u_t Y_t^u)^2 dt \int_{t_0}^T 1^2 dt \right] \\ &= (\mu - r)^2 (T - t_0) \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T (u_t Y_t^u)^2 dt \right] \\ &< \infty. \end{aligned}$$

By the Itô isometry it holds that

$$(3.9) \quad \mathbb{E}_{t_0, x_0} \left[\left(\int_{t_0}^T \sigma u_t Y_t^u dW_t \right)^2 \right] = \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T (\sigma u_t Y_t^u)^2 dt \right] < \infty.$$

Thus we obtain

$$\begin{aligned} & \mathbb{E}_{t_0, x_0} [(Y_T^u)^2] \\ & \stackrel{(3.7)+(3.5)}{\leq} 3\mathbb{E}_{t_0, x_0} \left[e^{-2rt_0} x_0^2 + \left(\int_{t_0}^T (\mu - r) u_t Y_t^u dt \right)^2 + \left(\int_{t_0}^T \sigma u_t Y_t^u dW_t \right)^2 \right] \\ & \stackrel{(3.8)+(3.9)}{<} \infty. \end{aligned}$$

This proves the first part of the lemma.

Now let $u \in \mathcal{A}_{t_0, x_0}$ and a, b, M as in the formulation of the lemma. We show that M is a $(\mathbb{P}_{t_0, x_0}, \mathbb{F})$ -martingale. Consider that it suffices to show that

$$(3.10) \quad \mathbb{E}_{t_0, x_0} \left[\sqrt{\int_{t_0}^T (a(s)X_s^u + b(s))^2 u_s^2 (X_s^u)^2 ds} \right] < \infty.$$

If we assume that the above inequality holds, then it follows that $\int_{t_0}^T (a(s)X_s^u + b(s))^2 u_s^2 (X_s^u)^2 ds < \infty$ $\mathbb{P}_{t_0, x_0}$ -a.s., i.e that M_t is well defined and a continuous local martingale with $M_{t_0} = 0$. Moreover, by the Burkholder-Davis-Gundy inequality we then obtain that

$$\begin{aligned} \mathbb{E}_{t_0, x_0} \left[\sup_{t_0 \leq t \leq T} |M_t| \right] & \leq \tilde{C} \mathbb{E}_{t_0, x_0} \left[[M, M]_T^{1/2} \right] \\ & = \tilde{C} \mathbb{E}_{t_0, x_0} \left[\sqrt{\int_{t_0}^T (a(s)X_s^u + b(s))^2 u_s^2 (X_s^u)^2 ds} \right] \\ & < \infty \end{aligned}$$

for some $\tilde{C} > 0$. This shows that M_t is a martingale as by the local martingale property, there exists a monotone increasing sequence of stopping times τ_n with $\lim_{n \rightarrow \infty} \tau_n = \infty$ such that the stopped process $M_t^{\tau_n}$ is a martingale. Therefore it holds for all $s \leq t$ by dominated convergence that

$$\mathbb{E}_{t_0, x_0} [M_t \mid \mathcal{F}_s] = \lim_{n \rightarrow \infty} \mathbb{E}_{t_0, x_0} [M_t^{\tau_n} \mid \mathcal{F}_s] = \lim_{n \rightarrow \infty} M_s^{\tau_n} = M_s \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

We prove (3.10). Note that a and b are bounded on $[t_0, T]$, say $|a|, |b| \leq C$. By (3.4), (3.5), Jensen and Cauchy-Schwarz we now can conclude that

$$\begin{aligned}
& \mathbb{E}_{t_0, x_0} \left[\sqrt{\int_{t_0}^T (a(s)X_s^u + b(s))^2 u_s^2 (X_s^u)^2 ds} \right] \\
& \stackrel{(3.4)+(3.5)}{\leq} \sqrt{2}C \left(\mathbb{E}_{t_0, x_0} \left[\sqrt{\int_{t_0}^t u_s^2 (X_s^u)^4 ds} \right] + \mathbb{E}_{t_0, x_0} \left[\sqrt{\int_{t_0}^t u_s^2 (X_s^u)^2 ds} \right] \right) \\
& \stackrel{Jensen}{\leq} \sqrt{2}C \left(\mathbb{E}_{t_0, x_0} \left[\sup_{t \leq s' \leq T} |X_{s'}^u| \sqrt{\int_{t_0}^t u_s^2 (X_s^u)^2 ds} \right] + \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^t u_s^2 (X_s^u)^2 ds \right]^{1/2} \right) \\
& \stackrel{CS}{\leq} \sqrt{2}C \left(\mathbb{E}_{t_0, x_0} \left[\left(\sup_{t \leq s' \leq T} |X_{s'}^u| \right)^2 \right]^{1/2} \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^t u_s^2 (X_s^u)^2 ds \right]^{1/2} \right. \\
& \quad \left. + \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^t u_s^2 (X_s^u)^2 ds \right]^{1/2} \right) \\
& = \underbrace{\sqrt{2}C \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^t u_s^2 (X_s^u)^2 ds \right]^{1/2}}_{< \infty} \left(\mathbb{E}_{t_0, x_0} \left[\left(\sup_{t \leq s' \leq T} |X_{s'}^u| \right)^2 \right]^{1/2} + 1 \right).
\end{aligned}$$

Note that $\mathbb{E}_{t_0, x_0} [\int_{t_0}^t u_s^2 (X_s^u)^2 ds]^{1/2} < \infty$ as $u \in \mathcal{A}_{t_0, x_0}$. Finally Consider that by the defining property of the wealth process (2.9) it holds that

$$X_t^u = x_0 + \int_{t_0}^t (r + (\mu - r)u_s) X_s^u ds + \int_{t_0}^t \sigma u_s X_s^u dW_s \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

We again use (3.5) and Cauchy-Schwarz to get $\mathbb{P}_{t_0, x_0}$ -a.s.

$$\begin{aligned}
(X_t^u)^2 & \stackrel{(3.5)}{\leq} 3x_0^2 + 3 \left(\int_{t_0}^t (r + (\mu - r)u_s) X_s^u ds \right)^2 + 3 \left(\int_{t_0}^t \sigma u_s X_s^u dW_s \right)^2 \\
& \stackrel{CS}{\leq} 3x_0^2 + 3 \int_{t_0}^t ((r + (\mu - r)u_s) X_s^u)^2 ds \int_{t_0}^t 1 ds + 3 \left(\int_{t_0}^t \sigma u_s X_s^u dW_s \right)^2 \\
& \stackrel{(3.5)}{\leq} 3x_0^2 + 6(T - t_0) \int_{t_0}^T (r^2 + (\mu - r)^2 u_s^2) (X_s^u)^2 ds + 3 \left(\int_{t_0}^t \sigma u_s X_s^u dW_s \right)^2.
\end{aligned}$$

Note that $\mathbb{E}_{t_0, x_0} [\int_{t_0}^T (r^2 + (\mu - r)^2 u_s^2) (X_s^u)^2 ds] < \infty$ as $u \in \mathcal{A}_{t_0, x_0}$. Now we use the Burkholder-Davis-Gundy inequality to obtain

$$\mathbb{E}_{t_0, x_0} \left[\sup_{t \leq s' \leq T} \left(\int_{t_0}^{s'} \sigma u_s X_s^u dW_s \right)^2 \right] \leq \bar{C} \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \sigma^2 u_s^2 (X_s^u)^2 ds \right] < \infty,$$

for some $\bar{C} > 0$, where the last inequality again holds since $u \in \mathcal{A}_{t_0, x_0}$. This shows that $\mathbb{E}_{t_0, x_0} \left[\left(\sup_{t \leq s' \leq T} |X_{s'}^u| \right)^2 \right] < \infty$. Therefore (3.10) holds and M is a martingale. $\square$

Consider now the control $u \equiv 0$. Investing with this control corresponds to holding all money in the bond. The corresponding wealth process X^u solves the SDE

$$dX_t^u = (r(1 - u(t, X_t^u)) + \mu u(t, X_t^u))X_t^u dt + \sigma u(t, X_t^u)X_t^u dW_t = rX_t^u dt.$$

This is actually an ordinary differential equation with the solution $X_t^u = x_0 e^{r(t-t_0)}$ $\mathbb{P}_{t_0, x_0}$ -a.s. as expected. Note that this wealth process is $\mathbb{P}_{t_0, x_0}$ -a.s. deterministic. One could ask, if there is any other control v satisfying $X_{t_0}^v = x_0$ and $X_T^v = x_0 e^{r(T-t_0)}$ for a given time horizon T . The answer is no.

Lemma 3.3. ² *No admissible control $u \in \mathcal{A}_{t_0, x_0}$ can move a given deterministic wealth x_0 at time $t_0 \in [0, T)$ to any other deterministic wealth at time T apart from $x_0 e^{r(T-t_0)}$ in which case u equals zero.*

Proof. Set $\delta = (\mu - r)/\sigma$ and define the measure $\tilde{\mathbb{P}}_{t_0, x_0}$ as

$$\frac{d\tilde{\mathbb{P}}_{t_0, x_0}}{d\mathbb{P}_{t_0, x_0}} := \exp \left\{ -\delta(W_T - W_{t_0}) - \frac{\delta^2}{2}(T - t_0) \right\}.$$

We show that $\tilde{W}_t := W_t - W_{t_0} + \delta(t - t_0)$ is a Brownian motion under $\tilde{\mathbb{P}}_{t_0, x_0}$ on $[t_0, T]$ starting in 0. Since the Radon-Nikodým derivative is strictly bigger than zero, Nullsets under $\tilde{\mathbb{P}}_{t_0, x_0}$ are the same as Nullsets under $\mathbb{P}_{t_0, x_0}$. So since $\tilde{W}$ has $\mathbb{P}_{t_0, x_0}$ -a.s. continuous trajectories, the process also has $\tilde{\mathbb{P}}_{t_0, x_0}$ -a.s. continuous trajectories. Note that $\tilde{W}_{t_0} = 0$.

Now we show that

$$(\tilde{W}_{t_1} - \tilde{W}_{t_0}, \tilde{W}_{t_2} - \tilde{W}_{t_1}, \dots, \tilde{W}_{t_n} - \tilde{W}_{t_{(n-1)}}) \sim \mathcal{N}((0, \dots, 0)^T, \text{diag}(t_1 - t_0, \dots, t_n - t_{(n-1)})).$$

Let $\lambda \in \mathbb{R}^n$ and $t_0 < t_1 < t_2 < \dots < t_n = T$ (the case $t_n < T$ follows from the case $\lambda_n = 0$). We then have

$$\tilde{\mathbb{E}}_{t_0, x_0} \left[\exp \left\{ i \sum_{k=1}^n (\tilde{W}_{t_k} - \tilde{W}_{t_{(k-1)}}) \lambda_k \right\} \right]$$

²This statement has been added to the paper as an answer after we contacted professor G. Peskir since we had a question regarding the proof of [2, Theorem 7, p.16] (see proof of Theorem 4.5 for more details).

$$\begin{aligned}
&= \mathbb{E}_{t_0, x_0} \left[\exp \left\{ i \sum_{k=1}^n (\tilde{W}_{t_k} - \tilde{W}_{t_{(k-1)}}) \lambda_k - \delta(W_T - W_{t_0}) - \frac{\delta^2}{2}(T - t_0) \right\} \right] \\
&= \mathbb{E}_{t_0, x_0} \left[\exp \left\{ i \sum_{k=1}^n (W_{t_k} - W_{t_{(k-1)}} + \delta(t_k - t_{(k-1)})) \lambda_k - \delta(W_T - W_{t_0}) - \frac{\delta^2}{2}(T - t_0) \right\} \right] \\
&= \mathbb{E}_{t_0, x_0} \left[\exp \left\{ \sum_{k=1}^n \left((W_{t_k} - W_{t_{(k-1)}} + \delta(t_k - t_{(k-1)})) i \lambda_k \right. \right. \right. \\
&\quad \left. \left. \left. - \delta(W_{t_k} - W_{t_{(k-1)}}) - \frac{\delta^2}{2}(t_k - t_{(k-1)}) \right) \right\} \right] \\
&= \mathbb{E}_{t_0, x_0} \left[\exp \left\{ \sum_{k=1}^n \left((W_{t_k} - W_{t_{(k-1)}}) (i \lambda_k - \delta) \right. \right. \right. \\
&\quad \left. \left. \left. + (t_k - t_{(k-1)}) \left(i \lambda_k \delta - \frac{\delta^2}{2} \right) \right) \right\} \right] \\
&= \exp \left\{ \sum_{k=1}^n \left(\frac{1}{2} (t_k - t_{(k-1)}) (i \lambda_k - \delta)^2 + (t_k - t_{(k-1)}) \left(i \lambda_k \delta - \frac{\delta^2}{2} \right) \right) \right\} \\
&= \exp \left\{ - \sum_{k=1}^n \frac{1}{2} (t_k - t_{(k-1)}) \lambda_k^2 \right\},
\end{aligned}$$

hence

$$(\tilde{W}_{t_1} - \tilde{W}_{t_0}, \tilde{W}_{t_2} - \tilde{W}_{t_1}, \dots, \tilde{W}_{t_n} - \tilde{W}_{t_{(n-1)}}) \sim \mathcal{N}((0, \dots, 0)^T, \text{diag}(t_1 - t_0, \dots, t_n - t_{(n-1)})).$$

Now consider that for any $u \in \mathcal{A}_{t_0, x_0}$ it holds that

$$\begin{aligned}
X_t^u e^{-r(T-t_0)} &\stackrel{(3.7)}{=} x_0 + \int_{t_0}^T \delta \sigma u_t X_t^u e^{-r(t-t_0)} dt + \int_{t_0}^T \sigma u_t X_t^u e^{-r(t-t_0)} dW_t \\
&= x_0 + \int_{t_0}^T \sigma u_t X_t^u e^{-r(t-t_0)} (\delta dt + dW_t) \\
&= x_0 + \int_{t_0}^T \sigma u_t X_t^u e^{-r(t-t_0)} d\tilde{W}_t \quad \tilde{\mathbb{P}}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

Note that the stochastic integral on the right-hand side $M_t := \int_{t_0}^T \sigma u_t X_t^u e^{-r(t-t_0)} d\tilde{W}_t$ is a $\tilde{\mathbb{P}}_{t_0, x_0}$ -martingale. This follows from the inequality

$$\begin{aligned}
&\tilde{\mathbb{E}}_{t_0, x_0} \left[\sqrt{[M, M]_T} \right] \\
&= \tilde{\mathbb{E}}_{t_0, x_0} \left[\sqrt{\int_{t_0}^T \sigma^2(u_s)^2 (X_s^u)^2 e^{-2r(s-t_0)} ds} \right]
\end{aligned}$$

$$\begin{aligned}
&= \mathbb{E}_{t_0, x_0} \left[\exp \left\{ -\delta(W_T - W_{t_0}) - \frac{\delta^2}{2}(T - t_0) \right\} \sqrt{\int_{t_0}^T \sigma^2(u_s)^2 (X_s^u)^2 e^{-2r(s-t_0)} ds} \right] \\
&\leq \left(\mathbb{E}_{t_0, x_0} \left[\exp \left\{ -2\delta(W_T - W_{t_0}) - \delta^2(T - t_0) \right\} \right] \right)^{1/2} \\
&\quad \times \left(\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \sigma^2(u_s)^2 (X_s^u)^2 e^{-2r(s-t_0)} ds \right] \right)^{1/2} \\
&< \infty,
\end{aligned}$$

since

$$\begin{aligned}
&\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \sigma^2(u_s)^2 (X_s^u)^2 e^{-2r(s-t_0)} ds \right] \\
&\leq \max\{1, e^{-2r(T-t_0)}\} \sigma^2 \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T (u_s)^2 (X_s^u)^2 ds \right] \\
&< \infty,
\end{aligned}$$

as $u \in \mathcal{A}_{t_0, x_0}$. So we can use the same argumentation as in Lemma 3.2 to show that M is a martingale under $\tilde{\mathbb{P}}_{t_0, x_0}$.

Assume now that X_T^u is deterministic. Then also M_T would be deterministic. But then for all $t \in [t_0, T]$

$$M_t = \tilde{\mathbb{E}}_{t_0, x_0}[M_T \mid \mathcal{F}_t] = M_T \quad \tilde{\mathbb{P}}_{t_0, x_0}\text{-a.s.},$$

hence M would be constant. So $M \equiv M_{t_0} = 0$ and therefore $X_T^u = x_0 e^{r(T-t_0)}$. Since this argument holds for all $t \in [t_0, T]$, we even have $X_t^u = x_0 e^{r(t-t_0)}$ for all $t \in [t_0, T]$ and therefore also $u \equiv 0$ (at least on the graph of X_t^u). $\square$

3.2 Mathematical formulation of the problems

Let $J(t, t_0, x_0; u) := \mathbb{E}_{t_0, x_0}[X_t^u] - c\text{Var}_{t_0, x_0}(X_t^u)$ for $u \in \mathcal{A}_{t_0, x_0}$, $t_0 \leq t \leq T$ and $c > 0$. Let $J(t_0, x_0; u) = J(T, t_0, x_0; u)$. Motivated by the introductory part in Chapter 1 and Lemma 3.2 we define the statically and dynamically optimal controls in the three problems:

- **Problem 1:** $\sup_{u \in \mathcal{A}_{t_0, x_0}} J(t_0, x_0; u)$, for $c > 0$ fixed.
- **Problem 2:** $\sup_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \text{Var}_{t_0, x_0}(X_T^u) \leq \alpha}} \mathbb{E}_{t_0, x_0}[X_T^u]$, for $\alpha > 0$ fixed.
- **Problem 3:** $\inf_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_T^u] \geq \beta}} \text{Var}_{t_0, x_0}(X_T^u)$, for $\beta \in \mathbb{R}$ fixed.

Definition 3.4 (Static optimality). ³ Fix $0 \leq t_0 < T < \infty$ and $x_0 \in \mathbb{R}$.

An admissible control $u_{1,c}^s$ on $[t_0, T]$ is *statically optimal in Problem 1* for (t_0, x_0) and $c > 0$, if $u_{1,c}^s \in \mathcal{A}_{t_0, x_0}$ and every other control $v \in \mathcal{A}_{t_0, x_0}$ satisfies

$$J(t_0, x_0; v) \leq J(t_0, x_0; u_{1,c}^s).$$

An admissible control $u_{2,\alpha}^s$ on $[t_0, T]$ is *statically optimal in Problem 2* for (t_0, x_0) and $\alpha > 0$, if $u_{2,\alpha}^s \in \mathcal{A}_{t_0, x_0}$, $\text{Var}_{t_0, x_0}(X_T^{u_{2,\alpha}^s}) \leq \alpha$ and every other control $v \in \mathcal{A}_{t_0, x_0}$ with $\text{Var}_{t_0, x_0}(X_T^v) \leq \alpha$ satisfies

$$\mathbb{E}_{t_0, x_0}[X_T^v] \leq \mathbb{E}_{t_0, x_0}[X_T^{u_{2,\alpha}^s}].$$

An admissible control $u_{3,\beta}^s$ on $[t_0, T]$ is *statically optimal in Problem 3* for (t_0, x_0) and $\beta \in \mathbb{R}$, if $u_{3,\beta}^s \in \mathcal{A}_{t_0, x_0}$, $\mathbb{E}_{t_0, x_0}[X_T^{u_{3,\beta}^s}] \geq \beta$ and every other control $v \in \mathcal{A}_{t_0, x_0}$ with $\mathbb{E}_{t_0, x_0}[X_T^v] \geq \beta$ satisfies

$$\text{Var}_{t_0, x_0}(X_T^v) \geq \text{Var}_{t_0, x_0}(X_T^{u_{3,\beta}^s}).$$

In fact a control is statically optimal if and only if the supremum, resp. the infimum of the corresponding expressions is attained in it.

³This definitions are slightly different from the ones given in J.L. Pedersen and G. Peskir [2, p. 5] as we only consider admissible controls in $\mathcal{A}_{t_0, x_0}$. This is needed in order to have enough properties to use Lemma 3.2. We give the precise explanation in the proof of Theorem 4.1.

As discussed in Chapter 1, the statically optimal controls inherit a *time inconsistency*⁴ that we wish to avoid. J.L. Pedersen and G. Peskir came up with a new notion of optimality, namely the dynamic optimality (see J.L. Pedersen and G. Peskir [2, p. 5-6]).

Definition 3.5 (Dynamic optimality). ⁵ Fix $0 < T < \infty$.

An admissible control $u_{1,c}^d$ on $[0, T[$ is *weakly dynamically optimal in Problem 1* for $c > 0$, if for every given and fixed $(t_0, x_0) \in [0, T[\times \mathbb{R}$ and every admissible control $v \in \mathcal{A}_{t_0, x_0}$ with $v(t_0, x_0)x_0 \neq u_{1,c}^d(t_0, x_0)x_0$, there exists a control $w \in \mathcal{A}_{t_0, x_0}$ satisfying $w(t_0, x_0)x_0 = u_{1,c}^d(t_0, x_0)x_0$ and

$$J(t_0, x_0; v) \leq J(t_0, x_0; w).$$

An admissible control $u_{2,\alpha}^d$ on $[0, T[$ is *weakly dynamically optimal in Problem 2* for $\alpha > 0$, if for every given and fixed $(t_0, x_0) \in [0, T[\times \mathbb{R}$ and every control $v \in \mathcal{A}_{t_0, x_0}$ with $v(t_0, x_0)x_0 \neq u_{2,\alpha}^d(t_0, x_0)x_0$ and $\text{Var}_{t_0, x_0}(X_T^v) \leq \alpha$, there exists a control $w \in \mathcal{A}_{t_0, x_0}$ satisfying $w(t_0, x_0)x_0 = u_{2,\alpha}^d(t_0, x_0)x_0$ with $\text{Var}_{t_0, x_0}(X_T^w) \leq \alpha$ and

$$\mathbb{E}_{t_0, x_0}[X_T^v] \leq \mathbb{E}_{t_0, x_0}[X_T^w].$$

An admissible control $u_{3,\beta}^d$ on $[0, T[$ is *weakly dynamically optimal in Problem 3* for $\beta \in \mathbb{R}$, if for every given and fixed $(t_0, x_0) \in [0, T[\times \mathbb{R}$ and every control $v \in \mathcal{A}_{t_0, x_0}$ with $v(t_0, x_0)x_0 \neq u_{3,\beta}^d(t_0, x_0)x_0$ and $\mathbb{E}_{t_0, x_0}[X_T^v] \geq \beta$, there exists a control $w \in \mathcal{A}_{t_0, x_0}$ satisfying $w(t_0, x_0)x_0 = u_{3,\beta}^d(t_0, x_0)x_0$ with $\mathbb{E}_{t_0, x_0}[X_T^w] \geq \beta$ and

$$\text{Var}(X_T^v) \geq \text{Var}(X_T^w).$$

We call a weakly dynamically optimal control strongly dynamically optimal if the corresponding inequality is strict.

Note that the condition $v(t_0, x_0)x_0 \neq u_{*,c}^d(t_0, x_0)x_0$ translates to $v(t_0, x_0) \neq u_{*,c}^d(t_0, x_0)$ if $x_0 \neq 0$ and the non trivial condition $\lim_{x \rightarrow 0} v(t_0, x)x \neq \lim_{x \rightarrow 0} u_{*,c}^d(t_0, x)x$ if $x_0 = 0$.

⁴See precise discussion in Remark 4.2.

⁵This definitions also differ from the ones given in J.L. Pedersen and G. Peskir [2, p. 5-6]. First of all we again restrict ourselves on $\mathcal{A}_{t_0, x_0}$. We also want to specify that the dynamically optimal controls are only admissible on $[t_0, T[$ (see Remark 4.4 and 4.6). Moreover we have changed the property $v(t_0, x_0) \neq u^d(t_0, x_0)$ that is not well defined for $x_0 = 0$ to $v(t_0, x_0)x_0 \neq u^d(t_0, x_0)x_0$ (well defined by the continuity property in Definition 2.10 and our convention $v(t, 0)0 = \lim_{x \rightarrow 0} v(t, x)x$). Also note that we introduce additionally the notion of *weakly* dynamical optimality (the *strongly* dynamically optimal controls here are the dynamically optimal controls in the paper). We will see that this is essential in order to solve Problem 2 dynamically as we will only be able to find a weakly dynamically optimal control for this problem if $\delta = 0$ (see proof of Theorem 4.5).

Chapter 4

Solution to the problems

Our goal is to find the closed form expressions of the optimal controls we defined before and their wealth processes. We will see that the main work is required in finding the statically and strongly dynamically optimal controls in Problem 1.

4.1 Optimal controls in Problem 1

Recall that Problem 1 corresponds to

$$\sup_{u \in \mathcal{A}_{t_0, x_0}} J(t_0, x_0; u)$$

for $c > 0$ given and fixed, where $J(t_0, x_0; u) := \mathbb{E}_{t_0, x_0}[X_T^u] - c \text{Var}_{t_0, x_0}(X_T^u)$.

Theorem 4.1. *Let X^u solve (2.9) with $X_{t_0}^u = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. for $(t_0, x_0) \in [0, T[\times \mathbb{R}$ given and fixed. Let B solve (2.8) and S solve (2.7). Set $\delta = (\mu - r)/\sigma$ for $\mu \in \mathbb{R}$, $r \in \mathbb{R}$ and $\sigma > 0$.*

(A) *Fix $(t_0, x_0) \in [0, T[\times \mathbb{R}$. A¹ statically optimal control in Problem 1 for $c > 0$ is given by*

$$(4.1) \quad u_{1,c}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \frac{1}{2c} e^{\delta^2(T-t_0)-r(T-t)} \right]$$

for $(t_0, x_0) \in [0, T[\times \mathbb{R}$. The corresponding wealth process is given by

$$(4.2) \quad X_t^s = x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t-W_{t_0})-\frac{\delta^2}{2}(t-t_0)} \right]$$

¹Note that we do not claim uniqueness here (see Remark 4.2)

$$(4.3) \quad = x_0 \frac{B_t}{B_{t_0}} + \frac{1}{2c} \left(\frac{B_t}{B_T} \right)^{1-\delta^2/r} \\ \times \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{B_t}{B_{t_0}} \right)^{\delta(1/\sigma + (\delta-\sigma)/(2r))} \left(\frac{S_t}{S_{t_0}} \right)^{-\delta/\sigma} \right]$$

$\mathbb{P}_{t_0, x_0}$ -a.s. for $t \in [t_0, T]$ (the second equality only holds for $r \neq 0$). The static value function² $V_s(t_0, x_0) := \lim_{t \uparrow T} J(t, t_0, x_0; u_{1,c}^s)$ is given by

$$(4.4) \quad V_s(t_0, x_0) = x_0 e^{r(T-t_0)} + \frac{1}{4c} \left[e^{\delta^2(T-t_0)} - 1 \right] = J(t_0, x_0; u_{1,c}^s).$$

(B) The unique strongly dynamically optimal control in Problem 1 for $c > 0$ is given by³

$$(4.5) \quad u_{1,c}^d(t, x) = \frac{\delta}{2c\sigma} \frac{1}{x} e^{(\delta^2-r)(T-t)}$$

for $(t, x) \in [0, T] \times (\mathbb{R} \setminus \{0\})$. Moreover, this control extends to an admissible control on $[0, T]$ with $u_{1,c}^d \in \mathcal{A}_{t_0, x_0}$ and the corresponding wealth process is given by

$$(4.6) \quad X_t^d = x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - 1 + \delta \int_{t_0}^t e^{\delta^2(t-s)} dW_s \right]$$

$$(4.7) \quad = x_0 \frac{B_t}{B_0} + \frac{\delta}{2c\sigma} \left(\frac{B_t}{B_T} \right)^{1-\delta^2/r} \left[\left(\frac{\sigma^2 - 2r}{2\delta^2} \right) \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - 1 \right] \right. \\ \left. + \log \left(\frac{S_t}{S_{t_0}} \right) + \delta^2 \int_{t_0}^t \left(\frac{B_t}{B_s} \right)^{\delta^2/r} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right]$$

$\mathbb{P}_{t_0, x_0}$ -a.s. for $t \in [t_0, T]$ (the second equality only holds for $r \neq 0$). The dynamic value function $V_d(t_0, x_0) := \lim_{t \uparrow T} J(t, t_0, x_0; u_{1,c}^d)$ is given by

$$(4.8) \quad V_d(t_0, x_0) = x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - \frac{1}{4} e^{2\delta^2(T-t_0)} - \frac{3}{4} \right] \\ = J(t_0, x_0; u_{1,c}^d)$$

for $(t_0, x_0) \in [0, T] \times \mathbb{R}$.

²Differently from J.L. Pedersen and G. Peskir [2, Theorem 3, p. 6] we define the value functions as limits $\lim_{t \uparrow T}$. The problem is that the dynamic value function for Problem 2 was not well defined ($X_T^d = X_T^{u_{2,\alpha}^d}$ does not exist there).

³Note that we do not have to distinguish the cases $\delta = 0$ and $\delta \neq 0$.

Proof. We prove this theorem in several steps. Let us first outline the proof here.

Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$. For the statically optimal control consider that the static value function can be written as:

$$\begin{aligned}
V_s(t_0, x_0) &= \sup_{u \in \mathcal{A}_{t_0, x_0}} (\mathbb{E}_{t_0, x_0}[X_T^u] - c \text{Var}(X_T^u)) \\
&= \sup_{u \in \mathcal{A}_{t_0, x_0}} \left(\mathbb{E}_{t_0, x_0}[X_T^u] + c \mathbb{E}_{t_0, x_0}[X_T^u]^2 - c \mathbb{E}_{t_0, x_0}[(X_T^u)^2] \right) \\
&= \sup_{M \in \mathbb{R}} \sup_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_{t_0}^u] = M}} \left(M + cM^2 - c \mathbb{E}_{t_0, x_0}[(X_T^u)^2] \right) \\
&= \sup_{M \in \mathbb{R}} [M + cM^2 - c \inf_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_{t_0}^u] = M}} \mathbb{E}_{t_0, x_0}[(X_T^u)^2]],
\end{aligned}$$

where the supremum in the last line is taken over all $M \in \mathbb{R}$ such that

$$\{u \in \mathcal{A}_{t_0, x_0} \mid \mathbb{E}_{t_0, x_0}[X_{t_0}^u] = M\} \neq \emptyset.$$

We will see, that this set is never empty if $\delta \neq 0$ and only non-empty for one $M \in \mathbb{R}$ if $\delta = 0$. We first consider the case $\delta \neq 0$ and then the case $\delta = 0$ separately. Hence in order to solve the static problem we can consider the problem

$$(4.9) \quad V_M(t_0, x_0) := \inf_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_{t_0}^u] = M}} \mathbb{E}_{t_0, x_0}[(X_T^u)^2]$$

for $M \in \mathbb{R}$ given and fixed and then maximize over all $M \in \mathbb{R}$. We solve (4.9) by applying the method of Lagrange multipliers. Define the Lagrangian

$$L_{t_0, x_0}(u, \lambda) = \mathbb{E}_{t_0, x_0}[(X_T^u)^2] - \lambda [\mathbb{E}_{t_0, x_0}[X_T^u] - M]$$

for $\lambda \in \mathbb{R}$ and $u \in \mathcal{A}_{t_0, x_0}$.

By the Hamilton-Jacobi-Bellman approach, we find in Step 1 for every $\lambda \in \mathbb{R}$ fixed a control $u_*^\lambda \in \mathcal{A}_{t_0, x_0}$ that solves the unconstrained problem

$$(4.10) \quad L_{t_0, x_0}(u_*^\lambda, \lambda) = \inf_{u \in \mathcal{A}_{t_0, x_0}} L_{t_0, x_0}(u, \lambda).$$

We also compute the corresponding wealth process $X^{u_*^\lambda}$ and additionally show that for $v \in \mathcal{A}_{t_0, x_0}$ with $u_*^\lambda(t_0, x_0)x_0 \neq v(t_0, x_0)x_0$ even the *strict* inequality

$$(4.11) \quad L_{t_0, x_0}(u_*^\lambda, \lambda) < L_{t_0, x_0}(v, \lambda)$$

holds. This will be needed to prove part (B) of the theorem.

In Step 2 we consider the case $\delta \neq 0$ and find $\lambda_* = \lambda_*(M, t_0, x_0) \in \mathbb{R}$ such that

$$\mathbb{E}_{t_0, x_0} \left[X_T^{u_*^{\lambda_*}} \right] = M.$$

Then it follows that for all admissible controls $u \in \mathcal{A}_{t_0, x_0}$ satisfying $\mathbb{E}_{t_0, x_0}(X_T^u) = M$ we have

$$\begin{aligned} E_{t_0, x_0} \left[\left(X_T^{u_*^{\lambda_*}} \right)^2 \right] &= L_{t_0, x_0}(u_*^{\lambda_*}, \lambda_*) \\ &\leq L_{t_0, x_0}(u, \lambda_*) \\ &= E_{t_0, x_0}[(X_T^u)^2]. \end{aligned}$$

Hence $u_*^{\lambda_*}$ is the minimizer in (4.9).

In Step 3 we recognize that the right-hand side of

$$V_s(t_0, x_0) = \sup_{M \in \mathbb{R}} [M + cM^2 - c\mathbb{E}_{t_0, x_0} \left[\left(X_t^{u_*^{\lambda_*}} \right)^2 \right]]$$

is a quadratic polynomial in M . We therefore can easily find the optimal $M = M_*$. Hence the statically optimal control in Problem 1 is $u_{1,c}^s = u_*^{\tilde{\lambda}}$ for $\tilde{\lambda} = \lambda_*(M_*, t_0, x_0)$ if $\delta \neq 0$.

In Step 4 we find the statically optimal control for the case $\delta = 0$. We first show that if $\delta = 0$, then for any $u \in \mathcal{A}_{t_0, x_0}$ it holds that $\mathbb{E}_{t_0, x_0}[X_t^u] = x_0 e^{r(t-t_0)}$ for all $t_0 \leq t \leq T$. It will then follow easily that the control $u \equiv 0$ is statically optimal in Problem 1. This will finish the proof of (A).

In Step 5 we show that the control $u_{1,c}^d$ defined through $u_{1,c}^d(t, x) = u_{1,c,t,x}^s(t, x)$ for all $(t, x) \in [0, T[\times (\mathbb{R} \setminus \{0\})$, where for all $(t_0, x_0) \in [0, T[\times \mathbb{R}$, u_{1,c,t_0,x_0}^s denotes the statically optimal control in Problem 1 for (t_0, x_0) found above, is strongly dynamically optimal in Problem 1. In fact this mainly follows from the additional property of u_*^{λ} proven in Step 1. We also compute the corresponding wealth process of $u_{1,c}^d$ in this step.

Finally in Step 6 we show uniqueness of the strongly dynamically optimal control. This will finish the proof of (B).

Step 1 Fix $(t_0, x_0) \in [0, T[\times \mathbb{R}$. The infimum in (4.10) is attained in u_*^{λ} with

$$u_*^{\lambda}(t, x) = -\frac{\delta}{\sigma} \frac{1}{x} \left[x - \frac{\lambda}{2} e^{-r(T-t)} \right], \quad \text{for } (t, x) \in [t_0, T] \times \mathbb{R}$$

with the corresponding wealth process

$$X_t^{u_*^\lambda} = e^{r(t-t_0)} \left[K - (K - x_0) e^{-\delta(W_t - W_{t_0}) - \frac{3}{2}\delta^2(t-t_0)} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.},$$

where $K := (\lambda/2)e^{-r(T-t_0)}$. In particular $u_*^\lambda(t, x) \in \mathcal{A}_{t_0, x_0}$ and for all $v \in \mathcal{A}_{t_0, x_0}$

$$(4.12) \quad L_{t_0, x_0}(v, \lambda) \geq L_{t_0, x_0}(u_*^\lambda, \lambda).$$

Moreover, for any control $v \in \mathcal{A}_{t_0, x_0}$ with $v(t_0, x_0)x_0 \neq u_*^\lambda(t_0, x_0)x_0$ we even have the strict inequality

$$(4.13) \quad L_{t_0, x_0}(v, \lambda) > L_{t_0, x_0}(u_*^\lambda, \lambda).$$

Recall that

$$L_{t_0, x_0}(u, \lambda) = \mathbb{E}_{t_0, x_0}[(X_T^u)^2] - \lambda[\mathbb{E}_{t_0, x_0}[X_T^u] - M].$$

In order to solve (4.10) we can therefore consider the equivalent optimal control problem

$$(4.14) \quad \tilde{V}^\lambda(t_0, x_0) = \inf_{u \in \mathcal{A}_{t_0, x_0}} \mathbb{E}_{t_0, x_0} \left[(X_T^u)^2 - \lambda X_T^u \right]$$

which gives us the same optimal control u_*^λ . This problem can now be solved with the Hamilton-Jacobi-Bellman (HJB) approach. The HJB system for this problem is

$$(4.15) \quad \inf_{u \in \mathbb{R}} \left[V_t^\lambda(t, x) + (r(1-u) + \mu u)x V_x^\lambda(t, x) + \frac{1}{2}\sigma^2 u^2 x^2 V_{xx}^\lambda(t, x) \right] = 0$$

$$(4.16) \quad V^\lambda(T, x) = x^2 - \lambda x,$$

where the first equality should hold for each $(t, x) \in]0, T[\times (\mathbb{R} \setminus \{0\})$ fixed. Note that these equations do not depend on (t_0, x_0) .

We define u_*^λ on $]0, T[\times (\mathbb{R} \setminus \{0\})$ such that for all $(t, x) \in]0, T[\times (\mathbb{R} \setminus \{0\})$ the infimum in (4.15) is uniquely attained in $u = u_*^\lambda(t, x)$. Moreover, it will be easy to see that V^λ extends to a continuous map on $[0, T] \times \mathbb{R}$. This will give us that u_*^λ is continuous on $]0, T[\times (\mathbb{R} \setminus \{0\})$ and even extends to a continuous map on $[0, T] \times (\mathbb{R} \setminus \{0\})$ for which $(t, x) \mapsto u_*^\lambda(t, x)x$ is continuously continuable on $[0, T] \times \mathbb{R}$

as required in Definition 2.10. Then we fix $(t_0, x_0) \in [0, T[\times \mathbb{R}$ and show that $u_*^\lambda \in \mathcal{A}_{t_0, x_0}$ and that for any admissible control $v \in \mathcal{A}_{t_0, x_0}$ the two identities

$$(4.17) \quad \mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_*^\lambda} \right)^2 - \lambda X_T^{u_*^\lambda} \right] = V^\lambda(t_0, x_0)$$

$$(4.18) \quad \mathbb{E}_{t_0, x_0} \left[(X_T^v)^2 - \lambda X_T^v \right] \geq V^\lambda(t_0, x_0).$$

hold, i.e. that $\tilde{V}^\lambda = V^\lambda$, where V^λ is the solution to the HJB system and $\tilde{V}^\lambda$ as in (4.14). Moreover, we show that for the case $u_*^\lambda(t_0, x_0)x_0 \neq v(t_0, x_0)x_0$ we even have the strict inequality in (4.18). This will then be sufficient for proving (4.12) and (4.13).

Step 1. 1 *The control*

$$(4.19) \quad u_*^\lambda(t, x) = -\frac{\delta}{\sigma} \frac{1}{x} \left[x - \frac{\lambda}{2} e^{-r(T-t)} \right]$$

and the map

$$(4.20) \quad V^\lambda(t, x) = e^{-(\delta^2 - 2r)(T-t)} x^2 - \lambda e^{-(\delta^2 - r)(T-t)} x - \frac{\lambda^2}{4} \left[1 - e^{\delta^2(T-t)} \right],$$

for $(t, x) \in [t_0, T] \times \mathbb{R}$ satisfy the above requirements, i.e. V^λ is a solution to the HJB system, for all $(t, x) \in]0, T[\times (\mathbb{R} \setminus \{0\})$ the infimum in (4.15) is uniquely attained in $u = u_*^\lambda(t, x)$ and u_*^λ is continuous on $]0, T[\times (\mathbb{R} \setminus \{0\})$ and extends to a continuous map on $[0, T] \times (\mathbb{R} \setminus \{0\})$ for which $(t, x) \mapsto u_*^\lambda(t, x)x$ is continuously continuable on $[0, T] \times \mathbb{R}$.

In order to solve the HJB system we make the ansatz $V_{xx}^\lambda > 0$. Reformulating (4.15) gives us the condition

$$(4.21) \quad \inf_{u \in \mathbb{R}} \left[\left(\frac{1}{2} \sigma^2 x^2 V_{xx}^\lambda \right) u^2 + \left((\mu - r)x V_x^\lambda \right) u + \left(V_t^\lambda + rx V_x^\lambda \right) \right] = 0.$$

As minimizing a quadratic function $f(z) = az^2 + bz + c$ for $a > 0$ leads to $f'(z^*) = 2az^* + b = 0$. Hence $z^* = -\frac{b}{2a}$ is the unique minimal point and therefore $f(z^*) = \frac{b^2}{4a} - \frac{b^2}{2a} + c = c - \frac{b^2}{4a}$. So (4.15) leads to the condition

$$0 = \left(V_t^\lambda + rxV_x^\lambda \right) - \frac{\left((\mu - r)xV_x^\lambda \right)^2}{4 \left(\frac{1}{2}\sigma^2 x^2 V_{xx}^\lambda \right)}$$

on $]0, T[\times (\mathbb{R} \setminus \{0\})$ and the definition

$$u_*^\lambda = - \frac{(\mu - r)xV_x^\lambda}{2 \left(\frac{1}{2}\sigma^2 x^2 V_{xx}^\lambda \right)}.$$

Recall that we set $\delta = (\mu - r)/\sigma$. So the equations read:

$$(4.22) \quad u_*^\lambda = - \frac{\delta}{\sigma} \frac{1}{x} \frac{V_x^\lambda}{V_{xx}^\lambda},$$

$$(4.23) \quad 0 = V_t^\lambda + rxV_x^\lambda - \frac{\delta^2 (V_x^\lambda)^2}{2 V_{xx}^\lambda}.$$

In order to solve the partial differential equation (4.23), we make the ansatz $V^\lambda(t, x) = a(t)x^2 + b(t)x + c(t)$ for $a, b, c \in C^1([0, T])$. Inserting this ansatz into the equation leads to

$$(a'(t)x^2 + b'(t)x + c'(t)) + rx(2a(t)x + b(t)) - \frac{\delta^2 (2a(t)x + b(t))^2}{2 \cdot 2a(t)} = 0.$$

Rearranging the equation we get

$$\begin{aligned} 0 &= \left(a'(t) + 2ra(t) - \frac{\delta^2 4a(t)^2}{2 \cdot 2a(t)} \right) x^2 + \left(b'(t) + rb(t) - \frac{\delta^2 4a(t)b(t)}{2 \cdot 2a(t)} \right) x + \\ &\quad \left(c'(t) - \frac{\delta^2 b(t)^2}{2 \cdot 2a(t)} \right) \\ &= \left(a'(t) + a(t)(2r - \delta^2) \right) x^2 + \left(b'(t) + b(t)(r - \delta^2) \right) x + \\ &\quad \left(c'(t) - \frac{\delta^2 b(t)^2}{2 \cdot 2a(t)} \right). \end{aligned}$$

As this has to hold for all $(t, x) \in]0, T[\times (\mathbb{R} \setminus \{0\})$, viewing the left-hand-side as a polynomial in x with coefficients depending on t , all coefficients have to be constantly zero. So we obtain

$$a'(t) = -a(t)(2r - \delta^2)$$

$$b'(t) = -b(t)(r - \delta^2)$$

$$c'(t) = \frac{\delta^2}{2} \frac{b(t)^2}{2a(t)} = -\frac{\delta^2}{4} \frac{b(t)^2}{a(t)}$$

and (4.16) gives:

$$a(T)x^2 + b(T)x + c(T) = x^2 - \lambda x$$

for all $x \in \mathbb{R}$. Hence we have

$$\begin{aligned} a(T) &= 1 \\ b(T) &= -\lambda \\ c(T) &= 0. \end{aligned}$$

So we obtain the solutions

$$\begin{aligned} a(t) &= e^{-(2r-\delta^2)(t-T)} = e^{-(\delta^2-2r)(T-t)} \\ b(t) &= -\lambda e^{-(r-\delta^2)(t-T)} = -\lambda e^{-(\delta^2-r)(T-t)} \\ c(t) &= c(T) - \int_t^T \frac{\delta^2}{2} \frac{b(y)^2}{2a(y)} dy \\ &= - \int_t^T \frac{\delta^2}{2} \frac{\left(-\lambda e^{-(\delta^2-r)(T-y)}\right)^2}{2e^{-(\delta^2-2r)(T-y)}} dy \\ &= -\frac{\lambda^2 \delta^2}{4} \int_t^T e^{-2(\delta^2-r)(T-y) + (\delta^2-2r)(T-y)} dy \\ &= -\frac{\lambda^2 \delta^2}{4} \int_t^T e^{\delta^2(T-y)} dy \\ &= -\frac{\lambda^2 \delta^2}{4} \left[\frac{1}{\delta^2} e^{\delta^2(T-y)} \right]_t^T \\ &= -\frac{\lambda^2}{4} \left[1 - e^{\delta^2(T-t)} \right]. \end{aligned}$$

Therefore we get the expression

$$V^\lambda(t, x) = e^{-(\delta^2-2r)(T-t)} x^2 - \lambda e^{-(\delta^2-r)(T-t)} x - \frac{\lambda^2}{4} \left[1 - e^{\delta^2(T-t)} \right]$$

for all $(t, x) \in]0, T[\times (\mathbb{R} \setminus \{0\})$ as claimed in (4.20).⁴ Note that as $a(t) > 0$ for all $0 \leq t \leq T$ it holds that $V_{xx}^\lambda > 0$ so these solutions are reasonable. Clearly V^λ extends to a continuous map on $[t_0, T] \times \mathbb{R}$ as claimed. Inserting these solutions into (4.22) we obtain for all $(t, x) \in [0, T] \times (\mathbb{R} \setminus \{0\})$

$$\begin{aligned} u_*^\lambda(t, x) &= -\frac{\delta}{\sigma} \frac{1}{x} \frac{V_x^\lambda(t, x)}{V_{xx}^\lambda(t, x)} \\ &= -\frac{\delta}{\sigma} \frac{1}{x} \frac{2a(t)x + b(t)}{2a(t)} \\ &= -\frac{\delta}{\sigma} \frac{1}{x} \left[x + \frac{b(t)}{2a(t)} \right] \\ &= -\frac{\delta}{\sigma} \frac{1}{x} \left[x - \frac{\lambda e^{-(\delta^2-r)(T-t)}}{2e^{-(\delta^2-2r)(T-t)}} \right] \\ &= -\frac{\delta}{\sigma} \frac{1}{x} \left[x - \frac{\lambda}{2} e^{-r(T-t)} \right], \end{aligned}$$

as claimed in (4.19). Note that $(t, x) \mapsto u_*^\lambda(t, x)x$ is continuously continuable on $(t, x) \in [0, T] \times \mathbb{R}$.

Step 1. 2 For all $(t, x) \in [t_0, T] \times \mathbb{R}$ and all $v \in \mathcal{A}_{t_0, x_0}$ with $v(t, x)x \neq u_*^\lambda(t, x)x$ we have⁵

$$(4.24) \quad \begin{aligned} 0 \leq & V_t^\lambda(t, x) + (r(1 - v(t, x)) + \mu v(t, x))x V_x^\lambda(t, x) \\ & + \frac{1}{2} \sigma^2 (v(t, x)x)^2 V_{xx}^\lambda(t, x) \end{aligned}$$

with equality if and only if $v(t, x)x = u_*^\lambda(t, x)x$. In particular this statement holds true for $x = 0$.

Recall that we use the convention $u_*^\lambda(t, 0) \cdot 0 = \lim_{x \rightarrow 0} u_*^\lambda(t, x) \cdot x$. By definition of u_*^λ we know that

$$(4.25) \quad 0 = V_t^\lambda(t, x) + (r(1 - u_*^\lambda(t, x)) + \mu u_*^\lambda(t, x))x V_x^\lambda(t, x)$$

⁴This actually only proves that if there is a solution to (4.23) of the form $a(t)x^2 + b(t)x + c(t)$ then the functions a, b and c have to be the ones given above. However, one should plug V^λ into (4.23) to show that this function really solves the equation.

⁵This is needed in order to extend the definition of dynamical optimality given in the paper to the definition we gave in this thesis.

$$+ \frac{1}{2} \sigma^2 (u_*^\lambda(t, x))^2 x^2 V_{xx}^\lambda(t, x)$$

holds for all $(t, x) \in]0, T[\times (\mathbb{R} \setminus \{0\})$ hence also for all $(t, x) \in [0, T] \times \mathbb{R}$. It is clear that

$$V_t^\lambda(t, x) + (r(1 - \tilde{v}) + \mu\tilde{v})xV_x^\lambda(t, x) + \frac{1}{2}\sigma^2\tilde{v}^2x^2V_{xx}^\lambda(t, x) \geq 0$$

for any $(t, x) \in [0, T] \times (\mathbb{R} \setminus \{0\})$ and $\tilde{v} \in \mathbb{R}$ with equality if and only if $\tilde{v} = u_*^\lambda(t, x)$. We show that it even holds that for any $v \in [0, T] \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$ with $(t, x) \mapsto v(t, x)x$ continuously continuable on $[0, T] \times \mathbb{R}$ and $v(t, 0) \cdot 0 \neq u_*^\lambda(t, 0) \cdot 0$ for some $t \in [0, T]$ fixed it holds that

$$\begin{aligned} & V_t^\lambda(t, 0) + (r(1 - v(t, 0)) + \mu v(t, 0)) \cdot 0 V_x^\lambda(t, 0) + \frac{1}{2} \sigma^2 (v(t, 0) \cdot 0)^2 V_{xx}^\lambda(t, 0) \\ &= V_t^\lambda(t, 0) + (\mu - r)v(t, 0) \cdot 0 V_x^\lambda(t, 0) + \frac{1}{2} \sigma^2 (v(t, 0) \cdot 0)^2 V_{xx}^\lambda(t, 0) \\ &> 0. \end{aligned}$$

But this is clear as the infimum in

$$\inf_{u_0 \in \mathbb{R}} [V_t^\lambda(t, 0) + (\mu - r)u_0 V_x^\lambda(t, 0) + \frac{1}{2} \sigma^2 u_0^2 V_{xx}^\lambda(t, 0)]$$

is uniquely attained in

$$u_0 = -\frac{\delta V_x^\lambda(t, 0)}{\sigma V_{xx}^\lambda(t, 0)} = u_*^\lambda(t, 0) \cdot 0$$

and is equal to zero by (4.25).

Step 1. 3 *The related wealth process for u_*^λ is given by*

$$(4.26) \quad X_t^{u_*^\lambda} = e^{r(t-t_0)} \left[K - (K - x_0) e^{-\delta(W_t - W_{t_0}) - \frac{3}{2}\delta^2(t-t_0)} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$. Recall that the wealth process solves the SDE

$$dX_t^{u_*^\lambda} = \left(r \left(1 - u_*^\lambda \left(t, X_t^{u_*^\lambda} \right) \right) + \mu u_*^\lambda \left(t, X_t^{u_*^\lambda} \right) \right) X_t^{u_*^\lambda} dt + \sigma u_*^\lambda \left(t, X_t^{u_*^\lambda} \right) X_t^{u_*^\lambda} dW_t$$

with $X_{t_0}^{u_\lambda} = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. This leads to the equation

$$\begin{aligned} dX_t^{u_\lambda} &\stackrel{(4.19)}{=} \left(r \left(1 + \frac{\delta}{\sigma} \frac{1}{X_t^{u_\lambda}} \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] \right) - \mu \frac{\delta}{\sigma} \frac{1}{X_t^{u_\lambda}} \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] \right) X_t^{u_\lambda} dt \\ &\quad - \sigma \frac{\delta}{\sigma} \frac{1}{X_t^{u_\lambda}} \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] X_t^{u_\lambda} dW_t, \\ &= \left(r X_t^{u_\lambda} + (r - \mu) \frac{\delta}{\sigma} \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] \right) dt - \delta \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] dW_t, \end{aligned}$$

hence

$$(4.27) \quad dX_t^{u_\lambda} = \left(r X_t^{u_\lambda} + \delta^2 \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] \right) dt - \delta \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] dW_t,$$

with $X_{t_0}^{u_\lambda} = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. Define the process

$$(4.28) \quad Z_t^\lambda := K - e^{-r(t-t_0)} X_t^{u_\lambda}$$

for $K := (\lambda/2)e^{-r(T-t_0)}$. Similar to the proof of Lemma 3.2 we obtain by Itô's formula

$$\begin{aligned} dZ_t^\lambda &= e^{-r(t-t_0)} dX_t^{u_\lambda} - r e^{-r(t-t_0)} X_t^{u_\lambda} dt \\ &\stackrel{(4.27)}{=} e^{-r(t-t_0)} \left(\left(r X_t^{u_\lambda} + \delta^2 \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] \right) dt - \delta \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] dW_t \right) - \cancel{r e^{-r(t-t_0)} X_t^{u_\lambda} dt} \\ &= e^{-r(t-t_0)} \left(\delta^2 \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] dt - \delta \left[X_t^{u_\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] dW_t \right) \\ &= \delta^2 \left[X_t^{u_\lambda} e^{-r(t-t_0)} - \frac{\lambda}{2} e^{-r(T-t_0)} \right] dt - \delta \left[X_t^{u_\lambda} e^{-r(t-t_0)} - \frac{\lambda}{2} e^{-r(T-t_0)} \right] dW_t \\ &= -\delta^2 Z_t dt - \delta Z_t dW_t. \end{aligned}$$

By Lemma 2.4 the unique strong solution of this linear SDE reads

$$\begin{aligned} Z_t^\lambda &= Z_{t_0} e^{(-\delta^2 - \frac{(-\delta)^2}{2})(t-t_0) - \delta(W_t - W_{t_0})} \\ &= (K - x_0) e^{-\frac{3\delta^2}{2}(t-t_0) - \delta(W_t - W_{t_0})} \quad \mathbb{P}_{t_0, x_0}\text{-a.s.} \end{aligned}$$

So it holds that

$$K - e^{-r(t-t_0)} X_t^{u_*^\lambda} \stackrel{(4.28)}{=} Z_t^\lambda = (K - x_0) e^{-\frac{3\delta^2}{2}(t-t_0) - \delta(W_t - W_{t_0})} \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

Hence we obtain the wealth process

$$(4.29) \quad X_t^{u_*^\lambda} = e^{r(t-t_0)} \left[K - (K - x_0) e^{-\delta(W_t - W_{t_0}) - \frac{3}{2}\delta^2(t-t_0)} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

for $t \in [t_0, T]$, as claimed in (4.26). Note that X^λ is a solution of the corresponding SDE if and only if Z^λ is a solution to the linear SDE given above. So X^λ is the unique solution of the corresponding SDE. This shows that u_*^λ is admissible on $[t_0, T]$.

Step 1. 4 *It holds that $u_*^\lambda \in \mathcal{A}_{t_0, x_0}$, i.e.*

$$\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T ((u_*^\lambda)_t^2 + 1) (X_t^{u_*^\lambda})^2 dt \right] < \infty.$$

Note that

$$\begin{aligned} \mathbb{E}_{t_0, x_0} \left[(X_t^{u_*^\lambda})^2 \right] &\stackrel{(4.29)}{=} \mathbb{E}_{t_0, x_0} \left[e^{2r(t-t_0)} \left[K - (K - x_0) e^{-\delta(W_t - W_{t_0}) - \frac{3}{2}\delta^2(t-t_0)} \right]^2 \right] \\ &= e^{2r(t-t_0)} \left[K^2 - 2K(K - x_0) \mathbb{E}_{t_0, x_0} \left[e^{-\delta(W_t - W_{t_0})} \right] e^{-\frac{3}{2}\delta^2(t-t_0)} \right. \\ &\quad \left. + (K - x_0)^2 \mathbb{E}_{t_0, x_0} \left[e^{-2\delta(W_t - W_{t_0})} \right] e^{-3\delta^2(t-t_0)} \right] \\ &= e^{2r(t-t_0)} \\ &\quad \times \left[K^2 - 2K(K - x_0) e^{-\delta^2(t-t_0)} + (K - x_0)^2 e^{-\delta^2(t-t_0)} \right] \\ (4.30) \quad &= e^{2r(t-t_0)} \left[K^2 + (x_0^2 - K^2) e^{-\delta^2(t-t_0)} \right]. \end{aligned}$$

Hence $t \mapsto \mathbb{E}_{t_0, x_0} \left[(X_t^{u_*^\lambda})^2 \right]$ is continuous and therefore bounded on $[t_0, T]$. By the definition $u_t = u(t, X_t^u)$ for u admissible we obtain

$$\begin{aligned} &\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T ((u_*^\lambda)_t^2 + 1) (X_t^{u_*^\lambda})^2 dt \right] \\ &= \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \left(\left(-\frac{\delta}{\sigma} \frac{1}{X_t^{u_*^\lambda}} \left[X_t^{u_*^\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right] \right)^2 + 1 \right) (X_t^{u_*^\lambda})^2 dt \right] \end{aligned}$$

$$\begin{aligned}
&= \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \left(\frac{\delta^2}{\sigma^2} \left[X_t^{u_*^\lambda} - \frac{\lambda}{2} e^{-r(T-t)} \right]^2 + (X_t^{u_*^\lambda})^2 \right) dt \right] \\
&\stackrel{(3.5)}{\leq} \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \left(2 \frac{\delta^2}{\sigma^2} \left[(X_t^{u_*^\lambda})^2 + \frac{\lambda^2}{4} e^{-2r(T-t)} \right] + (X_t^{u_*^\lambda})^2 \right) dt \right] \\
&= \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \left(\frac{\delta^2}{\sigma^2} \frac{\lambda^2}{2} e^{-2r(T-t)} + \left(2 \frac{\delta^2}{\sigma^2} + 1 \right) (X_t^{u_*^\lambda})^2 \right) dt \right] \\
&= \int_{t_0}^T \left(\frac{\delta^2}{\sigma^2} \frac{\lambda^2}{2} e^{-2r(T-t)} + \left(2 \frac{\delta^2}{\sigma^2} + 1 \right) \mathbb{E}_{t_0, x_0} \left[(X_t^{u_*^\lambda})^2 \right] \right) dt \\
&< \infty,
\end{aligned}$$

since the integrand is bounded on $[t_0, T]$.

Step 1. 5 Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$. Then (4.17) and (4.18) hold, i.e.

$$\begin{aligned}
\mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_*^\lambda} \right)^2 - \lambda X_T^{u_*^\lambda} \right] &= V^\lambda(t_0, x_0) \\
\mathbb{E}_{t_0, x_0} \left[(X_T^v)^2 - \lambda X_T^v \right] &\geq V^\lambda(t_0, x_0).
\end{aligned}$$

hold for every $v \in \mathcal{A}_{t_0, x_0}$. This means that $\tilde{V}^\lambda = V^\lambda$, where V^λ is the solution to the HJB system and

$$\tilde{V}^\lambda(t_0, x_0) = \inf_{u \in \mathcal{A}_{t_0, x_0}} \mathbb{E}_{t_0, x_0} \left[(X_T^u)^2 - \lambda X_T^u \right]$$

as in (4.14). Moreover, we show that for the case $u_*^\lambda(t_0, x_0)x_0 \neq v(t_0, x_0)x_0$ we even have the strict inequality in (4.18). In particular (4.12) and (4.13) hold true.

Fix any $v \in \mathcal{A}_{t_0, x_0}$. Note that by the second HJB equation (4.16) we have $(X_T^v)^2 - \lambda X_T^v = V^\lambda(T, X_T^v)$. Hence by Itô's formula we obtain

$$\begin{aligned}
(X_T^v)^2 - \lambda X_T^v &= V^\lambda(T, X_T^v) \\
&= V^\lambda(t_0, x_0) + \int_{t_0}^T V_t^\lambda(s, X_s^v) ds + \int_{t_0}^T V_x^\lambda(s, X_s^v) dX_s^v \\
&\quad + \int_{t_0}^T \frac{1}{2} V_{xx}^\lambda(s, X_s^v) d[X^v, X^v]_s \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

Recall that $dX_s^v = (r(1-v(s, X_s^v)) + \mu v(s, X_s^v))X_s^v ds + \sigma v(s, X_s^v)X_s^v dW_s$ and hence also $d[X^v, X^v]_s = \sigma^2 v^2(s, X_s^v)(X_s^v)^2 ds$. Therefore we obtain $\mathbb{P}_{t_0, x_0}$ -a.s.

(4.31)

$$\begin{aligned} (X_T^v)^2 - \lambda X_T^v &= V^\lambda(t_0, x_0) \\ &+ \int_{t_0}^T \left[V_t^\lambda(s, X_s^v) + V_x^\lambda(s, X_s^v)((r(1-v(s, X_s^v)) \right. \\ &\quad \left. + \mu v(s, X_s^v))X_s^v) + \frac{1}{2} V_{xx}^\lambda(s, X_s^v) \sigma^2 v^2(s, X_s^v)(X_s^v)^2 \right] ds + M_T, \end{aligned}$$

where $M_t := \int_{t_0}^t V_x^\lambda(s, X_s^v) \sigma v_s X_s^v dW_s$. Note that M is a martingale by Lemma 3.2⁶ since $V_x^\lambda(t, x)$ is a linear polynomial in x with continuous (in t) coefficients. Now applying $\mathbb{E}_{t_0, x_0}$ on both sides of (4.31) we obtain

$$\begin{aligned} (4.32) \quad \mathbb{E}_{t_0, x_0} \left[(X_T^v)^2 - \lambda X_T^v \right] &= V^\lambda(t_0, x_0) \\ &+ \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \left[V_t^\lambda(s, X_s^v) + V_x^\lambda(s, X_s^v)((r(1-v_s) \right. \right. \\ &\quad \left. \left. + \mu v_s)X_s^v) + \frac{1}{2} V_{xx}^\lambda(s, X_s^v) \sigma^2 v_s^2 (X_s^v)^2 \right] ds \right]. \end{aligned}$$

Note that by (4.24) for any $v \in \mathcal{A}_{t_0, x_0}$ the expectation on the right-hand side of (4.32) is positive, i.e.

$$(4.33) \quad \mathbb{E}_{t_0, x_0} \left[(X_T^v)^2 - \lambda X_T^v \right] \geq V^\lambda(t_0, x_0),$$

as claimed in (4.18) and for $v = u_*^\lambda$ the expectation even vanishes, so

$$\mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_*^\lambda} \right)^2 - \lambda X_T^{u_*^\lambda} \right] = V^\lambda(t_0, x_0),$$

as claimed in (4.17).

Now assume that $u_*^\lambda(t_0, x_0)x_0 \neq v(t_0, x_0)x_0$. We show that in this case even the strict inequality holds. As both $(t, x) \mapsto u_*^\lambda(t, x)x$ and $(t, x) \mapsto v(t, x)x$ are

⁶Here we want to explain why we only optimize over $\mathcal{A}_{t_0, x_0}$ and not over all admissible controls v satisfying $\mathbb{E}_{t_0, x_0}[\int_{t_0}^T v^2(X_t^v)^2] < \infty$ as in the paper. The problem is that $\mathbb{E}_{t_0, x_0}[\sqrt{[M, M]_T}] < \infty$ would not hold in general. So this would cause problems in proving that M is a martingale. However, in agreement with G. Peskir we decided to introduce the subset $\mathcal{A}_{t_0, x_0}$ of admissible controls.

continuous (in particular also the difference), there exists an neighborhood $R_\epsilon := [t_0, t_0 + \epsilon] \times [x_0 - \epsilon, x_0 + \epsilon]$ for some $\epsilon > 0$ such that $t_0 + \epsilon \leq T$ and $u_*^\lambda(t, x)x \neq v(t, x)x$ for all $(t, x) \in R_\epsilon$. Note that the map

$$(t, x) \mapsto V_t^\lambda(t, x) + V_x^\lambda(t, x)((r(1 - v(t, x)) + \mu v(t, x))x) + \frac{1}{2}V_{xx}^\lambda(t, x)\sigma^2 v^2(t, x)x^2$$

is continuous on $[t_0, T] \times \mathbb{R}$. As R_ϵ is compact and by again (4.24), we can define

$$0 < \beta := \min_{(t, x) \in R_\epsilon} \left[V_t^\lambda(t, x) + V_x^\lambda(t, x)((r(1 - v(t, x)) + \mu v(t, x))x) + \frac{1}{2}V_{xx}^\lambda(t, x)\sigma^2 v^2(t, x)x^2 \right].$$

Set $\tau_\epsilon := \inf\{t \in [t_0, t_0 + \epsilon] \mid (t, X_t^v) \notin R_\epsilon\}$. Note that by continuity of $t \mapsto X_t^v(\omega)$ for $\mathbb{P}_{t_0, x_0}$ -a.e. $\omega \in \Omega$ as strong solution of an SDE, $\tau_\epsilon > t_0$, $\mathbb{P}_{t_0, x_0}$ -a.s. So we obtain

$$\begin{aligned} & \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \left[V_t^\lambda(s, X_s^v) + V_x^\lambda(s, X_s^v)((r(1 - v_s) + \mu v_s)X_s^v) \right. \right. \\ & \quad \left. \left. + \frac{1}{2}V_{xx}^\lambda(s, X_s^v)\sigma^2 v_s^2(X_s^v)^2 \right] ds \right] \\ & \geq \beta \mathbb{E}_{t_0, x_0}[\tau_\epsilon - t_0] > 0. \end{aligned}$$

Hence it follows from (4.32) that

$$(4.34) \quad \mathbb{E}_{t_0, x_0} \left[(X_T^v)^2 - \lambda X_T^v \right] > V^\lambda(t_0, x_0) = \mathbb{E}_{t_0, x_0} \left[(X_T^{u_*^\lambda})^2 - \lambda X_T^{u_*^\lambda} \right].$$

Now recall that

$$L_{t_0, x_0}(u, \lambda) = \mathbb{E}_{t_0, x_0}[(X_T^u)^2] - \lambda[\mathbb{E}_{t_0, x_0}[X_T^u] - M]$$

and realize that (4.12) and (4.13) correspond exactly to (4.33) and (4.34).

Step 2 *Let $\delta \neq 0$. Fix $M \in \mathbb{R}$ and define*

$$\lambda_* = \lambda_*(M, t_0, x_0) = 2 \frac{M - x_0 e^{-(\delta^2 - r)(T - t_0)}}{1 - e^{-\delta^2(T - t_0)}}.$$

Then λ_ is the unique $\lambda \in \mathbb{R}$ with*

$$\mathbb{E}_{t_0, x_0} \left[X_T^{u_*^\lambda} \right] = M.$$

Moreover, the infimum in

$$(4.35) \quad V_M(t_0, x_0) := \inf_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_T^u] = M}} \mathbb{E}_{t_0, x_0}[(X_T^u)^2]$$

is attained in $u_*^{\lambda_*}$.

We need to find $\lambda \in \mathbb{R}$ such that $\mathbb{E}_{t_0, x_0}[X_T^{\lambda}] = M$ holds. This means

$$\begin{aligned} M &= \mathbb{E}_{t_0, x_0}[X_T^{\lambda}] \\ &= e^{r(T-t_0)} \left[K - (K - x_0) \mathbb{E}_{t_0, x_0}[e^{-\delta(W_T - W_{t_0})}] e^{-\frac{3}{2}\delta^2(T-t_0)} \right] \\ &= e^{r(T-t_0)} \left[K - (K - x_0) e^{\frac{1}{2}\delta^2(T-t_0)} e^{-\frac{3}{2}\delta^2(T-t_0)} \right] \\ &= e^{r(T-t_0)} \left[x_0 e^{-\delta^2(T-t_0)} + K(1 - e^{-\delta^2(T-t_0)}) \right] \\ &= e^{r(T-t_0)} \left[x_0 e^{-\delta^2(T-t_0)} + \frac{\lambda}{2} e^{-r(T-t_0)} (1 - e^{-\delta^2(T-t_0)}) \right] \\ &= x_0 e^{-(\delta^2-r)(T-t_0)} + \frac{\lambda}{2} [1 - e^{-\delta^2(T-t_0)}]. \end{aligned}$$

Since we assumed $\delta \neq 0$, there exists a unique λ_* with $M = \mathbb{E}_{t_0, x_0}[X_T^{\lambda_*}]$ namely

$$\lambda_* = 2 \frac{M - x_0 e^{-(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}}.$$

We repeat the argument from the beginning of the proof in order to show that the infimum in (4.35) is attained in $u_*^{\lambda_*}$: For all admissible controls $u \in \mathcal{A}_{t_0, x_0}$ satisfying $\mathbb{E}_{t_0, x_0}(X_T^u) = M$ we have

$$\begin{aligned} E_{t_0, x_0} \left[\left(X_T^{u_*^{\lambda_*}} \right)^2 \right] &= \mathbb{E}_{t_0, x_0}[(X_T^{u_*^{\lambda_*}})^2] - \lambda \underbrace{[\mathbb{E}_{t_0, x_0}[X_T^{u_*^{\lambda_*}}] - M]}_0 \\ &= L_{t_0, x_0}(u_*^{\lambda_*}, \lambda_*) \\ &\stackrel{(4.12)}{\leq} L_{t_0, x_0}(u, \lambda_*) \\ &= \mathbb{E}_{t_0, x_0}[(X_T^u)^2] - \lambda \underbrace{[\mathbb{E}_{t_0, x_0}[X_T^u] - M]}_0 \\ &= E_{t_0, x_0}[(X_T^u)^2]. \end{aligned}$$

Step 3 *The statically optimal control in Problem 1 for the case $\delta \neq 0$ is $u_{1,c}^s = u_{*}^{\tilde{\lambda}}$, where $\tilde{\lambda} = \lambda_{*}(M_{*}, t_0, x_0)$ with*

$$M_{*} = x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - 1 \right].$$

Moreover, we show that (4.2), (4.3) and (4.4) hold.

By Step 2 we can reformulate Problem 1 to

$$\begin{aligned} V_s(t_0, x_0) &= \sup_{u \in \mathcal{A}_{t_0, x_0}} [\mathbb{E}_{t_0, x_0}[X_T^u] - c \text{Var}(X_T^u)] \\ &= \sup_{M \in \mathbb{R}} [M + cM^2 - c \inf_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_T^u] = M}} \mathbb{E}_{t_0, x_0}[(X_T^u)^2]] \\ (4.36) \quad &= \sup_{M \in \mathbb{R}} \left[M + cM^2 - c \mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_{*}^{\lambda_{*}}} \right)^2 \right] \right]. \end{aligned}$$

We compute the second moment of the terminal wealth for general λ

$$\begin{aligned} &\mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_{*}^{\lambda}} \right)^2 \right] \\ &\stackrel{(4.30)}{=} e^{2r(T-t_0)} \left(K^2 + (x_0^2 - K^2) e^{-\delta^2(T-t_0)} \right) \\ &= e^{2r(T-t_0)} \left(\left(\frac{\lambda}{2} e^{-r(T-t_0)} \right)^2 + \left(x_0^2 - \left(\frac{\lambda}{2} e^{-r(T-t_0)} \right)^2 \right) e^{-\delta^2(T-t_0)} \right) \\ &= x_0^2 e^{-(\delta^2-2r)(T-t_0)} + \frac{\lambda^2}{4} [1 - e^{-\delta^2(T-t_0)}]. \end{aligned}$$

For $\lambda = \lambda_{*}(M, t_0, x_0)$ we obtain

$$\mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_{*}^{\lambda_{*}}} \right)^2 \right] = x_0^2 e^{-(\delta^2-2r)(T-t_0)} + \frac{(M - x_0 e^{-(\delta^2-r)(T-t_0)})^2}{1 - e^{-\delta^2(T-t_0)}}.$$

So we can reformulate (4.36) to

$$\begin{aligned} V_s(t_0, x_0) &= \sup_{M \in \mathbb{R}} \left[M + cM^2 \right. \\ &\quad \left. - c \left(x_0^2 e^{-(\delta^2-2r)(T-t_0)} + \frac{(M - x_0 e^{-(\delta^2-r)(T-t_0)})^2}{1 - e^{-\delta^2(T-t_0)}} \right) \right] \end{aligned}$$

$$(4.37) \quad = \sup_{M \in \mathbb{R}} \left[c \left(1 - \frac{1}{1 - e^{-\delta^2(T-t_0)}} \right) M^2 + \left(1 + \frac{2cx_0 e^{-(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \right) M - cx_0^2 \left(e^{-(\delta^2-2r)(T-t_0)} + \frac{e^{-2(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \right) \right].$$

Since it holds that $c > 0$ and

$$\left(1 - \frac{1}{1 - e^{-\delta^2(T-t_0)}} \right) < 0,$$

we can find the optimal M_* similarly to (4.21) directly. We obtain the unique solution

$$(4.38) \quad \begin{aligned} M_* &= - \frac{\left(1 + \frac{2cx_0 e^{-(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \right)}{2c \left(1 - \frac{1}{1 - e^{-\delta^2(T-t_0)}} \right)} \\ &= - \frac{\left(1 - e^{-\delta^2(T-t_0)} + 2cx_0 e^{-(\delta^2-r)(T-t_0)} \right)}{2c (1 - e^{-\delta^2(T-t_0)} - 1)} \\ &= \frac{1 - e^{-\delta^2(T-t_0)} + 2cx_0 e^{-(\delta^2-r)(T-t_0)}}{2ce^{-\delta^2(T-t_0)}} \\ &= x_0 e^{r(T-t_0)} + \frac{1}{2c} [e^{\delta^2(T-t_0)} - 1] \end{aligned}$$

and

$$\begin{aligned} V_s(t_0, x_0) &= - cx_0^2 \left(e^{-(\delta^2-2r)(T-t_0)} + \frac{e^{-2(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \right) - \frac{\left(1 + \frac{2cx_0 e^{-(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \right)^2}{4c \left(1 - \frac{1}{1 - e^{-\delta^2(T-t_0)}} \right)} \\ &= - \frac{1}{1 - e^{-\delta^2(T-t_0)}} \\ &\quad \times \left(cx_0^2 \left(e^{-(\delta^2-2r)(T-t_0)} - \cancel{e^{-(\delta^2-2r)(T-t_0) = \delta^2(T-t_0)}} + \cancel{e^{-2(\delta^2-r)(T-t_0)}} \right) \right. \\ &\quad \left. + \frac{\left(1 - e^{-\delta^2(T-t_0)} + 2cx_0 e^{-(\delta^2-r)(T-t_0)} \right)^2}{4c (1 - e^{-\delta^2(T-t_0)} - 1)} \right) \\ &= - \frac{1}{1 - e^{-\delta^2(T-t_0)}} \left(\cancel{cx_0^2 (e^{-(\delta^2-2r)(T-t_0)})} \right) \\ &\quad + \frac{1}{-4ce^{-\delta^2(T-t_0)}} \left(\left(1 - e^{-\delta^2(T-t_0)} \right)^2 + \cancel{4c^2 x_0^2 e^{-2(\delta^2-r)(T-t_0)}} \right) \end{aligned}$$

$$\begin{aligned}
& + 2 \left(1 - e^{-\delta^2(T-t_0)} \right) 2cx_0 e^{-(\delta^2-r)(T-t_0)} \Big) \Big) \\
& = \frac{1}{(1 - e^{-\delta^2(T-t_0)}) (4ce^{-\delta^2(T-t_0)})} \left(\left(1 - e^{-\delta^2(T-t_0)} \right)^2 \right. \\
& \quad \left. + 2 \left(1 - e^{-\delta^2(T-t_0)} \right) 2cx_0 e^{-(\delta^2-r)(T-t_0)} \right) \\
& = x_0 e^{r(T-t_0)} + \frac{1}{4c} \left[e^{\delta^2(T-t_0)} - 1 \right].
\end{aligned}$$

This already proves the first part of (4.4). Note that by (4.36) we obtain the optimal control $u_*^{\tilde{\lambda}}$ by setting $\tilde{\lambda} = \lambda_*(M_*, t_0, x_0)$, i.e.

$$\begin{aligned}
\tilde{\lambda} &= 2 \frac{M_* - x_0 e^{-(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \\
&= 2 \frac{\left(x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - 1 \right] \right) - x_0 e^{-(\delta^2-r)(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \\
&= 2 \frac{x_0 \left(e^{r(T-t_0)} - e^{-(\delta^2-r)(T-t_0)} \right) + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - 1 \right]}{1 - e^{-\delta^2(T-t_0)}} \\
&= 2x_0 e^{r(T-t_0)} \frac{1 - e^{-\delta^2(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} + \frac{1}{c} e^{\delta^2(T-t_0)} \frac{1 - e^{-\delta^2(T-t_0)}}{1 - e^{-\delta^2(T-t_0)}} \\
&= 2x_0 e^{r(T-t_0)} + \frac{1}{c} e^{\delta^2(T-t_0)}.
\end{aligned}$$

This leads to the statically optimal control $u_{1,c}^s = u_*^{\tilde{\lambda}}$

$$\begin{aligned}
u_{1,c}^s(t, x) &= -\frac{\delta}{\sigma} \frac{1}{x} \left[x - \frac{\tilde{\lambda}}{2} e^{-r(T-t)} \right] \\
&= -\frac{\delta}{\sigma} \frac{1}{x} \left[x - \frac{2x_0 e^{r(T-t_0)} + \frac{1}{c} e^{\delta^2(T-t_0)}}{2} e^{-r(T-t)} \right] \\
&= \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \frac{1}{2c} e^{\delta^2(T-t_0)-r(T-t)} \right]
\end{aligned}$$

as claimed in (4.1) and its wealth process $X^s = X^{u_{1,c}^s}$

$$\begin{aligned}
X_t^s &= e^{r(t-t_0)} \left[\tilde{K} - (\tilde{K} - x_0) e^{-\delta(W_t - W_{t_0}) - \frac{3}{2}\delta^2(t-t_0)} \right] \\
&= e^{r(t-t_0)} \left[\frac{\tilde{\lambda}}{2} e^{-r(T-t_0)} - \left(\frac{\tilde{\lambda}}{2} e^{-r(T-t_0)} - x_0 \right) e^{-\delta(W_t - W_{t_0}) - \frac{3}{2}\delta^2(t-t_0)} \right]
\end{aligned}$$

$$\begin{aligned}
&= e^{r(t-t_0)} \left[\frac{1}{2} \left(2x_0 e^{r(T-t_0)} + \frac{1}{c} e^{\delta^2(T-t_0)} \right) e^{-r(T-t_0)} \right. \\
&\quad \left. - \left(\frac{1}{2} \left(2x_0 e^{r(T-t_0)} + \frac{1}{c} e^{\delta^2(T-t_0)} \right) e^{-r(T-t_0)} - x_0 \right) e^{-\delta(W_t-W_{t_0})-\frac{3}{2}\delta^2(t-t_0)} \right] \\
&= e^{r(t-t_0)} \left[\frac{1}{2} \left(2x_0 + \frac{1}{c} e^{(\delta^2-r)(T-t_0)} \right) \right. \\
&\quad \left. - \left(\frac{1}{2} \left(2x_0 + \frac{1}{c} e^{(\delta^2-r)(T-t_0)} \right) - x_0 \right) e^{-\delta(W_t-W_{t_0})-\frac{3}{2}\delta^2(t-t_0)} \right] \\
&= e^{r(t-t_0)} \left[x_0 + \frac{1}{2c} e^{(\delta^2-r)(T-t_0)} \right. \\
&\quad \left. - \frac{1}{2c} e^{(\delta^2-r)(T-t_0)-\delta(W_t-W_{t_0})-\frac{3}{2}\delta^2(t-t_0)} \right] \\
&= e^{r(t-t_0)} \left[x_0 + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left(e^{(\delta^2-r)(t-t_0)} - e^{(\delta^2-r)(t-t_0)-\delta(W_t-W_{t_0})-\frac{3}{2}\delta^2(t-t_0)} \right) \right] \\
&= x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - e^{\delta^2(t-t_0)-\delta(W_t-W_{t_0})-\frac{3}{2}\delta^2(t-t_0)} \right] \\
&= x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t-W_{t_0})-\frac{\delta^2}{2}(t-t_0)} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.},
\end{aligned}$$

as claimed in (4.2). Note that

$$(4.39) \quad \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^s] = \mathbb{E}_{t_0, x_0}[X_T^s]$$

by continuity of $t \mapsto \mathbb{E}_{t_0, x_0}[X_t^s] = x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} [e^{\delta^2(t-t_0)} - 1]$ and then also

$$(4.40) \quad \lim_{t \uparrow T} \text{Var}_{t_0, x_0}(X_t^s) = \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[(X_t^s)^2] - \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^s]^2 = \text{Var}_{t_0, x_0}(X_T^s)$$

by continuity of $t \mapsto \mathbb{E}_{t_0, x_0}[(X_t^s)^2] = \mathbb{E}_{t_0, x_0} \left[\left(X_t^{u_{\star}^{\lambda}} \right)^2 \right]$ (see (4.30)). Hence

$$\begin{aligned}
V_s(t_0, x_0) &= \lim_{t \uparrow T} [\mathbb{E}_{t_0, x_0}[X_t^s] - c \text{Var}_{t_0, x_0}(X_t^s)] \\
&= \mathbb{E}_{t_0, x_0}[X_T^s] - c \text{Var}_{t_0, x_0}(X_T^s) \\
&= J(t_0, x_0; u_{1,c}^s),
\end{aligned}$$

which finishes the proof of (4.4).

Now we rewrite X_t^s in terms of $B_t = be^{rt}$ and $S_t = S_0 \exp(\sigma W_t + (\mu - \frac{1}{2}\sigma^2)t)$ for $r \neq 0$ in order to prove (4.3).

$$X_t^s = x_0 \frac{be^{rt}}{be^{rt_0}} + \frac{1}{2c} \left(\frac{be^{rt}}{be^{rT}} \right)^{\frac{\delta^2-r}{-r}}$$

$$\begin{aligned}
& \times \left[\left(\frac{be^{rt}}{be^{rt_0}} \right)^{\delta^2/r} - \left(\frac{S_0 \exp(\sigma W_t + (\mu - \frac{1}{2}\sigma^2)t)}{S_0 \exp(\sigma W_{t_0} + (\mu - \frac{1}{2}\sigma^2)t_0)} \right)^{-\delta/\sigma} e^{\frac{\delta}{\sigma}(\mu - \frac{1}{2}\sigma^2)(t-t_0) - \frac{\delta^2}{2}(t-t_0)} \right] \\
& = x_0 \frac{B_t}{B_{t_0}} + \frac{1}{2c} \left(\frac{B_t}{B_{t_0}} \right)^{\frac{\delta^2-r}{-r}} \\
& \times \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{S_t}{S_{t_0}} \right)^{-\delta/\sigma} \left(\frac{e^{rt}}{e^{rt_0}} \right)^{\frac{1}{r}(\frac{\delta}{\sigma}(\mu - \frac{1}{2}\sigma^2) - \frac{\delta^2}{2})} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

With the identity $\delta = (\mu - r)/\sigma$ we obtain that

$$\begin{aligned}
\frac{1}{r} \left(\frac{\delta}{\sigma} \left(\mu - \frac{1}{2}\sigma^2 \right) - \frac{\delta^2}{2} \right) &= \frac{\delta}{r} \left(\frac{1}{\sigma} \left(\mu - \frac{1}{2}\sigma^2 \right) - \frac{\delta}{2} \right) \\
&= \frac{\delta}{r} \left(\frac{1}{\sigma} \left(\mu - \frac{1}{2}\sigma^2 \right) - \frac{\mu - r}{2\sigma} \right) \\
&= \frac{\delta}{2\sigma r} (\mu - \sigma^2 + r) \\
&= \delta \left(\frac{1}{\sigma} + \frac{\mu - \sigma^2 - r}{2\sigma r} \right) \\
&= \delta \left(\frac{1}{\sigma} + \frac{\delta - \sigma}{2r} \right).
\end{aligned}$$

So the wealth process of the statically optimal portfolio reads as stated in (4.3)

$$\begin{aligned}
X_t^s &= x_0 \frac{B_t}{B_{t_0}} + \frac{1}{2c} \left(\frac{B_t}{B_{t_0}} \right)^{\frac{\delta^2-r}{-r}} \\
&\times \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{S_t}{S_{t_0}} \right)^{-\delta/\sigma} \left(\frac{e^{rt}}{e^{rt_0}} \right)^{\delta(\frac{1}{\sigma} + \frac{\delta - \sigma}{2r})} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

Step 4 *In the case $\delta = 0$, for all $u \in \mathcal{A}_{t_0, x_0}$ it holds that $\mathbb{E}_{t_0, x_0}[X_T^u] = x_0 e^{r(T-t_0)}$. Moreover, in this case $u_{1,c}^s \equiv 0$ is a statically optimal control in Problem 1. Its wealth process X^s is $X_t^s = x_0 e^{r(t-t_0)}$ $\mathbb{P}_{t_0, x_0}$ -a.s. and the static value function for this case is $V_s(t_0, x_0) = x_0 e^{r(T-t_0)}$. Moreover, all three expressions can be obtained by setting $\delta = 0$ in the expressions found for $\delta \neq 0$.*

Note that after we have proven this step, we no longer have to distinguish between the case $\delta = 0$ and $\delta \neq 0$ for the control $u_{1,c}^s$. In particular the formulation of the theorem stating that

$$u_{1,c}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \frac{1}{2c} e^{\delta^2(T-t_0)-r(T-t)} \right]$$

and

$$\begin{aligned} X_t^s &= x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t-W_{t_0})-\frac{\delta^2}{2}(t-t_0)} \right] \\ &= x_0 \frac{B_t}{B_{t_0}} + \frac{1}{2c} \left(\frac{B_t}{B_T} \right)^{1-\delta^2/r} \\ &\quad \times \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{B_t}{B_{t_0}} \right)^{\delta(1/\sigma+(\delta-\sigma)/(2r))} \left(\frac{S_t}{S_{t_0}} \right)^{-\delta/\sigma} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.} \end{aligned}$$

is valid for any choice of δ .

We first show that in the case $\delta = 0$ the expected terminal wealth of all controls $u \in \mathcal{A}_{t_0, x_0}$ under $\mathbb{P}_{t_0, x_0}$ is the same, namely

$$(4.41) \quad \mathbb{E}_{t_0, x_0}[X_T^u] = e^{r(T-t_0)} x_0.$$

Fix $u \in \mathcal{A}_{t_0, x_0}$ and define as in the proof of Lemma 3.2

$$Y_t^u = e^{-rt} X_t^u.$$

Then by (3.6) and the identity $\mu = r$ we obtain

$$(4.42) \quad \begin{aligned} dY_t^u &= (\mu - r)u_t Y_t^u dt + \sigma u_t Y_t^u dW_t \\ &= \sigma u_t Y_t^u dW_t, \end{aligned}$$

or in the integral form

$$\begin{aligned} Y_t^u &= Y_{t_0}^u + \int_{t_0}^t \sigma u_s Y_s^u dW_s \\ &= e^{-rt_0} x_0 + \int_{t_0}^t \sigma u_s Y_s^u dW_s \quad \mathbb{P}_{t_0, x_0}\text{-a.s.} \end{aligned}$$

By Lemma 3.2 with $a \equiv 0$ and $b : s \mapsto \sigma e^{-rs}$ we know that $M_t := \int_{t_0}^t \sigma u_s Y_s^u dW_s$ is a $\mathbb{P}_{t_0, x_0}$ -martingale. By the martingale property we obtain

$$\mathbb{E}_{t_0, x_0}[Y_T^u] = e^{-rt_0} x_0.$$

Therefore we have that for all $u \in \mathcal{A}_{t_0, x_0}$

$$\mathbb{E}_{t_0, x_0}[X_T^u] = e^{r(T-t_0)}x_0.$$

So in the case $\delta = 0$, Problem 1 reduces to finding an admissible control $u \in \mathcal{A}_{t_0, x_0}$ with minimal Variance $\text{Var}_{t_0, x_0}[X_T^u]$. Consider the control $u_{1,c}^s(t, x) := 0$ for $(t, x) \in [t_0, T] \times (\mathbb{R} \setminus \{0\})$. This control corresponds to holding all the money in the risk less bond. The corresponding wealth process solves (2.9), i.e. $dX_t^s = rX_t^s$ so $X_t^s = x_0 e^{r(t-t_0)}$ $\mathbb{P}_{t_0, x_0}$ -a.s. which is deterministic, so it has variance 0. This shows that for the case $\delta = 0$, $u_{1,c}^s$ is statically optimal in Problem 1. Hence

$$V_s(t_0, x_0) = \mathbb{E}_{t_0, x_0}[X_T^s] - c\text{Var}_{t_0, x_0}[X_T^s] = x_0 e^{r(T-t_0)}.$$

Note that the result for $\delta = 0$ follows by setting $\delta = 0$ in the expressions found for $\delta \neq 0$ since

$$\begin{aligned} u_{1,c}^s(t, x) \big|_{\delta=0} &= \left(\frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \frac{1}{2c} e^{\delta^2(T-t_0)-r(T-t)} \right] \right) \bigg|_{\delta=0} = 0, \\ V_s(t_0, x_0) \big|_{\delta=0} &= \left(x_0 e^{r(T-t_0)} + \frac{1}{4c} \left[e^{\delta^2(T-t_0)} - 1 \right] \right) \bigg|_{\delta=0} = x_0 e^{r(T-t_0)} \end{aligned}$$

and $\mathbb{P}_{t_0, x_0}$ -a.s.

$$X_t^s \big|_{\delta=0} = \left(x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t-W_{t_0})-\frac{\delta^2}{2}(t-t_0)} \right] \right) \bigg|_{\delta=0} = x_0 e^{r(t-t_0)}$$

as claimed. This completes the proof of step 4 and also (A).

Step 5 Denote by u_{1,c,t_0,x_0}^s the statically optimal control in Problem 1 for $(t_0, x_0) \in [0, T[\times \mathbb{R}$. We define the control $u_{1,c}^d(t, x)$ by $u_{1,c}^d(t, x) = u_{1,c,t,x}^s(t, x)$ for all $(t, x) \in [0, T[\times (\mathbb{R} \setminus \{0\})$, where for all $(t_0, x_0) \in [0, T[\times \mathbb{R}$ u_{1,c,t_0,x_0}^s denotes the statically optimal control in Problem 1 for (t_0, x_0) found above. This control takes in each (t, x) the starting value of $u_{1,c,t,x}^s$. Writing this out, we get

$$u_{1,c}^d(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \frac{1}{2c} e^{(\delta^2-r)(T-t)}.$$

This control is strongly dynamically optimal in Problem 1. Its wealth process $X^d = X^{u_{1,c}^d}$ and the value function $V_d(t_0, x_0)$ satisfy (4.6), (4.7) and (4.8). This control extends to an admissible control on $[0, T]$ with $u_{1,c}^d \in \mathcal{A}_{t_0, x_0}$ for all $(t_0, x_0) \in [0, T[\times \mathbb{R}$.

Note that the control $u_{1,c}^d$ defined above does not depend on the starting point (t_0, x_0) and the map

$$[0, T] \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$$

$$(t, x) \mapsto u_{1,c}^d(t, x)x = \frac{\delta}{\sigma} \frac{1}{2c} e^{(\delta^2 - r)(T-t)}$$

is continuously continuable to a map on $[0, T] \times \mathbb{R} \rightarrow \mathbb{R}$.

We show that this control is strongly dynamically optimal. Fix $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Let v be an admissible control with $v(t_0, x_0)x_0 \neq u_{1,c}^d(t_0, x_0)x_0$. We take the control w in the definition of the strongly dynamical optimality to be $w := u_{1,c,t_0,x_0}^s$. Then we have that $w(t_0, x_0) = u_{1,c,t_0,x_0}^s(t_0, x_0) = u_{1,c}^d(t_0, x_0)$ by definition of $u_{1,c}^d$. We have to show that the strict inequality

$$(4.43) \quad J(t_0, x_0; v) < J(t_0, x_0; w)$$

holds. As u_{1,c,t_0,x_0}^s is the statically optimal control in Problem 1 for (t_0, x_0) , we already know that the weak inequality holds. Hence this construction immediately gives us a weakly dynamically optimal control $u_{1,c}^d$ (under the assumption that $u_{1,c}^d$ is admissible on $[0, T[$).

We first consider the case $\delta \neq 0$. If $M_v := \mathbb{E}_{t_0,x_0}[X_T^v] \neq \mathbb{E}_{t_0,x_0}[X_T^{u_{1,c,t_0,x_0}^s}] = M_*$ then

$$\begin{aligned} \mathbb{E}_{t_0,x_0}[X_T^w] - c\text{Var}(X_T^w) &= M_* + cM_*^2 - c\mathbb{E}_{t_0,x_0}\left[\left(X_T^{u_{1,c,t_0,x_0}^s}\right)^2\right] \\ &= \sup_{M \in \mathbb{R}} [M + cM^2 - c \inf_{\substack{u \in \mathcal{A}_{t_0,x_0} \\ \mathbb{E}_{t_0,x_0}[X_{t_0}^u] = M}} \mathbb{E}_{t_0,x_0}[(X_T^u)^2]] \\ &\stackrel{(4.37)}{>} M_v + cM_v^2 - c \inf_{\substack{u \in \mathcal{A}_{t_0,x_0} \\ \mathbb{E}_{t_0,x_0}[X_{t_0}^u] = M_v}} \mathbb{E}_{t_0,x_0}[(X_T^u)^2] \\ &\geq M_v + cM_v^2 - c\mathbb{E}_{t_0,x_0}[(X_T^v)^2] \\ &= \mathbb{E}_{t_0,x_0}[X_T^v] - c\text{Var}(X_T^v), \end{aligned}$$

where the strict inequality comes from the uniqueness of the maximal point M_* in (4.37).

Assume now that $\mathbb{E}_{t_0,x_0}[X_T^v] = \mathbb{E}_{t_0,x_0}[X_T^{u_{1,c,t_0,x_0}^s}] = M_*$. We claim that

$$\mathbb{E}_{t_0,x_0}\left[\left(X_T^{u_{1,c,t_0,x_0}^s}\right)^2\right] < \mathbb{E}_{t_0,x_0}[(X_T^v)^2].$$

But again this has already been shown since for $v(t_0, x_0)x_0 \neq u_{1,c,t_0,x_0}^s(t_0, x_0)x_0$ we know by Step 1 (in particular by (4.34) as $u_{1,c,t_0,x_0}^s = u_*^{\tilde{\lambda}}$) that

$$\begin{aligned} \mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_{1,c,t_0,x_0}^s} \right)^2 \right] - \tilde{\lambda} M_* &= \mathbb{E}_{t_0, x_0} \left[\left(X_T^{u_{1,c,t_0,x_0}^s} \right)^2 - \tilde{\lambda} X_T^{u_{1,c,t_0,x_0}^s} \right] \\ &\stackrel{(4.34)}{<} \mathbb{E}_{t_0, x_0} \left[(X_T^v)^2 - \tilde{\lambda} X_T^v \right] \\ &= \mathbb{E}_{t_0, x_0} \left[(X_T^v)^2 \right] - \tilde{\lambda} M_*. \end{aligned}$$

This shows that $u_{1,c}^d$ is strongly dynamically optimal in Problem 1.

Note that for $\delta = 0$ we know that $\mathbb{E}_{t_0, x_0}[X_T^u] = x_0 e^{r(T-t)}$ for all $u \in \mathcal{A}_{t_0, x_0}$. Hence dynamic optimality of the trivial control $u_{1,c}^d \equiv 0$ follows from Lemma 3.3.

We still have to compute its wealth process $X_t^d = X_t^{u_{1,c}^d}$ and the dynamic value function V_d . For the wealth process, we use a similiar trick as in the static case. Fix $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Recall that the wealth process solves the SDE

$$dX_t^d = \left(r + (\mu - r)u_{1,c}^d(t, X_t) \right) X_t^d dt + \sigma u_{1,c}^d(t, X_t) X_t^d dW_t.$$

with $X_{t_0}^d = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. Plugging in the definition of $u_{1,c}^d$ leads to the equation

$$\begin{aligned} dX_t^d &= \left(r + (\mu - r) \frac{\delta}{\sigma} \frac{1}{X_t^d} \frac{1}{2c} e^{(\delta^2 - r)(T-t)} \right) X_t^d dt + \sigma \frac{\delta}{\sigma} \frac{1}{X_t^d} \frac{1}{2c} e^{(\delta^2 - r)(T-t)} X_t^d dW_t \\ &= \left(r X_t^d + \frac{\delta^2}{2c} e^{(\delta^2 - r)(T-t)} \right) dt + \delta \frac{1}{2c} e^{(\delta^2 - r)(T-t)} dW_t. \end{aligned}$$

Define the process

$$\tilde{Z}_t = e^{r(T-t)} X_t^d.$$

Applying Itô's formula leads to

$$\begin{aligned} d\tilde{Z}_t &= -r e^{r(T-t)} X_t^d dt + e^{r(T-t)} dX_t^d + 0[X^d, X^d]_t \\ &= \cancel{-r e^{r(T-t)} X_t^d dt} + e^{r(T-t)} \left(\left(\cancel{r X_t^d} + \frac{\delta^2}{2c} e^{(\delta^2 - r)(T-t)} \right) dt + \delta \frac{1}{2c} e^{(\delta^2 - r)(T-t)} dW_t \right) \\ &= \frac{\delta^2}{2c} e^{\delta^2(T-t)} dt + \frac{\delta}{2c} e^{\delta^2(T-t)} dW_t. \end{aligned}$$

Hence it holds that

$$\tilde{Z}_t = \tilde{Z}_{t_0} + \int_{t_0}^t \frac{\delta^2}{2c} e^{\delta^2(T-s)} ds + \int_{t_0}^t \frac{\delta}{2c} e^{\delta^2(T-s)} dW_s$$

$$\begin{aligned}
&= x_0 e^{r(T-t_0)} + \frac{\delta^2}{2c} \left(-\frac{1}{\delta^2} \right) \left[e^{\delta^2(T-t)} - e^{\delta^2(T-t_0)} \right] + \frac{\delta}{2c} e^{\delta^2(T-t)} \int_{t_0}^t e^{\delta^2(t-s)} dW_s \\
&= x_0 e^{r(T-t_0)} + \frac{1}{2c} e^{\delta^2(T-t)} \left[e^{\delta^2(t-t_0)} - 1 + \delta \int_{t_0}^t e^{\delta^2(t-s)} dW_s \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

With $X_t^d = e^{-r(T-t)} \tilde{Z}_t$ we obtain

$$X_t^d = x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - 1 + \delta \int_{t_0}^t e^{\delta^2(t-s)} dW_s \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

So we have proved (4.6).

Now we want to rewrite this expression in terms of $B_t = be^{rt}$ and $S_t = S_0 \exp\left(\sigma W_t + \left(\mu - \frac{1}{2}\sigma^2\right)t\right)$ for $r \neq 0$. We use integration by parts. Fix $t_0 \leq t < T$. By Itô's formula for every $f \in C^2([t_0, t])$

$$f(t)W_t = f(t_0)W_{t_0} + \int_{t_0}^t f'(s)W_s ds + \int_{t_0}^t f(s)dW_s \quad \mathbb{P}_{t_0, x_0}\text{-a.s.},$$

hence

$$\int_{t_0}^t f(s)dW_s = f(t)W_t - f(t_0)W_{t_0} - \int_{t_0}^t f'(s)W_s ds \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

Consider also that

$$\log\left(\frac{S_t}{S_{t_0}}\right) = \sigma(W_t - W_{t_0}) + \left(\mu - \frac{1}{2}\sigma^2\right)(t - t_0),$$

hence

$$W_t = W_{t_0} + \frac{1}{\sigma} \left(\log\left(\frac{S_t}{S_{t_0}}\right) - \left(\mu - \frac{1}{2}\sigma^2\right)(t - t_0) \right).$$

Putting this together we obtain that for every $f \in C^2([t_0, t])$, $\mathbb{P}_{t_0, x_0}$ -a.s.

$$\begin{aligned}
&\int_{t_0}^t f(s)dW_s \\
&= f(t)W_t - f(t_0)W_{t_0} \\
&\quad - \int_{t_0}^t f'(s) \left(W_{t_0} + \frac{1}{\sigma} \left(\log\left(\frac{S_s}{S_{t_0}}\right) - \left(\mu - \frac{1}{2}\sigma^2\right)(s - t_0) \right) \right) ds \\
&= f(t)W_t - \cancel{f(t_0)W_{t_0}} - W_{t_0}(f(t) - \cancel{f(t_0)}) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log\left(\frac{S_s}{S_{t_0}}\right) ds
\end{aligned}$$

$$\begin{aligned}
& + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t f'(s)(s - t_0)ds \\
& = f(t)(W_t - W_{t_0}) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log \left(\frac{S_s}{S_{t_0}} \right) ds \\
& \quad + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t f'(s)(s - t_0)ds \\
& \stackrel{\text{PI}}{=} f(t)(W_t - W_{t_0}) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log \left(\frac{S_s}{S_{t_0}} \right) ds \\
& \quad + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \left([f(s)(s - t_0)]_{s=t_0}^{s=t} - \int_{t_0}^t f(s)ds \right) \\
& = f(t)(W_t - W_{t_0}) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log \left(\frac{S_s}{S_{t_0}} \right) ds \\
& \quad + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \left(f(t)(t - t_0) - \int_{t_0}^t f(s)ds \right) \\
& = f(t) \left(W_t - W_{t_0} + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} (t - t_0) \right) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log \left(\frac{S_s}{S_{t_0}} \right) ds \\
& \quad - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t f(s)ds \\
(4.44) \quad & = f(t) \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log \left(\frac{S_s}{S_{t_0}} \right) ds - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t f(s)ds.
\end{aligned}$$

We will use this formula several times. Consider that for $f : s \mapsto e^{\delta^2(t-s)}$ (note that $f(t) = 1$) we obtain

$$\begin{aligned}
& \int_{t_0}^t e^{\delta^2(t-s)} dW_s \\
& = \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^2}{\sigma} \int_{t_0}^t e^{\delta^2(t-s)} \log \left(\frac{S_s}{S_{t_0}} \right) ds - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t e^{\delta^2(t-s)} ds \\
& = \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^2}{\sigma} \int_{t_0}^t e^{\delta^2(t-s)} \log \left(\frac{S_s}{S_{t_0}} \right) ds - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma\delta^2} (e^{\delta^2(t-t_0)} - 1) \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

With this result, (4.6) and $\delta = \frac{\mu-r}{\sigma}$ we obtain

$$X_t^d = x_0 \frac{be^{rt}}{be^{rt_0}} + \frac{1}{2c} \left(\frac{be^{rt}}{be^{rT}} \right)^{\frac{\delta^2-r}{-r}} \left[e^{\delta^2(t-t_0)} - 1 + \delta \int_{t_0}^t e^{\delta^2(t-s)} dW_s \right]$$

$$\begin{aligned}
&= x_0 \frac{B_t}{B_{t_0}} + \frac{\delta}{2c\sigma} \left(\frac{B_t}{B_T} \right)^{\frac{\delta^2-r}{-r}} \left[\frac{\sigma}{\delta} \left(e^{\delta^2(t-t_0)} - 1 \right) + \log \left(\frac{S_t}{S_{t_0}} \right) \right. \\
&\quad \left. + \delta^2 \int_{t_0}^t e^{\delta^2(t-s)} \log \left(\frac{S_s}{S_{t_0}} \right) ds - \frac{\mu - \frac{1}{2}\sigma^2}{\delta^2} \left(e^{\delta^2(t-t_0)} - 1 \right) \right] \\
&= x_0 \frac{B_t}{B_{t_0}} + \frac{\delta}{2c\sigma} \left(\frac{B_t}{B_T} \right)^{\frac{\delta^2-r}{-r}} \left[\left(\frac{\sigma}{\delta} - \frac{\mu - \frac{1}{2}\sigma^2}{\delta^2} \right) \left(e^{\delta^2(t-t_0)} - 1 \right) \right. \\
&\quad \left. + \log \left(\frac{S_t}{S_{t_0}} \right) + \delta^2 \int_{t_0}^t \left(\frac{B_t}{B_s} \right)^{\frac{\delta^2}{r}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right] \\
&= x_0 \frac{B_t}{B_{t_0}} + \frac{\delta}{2c\sigma} \left(\frac{B_t}{B_T} \right)^{\frac{\delta^2-r}{-r}} \left[\frac{\frac{1}{2}\sigma^2 - r}{\delta^2} \left(e^{\delta^2(t-t_0)} - 1 \right) \right. \\
&\quad \left. + \log \left(\frac{S_t}{S_{t_0}} \right) + \delta^2 \int_{t_0}^t \left(\frac{B_t}{B_s} \right)^{\frac{\delta^2}{r}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right] \\
&= x_0 \frac{B_t}{B_{t_0}} + \frac{\delta}{2c\sigma} \left(\frac{B_t}{B_T} \right)^{\frac{\delta^2-r}{-r}} \left[\frac{\sigma^2 - 2r}{2\delta^2} \left(\left(\frac{B_t}{B_{t_0}} \right)^{\frac{\delta^2}{r}} - 1 \right) \right. \\
&\quad \left. + \log \left(\frac{S_t}{S_{t_0}} \right) + \delta^2 \int_{t_0}^t \left(\frac{B_t}{B_s} \right)^{\frac{\delta^2}{r}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.},
\end{aligned}$$

as claimed in (4.7).

Finally we can compute the dynamic value function $V_d(t_0, x_0) = \mathbb{E}_{t_0, x_0}[X_T^d] - c\text{Var}_{t_0, x_0}(X_T^d)$ and show that $\mathbb{E}_{t_0, x_0}[X_T^d] - c\text{Var}_{t_0, x_0}(X_T^d) = \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^d] - c\text{Var}_{t_0, x_0}(X_t^d)$. Consider that

$$\begin{aligned}
\mathbb{E}_{t_0, x_0}[X_t^d] &\stackrel{(4.6)}{=} x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} e^{\delta^2(t-t_0)} \\
(4.45) \quad &= x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{\delta^2(T-t_0)-r(T-t)}
\end{aligned}$$

and

$$\begin{aligned}
\text{Var}_{t_0, x_0}[X_t^d] &\stackrel{(4.6)}{=} \frac{\delta^2}{4c^2} e^{2(\delta^2-r)(T-t)} \text{Var}_{t_0, x_0} \left[\int_{t_0}^t e^{\delta^2(T-s)} dW_s \right] \\
&= \frac{\delta^2}{4c^2} e^{2(\delta^2-r)(T-t)} \int_{t_0}^t e^{2(\delta^2(T-s))} ds \\
&= \frac{\delta^2}{4c^2} e^{2(\delta^2-r)(T-t)} \frac{1}{-2\delta^2} \left[e^{2(\delta^2-r)(T-t)} - e^{2(\delta^2(T-t_0))} \right]
\end{aligned}$$

$$= \frac{1}{8c^2} e^{2(\delta^2-r)(T-t)} \left[e^{2(\delta^2(T-t_0))} - e^{2(\delta^2-r)(T-t)} \right].$$

Hence $t \mapsto J(t, t_0, x_0; u_{1,c}^d) = \mathbb{E}_{t_0, x_0}[X_t^d] - c \text{Var}_{t_0, x_0}(X_t^d)$ is continuous on $[t_0, T]$, i.e.

$$V_d(t_0, x_0) = \lim_{t \uparrow T} J(t, x_0; u_{1,c}^d) = J(t_0, x_0; u_{1,c}^d)$$

and the dynamic value function reads

$$\begin{aligned} V_d(t_0, x_0) &= \mathbb{E}_{t_0, x_0}[X_T^d] - c \text{Var}_{t_0, x_0}(X_T^d) \\ &= x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - 1 \right] - \frac{1}{8c} \left[e^{2(\delta^2(T-t_0))} - 1 \right] \\ &= x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - \frac{1}{4} e^{2(\delta^2(T-t_0))} - \frac{3}{4} \right], \end{aligned}$$

just as claimed in (4.8).

Finally we show that $u_{1,c}^d \in \mathcal{A}_{t_0, x_0}$. To see this, note that $u_{1,c}^d$ is admissible on $[0, T]$ (X_T^d is well defined and the continuity property holds) and consider that similarly to the computation for the statically optimal control for any $(t_0, x_0) \in [0, T] \times \mathbb{R}$ fixed we have

$$\begin{aligned} &\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T ((u_{1,c}^d)_t^2 + 1)(X_t^d)^2 dt \right] \\ &= \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T \left(\left(\frac{\delta}{\sigma} \frac{1}{X_t^d} \frac{1}{2c} e^{(\delta^2-r)(T-t)} \right)^2 + 1 \right) (X_t^{u_{1,c}^d})^2 dt \right] \\ &= \int_{t_0}^T \left(\frac{\delta^2}{\sigma^2} \frac{1}{4c^2} e^{2(\delta^2-r)(T-t)} + \mathbb{E}_{t_0, x_0}[(X_t^d)^2] \right) dt < \infty, \end{aligned}$$

as $t \mapsto \mathbb{E}_{t_0, x_0}[(X_t^d)^2] = \text{Var}_{t_0, x_0}(X_t^d) + \mathbb{E}_{t_0, x_0}[X_t^d]^2$ is continuous hence bounded on $[t_0, T]$.

Step 6 *The control $u_{1,c}^d$ found in Step 5 is the unique strongly dynamically optimal control in Problem 1.*

Assume by contradiction that there exist a second strongly dynamically optimal control $\tilde{u}_{1,c}^d$ and $(t_0, x_0) \in [0, T] \times (\mathbb{R} \setminus \{0\})$ such that $\tilde{u}_{1,c}^d(t_0, x_0) \neq u_{1,c}^d(t_0, x_0)$. Then $u_{1,c}^s(t_0, x_0) = u_{1,c}^d(t_0, x_0) \neq \tilde{u}_{1,c}^d(t_0, x_0)$. But then by the strongly dynamic

optimality of $u_{1,c}^d$ (in particular by (4.43)) it holds that for any $w \in \mathcal{A}_{t_0, x_0}$ with $w(t_0, x_0) = \tilde{u}_{1,c}^d(t_0, x_0) \neq u_{1,c,t_0,x_0}^s(t_0, x_0)$ we have

$$J(t_0, x_0; w) < J(t_0, x_0; u_{1,c,t_0,x_0}^s).$$

On the other hand since $u_{1,c,t_0,x_0}^s \in \mathcal{A}_{t_0, x_0}$ and $\tilde{u}_{1,c}^d(t_0, x_0) \neq u_{1,c,t_0,x_0}^s(t_0, x_0)$ there should exist an $\tilde{w} \in \mathcal{A}_{t_0, x_0}$ with $\tilde{w}(t_0, x_0) = \tilde{u}_{1,c}^d(t_0, x_0)$ such that

$$J(t_0, x_0; \tilde{w}) > J(t_0, x_0; u_{1,c,t_0,x_0}^s)$$

by the strongly dynamic optimality of $\tilde{u}_{1,c}^d$. This is a contradiction and finishes the proof of Step 6, (B) and Theorem 4.1. $\square$

Remark 4.2. We would like to compare the two strategies with each other. Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$. Then

$$\begin{aligned} V_d(t_0, x_0) &= x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - \frac{1}{4} e^{2\delta^2(T-t_0)} - \frac{3}{4} \right] \\ &\leq x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - \frac{1}{4} - \frac{3}{4} \right] \\ &= x_0 e^{r(T-t_0)} + \frac{1}{2c} \left[e^{\delta^2(T-t_0)} - 1 \right] \\ &= V_s(t_0, x_0). \end{aligned}$$

In fact, as $u_{1,c}^d \in \mathcal{A}_{t_0, x_0}$,

$$V_d(t_0, x_0) = J(t_0, x_0; u_{1,c}^d) \leq J(t_0, x_0; u_{1,c}^s) = V_s(t_0, x_0)$$

should hold by definition of static optimality anyway. So comparing V_s and V_d pointwise isn't useful.

In order to compare the strategies more dynamically, we compare the values $\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_s(t, X_t^s)]$ and $\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_d(t, X_t^d)]$ with each other.⁷ This corresponds to comparing the expected values of the processes at the terminal time starting from the state (t_0, x_0) . Note that

$$\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_s(t, X_t^s)] = \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} \left[X_t^s e^{r(T-t)} + \frac{1}{2c} \left[e^{\delta^2(T-t)} - 1 \right] \right]$$

⁷Different from J.L. Pedersen and G. Peskir [2, Remark 4, p. 12] we do not directly compare the values $V_d(T, X_T^d)$ and $V_d(T, X_T^s)$ since for Problem 2 and 3 this expression is not well defined (again $X_T^{u_{2,\alpha}^d}$ and $X_T^{u_{3,\beta}^d}$ are not well defined).

$$\begin{aligned}
&= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [X_t^s] \\
&\stackrel{(4.39)}{=} \mathbb{E}_{t_0, x_0} [X_T^s]
\end{aligned}$$

and similarly

$$\begin{aligned}
\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_d(t, X_t^d)] &= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} \left[X_t^d e^{r(T-t)} + \frac{1}{2c} [e^{\delta^2(T-t_0)} - \frac{1}{4} e^{2\delta^2(T-t_0)} - \frac{3}{4}] \right] \\
&= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [X_t^d] \\
&\stackrel{(4.45)}{=} \mathbb{E}_{t_0, x_0} [X_T^d].
\end{aligned}$$

As $V_s(T, x) = x = V_d(T, x)$ for all $x \in \mathbb{R}$, we have that $V_s(T, X_T^s) = X_T^s$ and $V_d(T, X_T^d) = X_T^d$. Consider that

$$\mathbb{E}_{t_0, x_0} [X_T^s] \stackrel{(4.38)}{=} x_0 e^{r(T-t_0)} + \frac{1}{2c} [e^{\delta^2(T-t_0)} - 1] \stackrel{(4.45)}{=} \mathbb{E}_{t_0, x_0} [X_T^d]$$

and therefore

$$\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_s(t, X_t^s)] = \mathbb{E}_{t_0, x_0} [X_T^s] = \mathbb{E}_{t_0, x_0} [X_T^d] = \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_d(t, X_t^d)].$$

So the strategies are under this dynamic aspect as good as each other.

As mentioned in the introductory chapter, the strongly dynamically optimal control is *time consistent*, but the statically optimal control is not. Denote by u_{1,c,t_0,x_0}^s the statically optimal control in Problem 1 for $(t_0, x_0) \in [0, T[\times \mathbb{R}$ and its wealth process by X^s . Assume that the investor acted statically optimal until $t_0 < t < T$. If he now decides to invest statically optimal in Problem 1 for the new starting point (t, X_t^s) , Theorem 4.1 would tell him to change his investment strategy. In particular

$$u_{1,c,t_0,x_0}^s(t, X_t^s) \neq u_{1,c,t,X_t^s}^s(t, X_t^s) \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

So the statically optimal controls are not useful in practice. Note that the dynamically optimal controls are time consistent in the above sense, as they do not depend on the initial time and wealth.

Furthermore consider that the amount of wealth held in the stock at time t

$$\begin{aligned}
xu_{1,d}^d(t, x) &= x \frac{\delta}{2c\sigma} \frac{1}{x} e^{(\delta^2-r)(T-t)} \\
&= \frac{\delta}{2c\sigma} e^{(\delta^2-r)(T-t)}
\end{aligned}$$

does not depend on the total wealth x for the strongly dynamically optimal control. Thus even if the total wealth x is large, it is dynamically optimal to hold the same amount in the risky stock, as if the total wealth is small.

Also consider that the statically optimal control in Problem 1 is not unique. The wealth processes of the statically optimal control is $\mathbb{P}_{t_0, x_0}$ -a.s. bounded from above as

$$\begin{aligned} \sup_{t_0 \leq t \leq T} X_t^s &= \sup_{t_0 \leq t \leq T} \left[x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t - W_{t_0}) - \frac{\delta^2}{2}(t-t_0)} \right] \right] \\ &\leq \sup_{t_0 \leq t \leq T} \left| \left[x_0 e^{r(t-t_0)} + \frac{1}{2c} e^{(\delta^2-r)(T-t)} \left[e^{\delta^2(t-t_0)} - 0 \right] \right] \right| =: C. \end{aligned}$$

But therefore the control

$$\tilde{u}_{1,c}^s(t, x) = \begin{cases} u_{1,c}^s(t, x) & , \text{ if } x < C \\ 0 & , \text{ else} \end{cases}$$

has the same wealth process X^s and is therefore also statically optimal in Problem 1. We will see that the wealth processes of the statically optimal controls we find in the other two problems are also bounded from above. So the statically optimal controls are also not unique in Problem 2 and 3. However, we suppose that one should be able to prove that the resulting statically optimal wealth processes X^s (in all three problems) are unique.

With similar argumentation as in Step 6 one can prove uniqueness of the strongly dynamically optimal controls in Problem 2 and 3. Note that the argumentation used to prove that the statically optimal controls are not unique cannot be used for the strongly dynamically optimal controls, as one would need uniform boundedness from above of the wealth process X^d over all $(t_0, x_0) \in [0, T[\times \mathbb{R}$, which is not the case ($X_{t_0}^d = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s.).

4.2 Optimal controls in Problem 2

We now turn our attention to the constrained problems. Recall that Problem 2 reads

$$\sup_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \text{Var}(X_T^u) \leq \alpha}} \mathbb{E}_{t_0, x_0}[X_T^u]$$

for $\alpha > 0$ fixed. Having solved Problem 1, it will be rather easy to solve Problem 2.

Corollary 4.3. *Let X^u solve (2.9) with $X_{t_0}^u = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. for $(t_0, x_0) \in [0, T] \times \mathbb{R}$ given and fixed. Let B solve (2.8) and S solve (2.7). Set $\delta = (\mu - r)/\sigma$ for $\mu \in \mathbb{R}$, $r \in \mathbb{R}$ and $\sigma > 0$.*

(A) *Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$. Let $\delta \neq 0$. A statically optimal control in Problem 2 for $\alpha > 0$ is given by*

$$(4.46) \quad u_{2, \alpha}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \sqrt{\alpha} \frac{e^{\delta^2(T-t_0)-r(T-t)}}{\sqrt{e^{\delta^2(T-t_0)} - 1}} \right]$$

for $(t, x) \in [t_0, T] \times (\mathbb{R} \setminus \{0\})$. The corresponding wealth process is given by

$$(4.47) \quad X_t^s = x_0 e^{r(t-t_0)} + \sqrt{\alpha} \frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t_0)} - 1}} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t - W_{t_0}) - \frac{\delta^2}{2}(t-t_0)} \right]$$

$$(4.48) \quad = x_0 \frac{B_t}{B_{t_0}} + \frac{\sqrt{\alpha}}{\sqrt{\left(\frac{B_T}{B_{t_0}}\right)^{\delta^2/r} - 1}} \left(\frac{B_t}{B_T} \right)^{1-\delta^2/r} \\ \times \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{B_t}{B_{t_0}} \right)^{\delta(1/\sigma + (\delta-\sigma)/(2r))} \left(\frac{S_t}{S_{t_0}} \right)^{-\delta/\sigma} \right]$$

$\mathbb{P}_{t_0, x_0}$ -a.s. (the second equality only holds for $r \neq 0$) for $(t, x) \in [t_0, T] \times (\mathbb{R} \setminus \{0\})$ for $t \in [t_0, T]$. The static value function $V_s^1(t_0, x_0) := \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^s] = \mathbb{E}_{t_0, x_0}[X_T^s]$ is given by

$$(4.49) \quad V_s^1(t_0, x_0) = x_0 e^{r(T-t_0)} + \sqrt{\alpha} \sqrt{e^{\delta^2(T-t_0)} - 1}$$

If $\delta = 0$ then any control $u \in \mathcal{A}_{t_0, x_0}$ satisfying $\text{Var}_{t_0, x_0}(X_T^u) \leq \alpha$ (e.g. $u \equiv 0$) is statically optimal in Problem 3.

(B) If $\delta \neq 0$ the unique strongly dynamically optimal control in Problem 2 for $\alpha > 0$ is given by

$$(4.50) \quad u_{2,\alpha}^d(t, x) = \sqrt{\alpha} \frac{\delta}{\sigma} \frac{1}{x} \frac{e^{(\delta^2 - r)(T-t)}}{\sqrt{e^{\delta^2(T-t)} - 1}}$$

for $(t, x) \in [0, T[\times (\mathbb{R} \setminus \{0\})$. The corresponding wealth process for $(t_0, x_0) \in [0, T[\times (\mathbb{R} \setminus \{0\})$ fixed is given by⁸

$$(4.51) \quad X_t^d = x_0 e^{r(t-t_0)} + 2\sqrt{\alpha} e^{-r(T-t)} \times \left[\sqrt{e^{\delta^2(T-t_0)} - 1} - \sqrt{e^{\delta^2(T-t)} - 1} + \frac{\delta}{2} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} dW_s \right]$$

$$(4.52) \quad = x_0 \frac{B_t}{B_{t_0}} + \frac{\delta \sqrt{\alpha}}{\sigma} \frac{B_t}{B_T} \left[\frac{2r - \sigma^2}{\delta^2} \left(\sqrt{\left(\frac{B_T}{B_t} \right)^{\delta^2/r} - 1} - \sqrt{\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - 1} \right) + \frac{\left(\frac{B_T}{B_t} \right)^{\delta^2/r}}{\sqrt{\left(\frac{B_T}{B_t} \right)^{\delta^2/r} - 1}} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^2}{2} \int_{t_0}^t \left(\frac{B_T}{B_s} \right)^{\delta^2/r} \frac{\left(\left(\frac{B_T}{B_s} \right)^{\delta^2/r} - 2 \right)}{\sqrt{\left(\frac{B_T}{B_s} \right)^{\delta^2/r} - 1}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right]$$

$\mathbb{P}_{t_0, x_0}$ -a.s. for $(t, x) \in [t_0, T[\times (\mathbb{R} \setminus \{0\})$ (the second equality only holds for $r \neq 0$). The dynamic value function $V_d^1(t_0, x_0) := \lim_{t \rightarrow T} \mathbb{E}_{t_0, x_0}(X_t^d)$ for this case is given by

$$(4.53) \quad V_d^1(t_0, x_0) = x_0 e^{r(T-t_0)} + 2\sqrt{\alpha} \sqrt{e^{\delta^2(T-t_0)} - 1}$$

for $(t_0, x_0) \in [0, T[\times \mathbb{R}$.

For the case $\delta = 0$ there does not exist any strongly dynamically optimal controls. Moreover, in this case $u_{2,\alpha}^d \equiv 0$ is weakly dynamically optimal in Problem 2.

Proof. (A): First note that for $\delta = 0$ any control $u \in \mathcal{A}_{t_0, x_0}$ satisfying $\text{Var}_{t_0, x_0}(X_T^u) \leq \alpha$ is statically optimal as for all $u \in \mathcal{A}_{t_0, x_0}$ all wealth processes have the same

⁸The second expression corrects a typo on in J.L. Pedersen and G. Peskir [2, Corollary 5, p. 14].

terminal expectation $x_0 e^{r(T-t_0)}$ under $\mathbb{P}_{t_0, x_0}$. Also note that $0 \equiv u \in \mathcal{A}_{t_0, x_0}$ with $\text{Var}_{t_0, x_0}(X_T^u) = \text{Var}_{t_0, x_0}(x_0 e^{r(T-t_0)}) = 0 < \alpha$.

Assume now that $\delta \neq 0$. Again we want to solve this problem by considering the corresponding Lagrangian. Here we have to modify the method slightly as the constraint here is an inequality. The method we are using here is called the method of Kuhn-Tucker.

Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$ and define the Lagrangian

$$L_{t_0, x_0}(u, c) := \mathbb{E}_{t_0, x_0}[X_T^u] - c[\text{Var}_{t_0, x_0}(X_T^u) - \alpha],$$

for $c > 0$ and $u \in \mathcal{A}_{t_0, x_0}$. Assume for each fixed $c > 0$ there exists an admissible control $u_*^c \in \mathcal{A}_{t_0, x_0}$ such that

$$(4.54) \quad L_{t_0, x_0}(u_*^c, c) = \sup_{u \in \mathcal{A}_{t_0, x_0}} L_{t_0, x_0}(u, c).$$

If we can additionally find an $c_* > 0$ such that

$$\text{Var}_{t_0, x_0}(X_T^{u_*^{c_*}}) = \alpha,$$

we see that for all controls $v \in \mathcal{A}_{t_0, x_0}$ with $\text{Var}_{t_0, x_0}(X_T^v) \leq \alpha$ we have that

$$\begin{aligned} \mathbb{E}_{t_0, x_0}[X_T^{u_*^{c_*}}] &= L_{t_0, x_0}(u_*^{c_*}, c_*) \\ &\geq L_{t_0, x_0}(v, c_*) \\ &= \mathbb{E}_{t_0, x_0}[X_T^v] - c_*[\text{Var}_{t_0, x_0}(X_T^v) - \alpha] \\ &\geq \mathbb{E}_{t_0, x_0}[X_T^v] \end{aligned}$$

Hence $u_{2, \alpha}^s := u_*^{c_*}$ is statically optimal in Problem 2.

Note that (as α is fixed) solving (4.54) is equivalent to solving

$$(4.55) \quad \mathbb{E}_{t_0, x_0}[X_T^{u_*^c}] - c[\text{Var}_{t_0, x_0}(X_T^{u_*^c})] = \sup_{u \in \mathcal{A}_{t_0, x_0}} \mathbb{E}_{t_0, x_0}[X_T^u] - c\text{Var}_{t_0, x_0}(X_T^u).$$

But this is exactly the problem of finding the statically optimal control in Problem 1. So with Theorem 4.1 we find

$$u_*^c(t, x) = u_{1, c}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \frac{1}{2c} e^{\delta^2(T-t_0) - r(T-t)} \right].$$

Next we use the expressions for M_*^c and $V_s^c(t_0, x_0)$ (the new superscript c emphasizes the dependence on c) found in the proof of Theorem 4.1 to obtain

$$\text{Var}_{t_0, x_0}(X_T^{u_{1, c}^s}) = \frac{1}{c} \left[\mathbb{E}_{t_0, x_0}[X_T^{u_{1, c}^s}] - \left(\mathbb{E}_{t_0, x_0}[X_T^{u_{1, c}^s}] - c\text{Var}_{t_0, x_0}(X_T^{u_{1, c}^s}) \right) \right]$$

$$\begin{aligned}
&= \frac{1}{c} [M_*^c - V_s^c(t_0, x_0)] \\
&\stackrel{(4.38)+(4.4)}{=} \frac{1}{c} \left[\left(x_0 e^{r(T-t_0)} + \frac{1}{2c} [e^{\delta^2(T-t_0)} - 1] \right) \right. \\
&\quad \left. - \left(x_0 e^{r(T-t_0)} + \frac{1}{4c} [e^{\delta^2(T-t_0)} - 1] \right) \right] \\
(4.56) \quad &= \frac{1}{4c^2} [e^{\delta^2(T-t_0)} - 1].
\end{aligned}$$

Hence the condition $\text{Var}_{t_0, x_0} \left(X_T^{u_{1,c}^s} \right) = \alpha$ leads to

$$c_* = \frac{1}{2\sqrt{\alpha}} \sqrt{e^{\delta^2(T-t_0)} - 1} > 0.$$

This leads to the expression

$$u_{2,\alpha}^s(t, x) = u_{1,c_*}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \sqrt{\alpha} \frac{e^{\delta^2(T-t_0)-r(T-t)}}{\sqrt{e^{\delta^2(T-t_0)} - 1}} \right],$$

as claimed in (4.46). Now we obtain the wealth process and the static value function directly by plugging in $c = c_*$ in the expressions found in Theorem 4.1, hence we obtain

$$\begin{aligned}
X_t^s &\stackrel{(4.2)}{=} x_0 e^{r(t-t_0)} + \sqrt{\alpha} \frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t_0)} - 1}} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t - W_{t_0}) - \frac{\delta^2}{2}(t-t_0)} \right] \\
&\stackrel{(4.3)}{=} x_0 \frac{B_t}{B_{t_0}} + \frac{\sqrt{\alpha}}{\sqrt{\left(\frac{B_T}{B_{t_0}}\right)^{\delta^2/r} - 1}} \left(\frac{B_t}{B_T} \right)^{1-\delta^2/r} \\
&\quad \times \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{B_t}{B_{t_0}} \right)^{\delta(1/\sigma + (\delta-\sigma)/(2r))} \left(\frac{S_t}{S_{t_0}} \right)^{-\delta/\sigma} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

and

$$\begin{aligned}
V_s^1(t_0, x_0) &= M_*^{c_*} \\
&\stackrel{(4.38)}{=} x_0 e^{r(T-t_0)} + \frac{1}{2c_*} [e^{\delta^2(T-t_0)} - 1] \\
&= x_0 e^{r(T-t_0)} + \sqrt{\alpha} \frac{e^{\delta^2(T-t_0)} - 1}{\sqrt{e^{\delta^2(T-t_0)} - 1}} \\
&= x_0 e^{r(T-t_0)} + \sqrt{\alpha} \sqrt{e^{\delta^2(T-t_0)} - 1}.
\end{aligned}$$

This shows that (4.47), (4.48) and (4.49) hold. Note that by (4.39) (replace c with c_*) $\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^s] = \mathbb{E}_{t_0, x_0}[X_T^s]$. This completes the proof for the static case.

(B): Let $\delta = 0$. As the expected terminal wealth of all $u \in \mathcal{A}_{t_0, x_0}$ is the same, there are no strongly dynamically optimal controls in Problem 2. Define $u_{2, \alpha}^d \equiv 0$. We show that this control is weakly dynamically optimal in Problem 2. Fix any $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Then any $u \in \mathcal{A}_{t_0, x_0}$ satisfies $\mathbb{E}_{t_0, x_0}[X_T^u] = x_0 e^{r(T-t_0)}$. Define $w \equiv 0$. Then $w(t_0, x_0)x_0 = 0 = u_{2, \alpha}^d(t_0, x_0)x_0$, $\text{Var}_{t_0, x_0}(X_T^w) = 0 < \alpha$ and $\mathbb{E}_{t_0, x_0}[X_T^w] = \mathbb{E}_{t_0, x_0}[X_T^v]$ for any $v \in \mathcal{A}_{t_0, x_0}$.

Assume now that $\delta \neq 0$. We use the same trick as in Theorem 4.1. Denote by $u_{2, \alpha, t_0, x_0}^s$ the statically optimal control in Problem 2 for $(t_0, x_0) \in [0, T[\times \mathbb{R}$. We define $u_{2, \alpha}^d(t, x) = u_{2, \alpha, t, x}^s(t, x)$ for all $(t, x) \in [0, T[\times (\mathbb{R} \setminus \{0\})$. Writing this out, we get

$$u_{2, \alpha}^d(t, x) = \sqrt{\alpha} \frac{\delta}{\sigma} \frac{1}{x} \frac{e^{(\delta^2 - r)(T-t)}}{\sqrt{e^{\delta^2(T-t)} - 1}},$$

as claimed in (4.50). Note that this control does not depend on the starting point (t_0, x_0) and that the map

$$[0, T[\times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$$

$$(t, x) \mapsto u_{2, \alpha}^d(t, x)x = \sqrt{\alpha} \frac{\delta}{\sigma} \frac{e^{(\delta^2 - r)(T-t)}}{\sqrt{e^{\delta^2(T-t)} - 1}}$$

is continuously continuable to a map on $[0, T[\times \mathbb{R}$. Also note that this control is not dynamically optimal in Problem 1 for $c = c_*$ (c_* also depends on t_0 which has been replaced by t). Again by construction we know that this control is weakly dynamically optimal, if we assume admissibility on $[0, T[$.

We show that this control is even strongly dynamically optimal. Fix any $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Let $v \in \mathcal{A}_{t_0, x_0}$ be any admissible control on $[t_0, T]$ with $v(t_0, x_0)x_0 \neq u_{2, \alpha}^d(t_0, x_0)x_0$ and $\text{Var}(X_T^v) \leq \alpha$. Take $w := u_{2, \alpha, t_0, x_0}^s$. We have to show that

$$\mathbb{E}_{t_0, x_0}[X_T^w] > \mathbb{E}_{t_0, x_0}[X_T^v].$$

Now as $u_{2, \alpha, t_0, x_0}^s$ is the statically optimal control in Problem 1 for $c = c_* > 0$, we already know by the proof Theorem 4.1 (in particular by (4.43)) that

$$\mathbb{E}_{t_0, x_0}[X_T^w] - c_* \text{Var}(X_T^w) > \mathbb{E}_{t_0, x_0}[X_T^v] - c_* \text{Var}(X_T^v).$$

But since by construction of $u_{2, \alpha, t_0, x_0}^s$, $\text{Var}_{t_0, x_0}(X_T^w) = \alpha$ holds, we have that

$$\mathbb{E}_{t_0, x_0}[X_T^w] = \mathbb{E}_{t_0, x_0}[X_T^w] - c_* (\text{Var}(X_T^w) - \alpha)$$

$$\begin{aligned}
&> \mathbb{E}_{t_0, x_0}[X_T^v] - c_* (\text{Var}(X_T^v) - \alpha) \\
&\geq \mathbb{E}_{t_0, x_0}[X_T^v].
\end{aligned}$$

This shows that $u_{2,\alpha}^d$ is dynamically optimal in Problem 2.

Next we compute the wealth process $X^d = X^{u_{2,\alpha}^d}$. We do this in the same way as in the proof of Theorem 4.1. Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$. Recall that the wealth process solves the SDE

$$dX_t^d = \left(r + (\mu - r)u_{2,\alpha}^d(t, X_t) \right) X_t^d dt + \sigma u_{2,\alpha}^d(t, X_t) X_t^d dW_t.$$

with $X_{t_0}^d = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. Plugging in the expression found for $u_{2,\alpha}^d$ leads to the equation

$$\begin{aligned}
dX_t^d &= \left(r + (\mu - r)\sqrt{\alpha}\frac{\delta}{\sigma}\frac{1}{X_t^d}\frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} \right) X_t^d dt + \sigma\sqrt{\alpha}\frac{\delta}{\sigma}\frac{1}{X_t^d}\frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} X_t^d dW_t \\
&= \left(rX_t^d + \sqrt{\alpha}\delta^2\frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} \right) dt + \sqrt{\alpha}\delta\frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} dW_t.
\end{aligned}$$

Define the process

$$\tilde{Z}_t = e^{r(T-t)} X_t^d.$$

Applying Itô's formula leads to

$$\begin{aligned}
d\tilde{Z}_t &= -re^{r(T-t)} X_t^d dt + e^{r(T-t)} dX_t^d + 0[X^d, X^d]_t \\
&= \cancel{-re^{r(T-t)} X_t^d dt} + e^{r(T-t)} \left(\left(rX_t^d + \sqrt{\alpha}\delta^2\frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} \right) dt \right. \\
&\quad \left. + \sqrt{\alpha}\delta\frac{e^{(\delta^2-r)(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} dW_t \right) \\
&= \sqrt{\alpha}\delta^2\frac{e^{\delta^2(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} dt + \sqrt{\alpha}\delta\frac{e^{\delta^2(T-t)}}{\sqrt{e^{\delta^2(T-t)}-1}} dW_t.
\end{aligned}$$

Hence we have

$$\begin{aligned}
\tilde{Z}_t &= \tilde{Z}_{t_0} + \int_{t_0}^t \sqrt{\alpha}\delta^2\frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)}-1}} ds + \int_{t_0}^t \sqrt{\alpha}\delta\frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)}-1}} dW_s \\
&= x_0 e^{r(T-t_0)} + \sqrt{\alpha}\delta^2 \left(-\frac{2}{\delta^2} \right) \left[\sqrt{e^{\delta^2(T-t)}-1} - \sqrt{e^{\delta^2(T-t_0)}-1} \right]
\end{aligned}$$

$$\begin{aligned}
& + \sqrt{\alpha}\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} dW_s \\
& = x_0 e^{r(T-t_0)} + 2\sqrt{\alpha} \left[\sqrt{e^{\delta^2(T-t_0)} - 1} - \sqrt{e^{\delta^2(T-t)} - 1} \right. \\
& \quad \left. + \frac{\delta}{2} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} dW_s \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

As $X_t^d = e^{-r(T-t)} \tilde{Z}_t$ we obtain

$$\begin{aligned}
(4.57) \quad X_t^d & = x_0 e^{r(t-t_0)} + 2\sqrt{\alpha} e^{-r(T-t)} \left[\sqrt{e^{\delta^2(T-t_0)} - 1} - \sqrt{e^{\delta^2(T-t)} - 1} \right. \\
& \quad \left. + \frac{\delta}{2} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} dW_s \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

This shows that (4.51) holds.

Again we want to rewrite this expression in terms of $B_t = be^{rt}$ and $S_t = S_0 \exp(\sigma W_t + (\mu - \frac{1}{2}\sigma^2)t)$ for $r \neq 0$ in order to show (4.52). Recall that by integration by parts it holds that for all $f \in C^2([t_0, T[; \mathbb{R})$ and $t \in [t_0, T[, \mathbb{P}_{t_0, x_0}\text{-a.s.}$

$$\int_{t_0}^t f(s) dW_s \stackrel{(4.44)}{=} f(t) \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log \left(\frac{S_s}{S_{t_0}} \right) ds - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t f(s) ds.$$

We use this for

$$f : s \mapsto \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}},$$

hence

$$\begin{aligned}
f'(s) & = \frac{(-\delta^2)e^{\delta^2(T-s)}\sqrt{e^{\delta^2(T-s)} - 1} - e^{\delta^2(T-s)} \frac{(-\delta^2)e^{\delta^2(T-s)}}{2\sqrt{e^{\delta^2(T-s)} - 1}}}{e^{\delta^2(T-s)} - 1} \\
& = -\frac{\delta^2}{2} e^{\delta^2(T-s)} \frac{2(e^{\delta^2(T-s)} - 1) - e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^{3/2}} \\
& = -\frac{\delta^2}{2} e^{\delta^2(T-s)} \frac{e^{\delta^2(T-s)} - 2}{(e^{\delta^2(T-s)} - 1)^{3/2}},
\end{aligned}$$

for $s \in]t_0, T[$ and get that

$$\begin{aligned}
& \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} dW_s \\
&= \frac{e^{\delta^2(T-t)}}{\sqrt{e^{\delta^2(T-t)} - 1}} \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^2}{2\sigma} \int_{t_0}^t e^{\delta^2(T-s)} \frac{e^{\delta^2(T-s)} - 2}{(e^{\delta^2(T-s)} - 1)^{3/2}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \\
&\quad - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} ds \\
&= \frac{e^{\delta^2(T-t)}}{\sqrt{e^{\delta^2(T-t)} - 1}} \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^2}{2\sigma} \int_{t_0}^t e^{\delta^2(T-s)} \frac{e^{\delta^2(T-s)} - 2}{(e^{\delta^2(T-s)} - 1)^{3/2}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \\
&\quad + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \frac{2}{\delta^2} \left[\sqrt{e^{\delta^2(T-t)} - 1} - \sqrt{e^{\delta^2(T-t_0)} - 1} \right] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}
\end{aligned}$$

Threfore with this expression and $\delta = \frac{\mu-r}{\sigma}$ we obtain $\mathbb{P}_{t_0, x_0}$ -a.s.

$$\begin{aligned}
X_t^d &\stackrel{(4.57)}{=} x_0 \frac{B_t}{B_{t_0}} + 2\sqrt{\alpha} \frac{B_t}{B_T} \left[\sqrt{e^{\delta^2(T-t_0)} - 1} - \sqrt{e^{\delta^2(T-t)} - 1} + \frac{\delta}{2} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} dW_s \right] \\
&= x_0 \frac{B_t}{B_{t_0}} + 2\sqrt{\alpha} \frac{B_t}{B_T} \left[\left(-1 + \frac{\delta}{2} \frac{2\mu - \sigma^2}{\sigma \delta^2} \right) \left(\sqrt{e^{\delta^2(T-t)} - 1} - \sqrt{e^{\delta^2(T-t_0)} - 1} \right) \right. \\
&\quad \left. + \frac{\delta}{2\sigma} \frac{e^{\delta^2(T-t)}}{\sqrt{e^{\delta^2(T-t)} - 1}} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta}{2} \frac{\delta^2}{2\sigma} \int_{t_0}^t e^{\delta^2(T-s)} \frac{e^{\delta^2(T-s)} - 2}{(e^{\delta^2(T-s)} - 1)^{3/2}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right] \\
&= x_0 \frac{B_t}{B_{t_0}} + \frac{\delta \sqrt{\alpha}}{\sigma} \frac{B_t}{B_T} \left[\left(-\frac{2\sigma}{\delta} + \frac{2\mu - \sigma^2}{\delta^2} \right) \left(\sqrt{e^{\delta^2(T-t)} - 1} - \sqrt{e^{\delta^2(T-t_0)} - 1} \right) \right. \\
&\quad \left. + \frac{e^{\delta^2(T-t)}}{\sqrt{e^{\delta^2(T-t)} - 1}} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^2}{2} \int_{t_0}^t e^{\delta^2(T-s)} \frac{e^{\delta^2(T-s)} - 2}{(e^{\delta^2(T-s)} - 1)^{3/2}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right] \\
&= x_0 \frac{B_t}{B_{t_0}} + \frac{\delta \sqrt{\alpha}}{\sigma} \frac{B_t}{B_T} \left[\frac{2r - \sigma^2}{\delta^2} \left(\sqrt{\left(\frac{B_T}{B_t} \right)^{\delta^2/r} - 1} - \sqrt{\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - 1} \right) \right. \\
&\quad + \frac{\left(\frac{B_T}{B_t} \right)^{\delta^2/r}}{\sqrt{\left(\frac{B_T}{B_t} \right)^{\delta^2/r} - 1}} \log \left(\frac{S_t}{S_{t_0}} \right) \\
&\quad \left. + \frac{\delta^2}{2} \int_{t_0}^t \left(\frac{B_T}{B_s} \right)^{\delta^2/r} \frac{\left(\frac{B_T}{B_s} \right)^{\delta^2/r} - 2}{\left(\left(\frac{B_T}{B_s} \right)^{\delta^2/r} - 1 \right)^{3/2}} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right],
\end{aligned}$$

as claimed in (4.52). Finally we can compute the dynamic value function $V_d(t_0, x_0) = \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^d]$. With the first expression for X^d we obtain

$$\begin{aligned} \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^d] &\stackrel{(4.57)}{=} \lim_{t \uparrow T} \left(x_0 e^{r(t-t_0)} + 2\sqrt{\alpha} e^{-r(T-t)} \left[\sqrt{e^{\delta^2(T-t_0)} - 1} - \sqrt{e^{\delta^2(T-t)} - 1} \right] \right) \\ &= x_0 e^{r(T-t_0)} + 2\sqrt{\alpha} \sqrt{e^{\delta^2(T-t_0)} - 1}. \end{aligned}$$

This proves (4.53). Note that uniqueness of the strongly dynamically optimal control can be proved exactly the same way as in the proof of Theorem 4.1. This completes the proof of (B) and Corollary 4.3. $\square$

Remark 4.4. We first show that the case $\delta = 0$ does not follow from the case $\delta \neq 0$ in any sense, i.e. that the corresponding part of [2, Corollary 5, p. 14] is not correct. Consider that by (4.1) the statically optimal control in Problem 1 for $(t_0, x_0) \in [0, T[\times (\mathbb{R} \setminus \{0\})$ and $\delta \neq 0$ is given by

$$u_{2,\alpha}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \sqrt{\alpha} \frac{e^{\delta^2(T-t_0)-r(T-t)}}{\sqrt{e^{\delta^2(T-t_0)} - 1}} \right]$$

Note that

$$\begin{aligned} \lim_{\delta \downarrow 0} \frac{\delta}{\sqrt{e^{\delta^2(T-t_0)} - 1}} &= \lim_{\delta \downarrow 0} \sqrt{\frac{\delta^2}{\sum_{n=1}^{\infty} \frac{(\delta^2(T-t_0))^n}{n!}}} \\ &= \lim_{\delta \downarrow 0} \sqrt{\frac{1}{(T-t_0) + \sum_{n=2}^{\infty} \frac{\delta^{2(n-1)}(T-t_0)^n}{n!}}} \\ &= \sqrt{\frac{1}{(T-t_0)}}, \end{aligned}$$

but

$$\begin{aligned} \lim_{\delta \uparrow 0} \frac{\delta}{\sqrt{e^{\delta^2(T-t_0)} - 1}} &= - \lim_{\delta \uparrow 0} \sqrt{\frac{\delta^2}{\sum_{n=1}^{\infty} \frac{(\delta^2(T-t_0))^n}{n!}}} \\ &= - \sqrt{\frac{1}{(T-t_0)}}. \end{aligned}$$

Hence the limit $\lim_{\delta \rightarrow 0} \frac{\delta}{\sqrt{e^{\delta^2(T-t_0)} - 1}}$ does not even exist. Moreover, $u_{2,\alpha}^s$ converges for $\delta \downarrow 0$, $\delta \neq 0$, resp. $\delta \uparrow 0$, $\delta \neq 0$ pointwise to the control

$$\tilde{u} : [t_0, T] \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$$

$$(t, x) \mapsto \frac{1}{x} \frac{1}{\sigma} \sqrt{\frac{\alpha}{(T - t_0)}},$$

respectively $-\tilde{u}$. It could be possible, that the control $\tilde{u}$ or $-\tilde{u}$ is meant to be statically optimal for the case $\delta = 0$. We show that this is not the case by showing that for $\delta = 0$ we have that $\text{Var}_{t_0, x_0}[(X_T^{\pm \tilde{u}})^2] > \alpha$, hence $\pm \tilde{u}$ are not statically optimal in Problem 2.

Let $\delta = 0$. Define $Y_t^{\pm \tilde{u}} := e^{-rt} X_t^{\pm \tilde{u}}$. By (4.42) we know that

$$dY_t^{\pm \tilde{u}} = \sigma u_t Y_t^{\pm \tilde{u}} dW_t = \pm \sqrt{\frac{\alpha}{T - t_0}} e^{-rt} dW_t.$$

Therefore we have

$$X_t^{\pm \tilde{u}} = x_0 \pm e^{rt} \sqrt{\frac{\alpha}{T - t_0}} \int_{t_0}^t e^{-rs} dW_s \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

Note that for the case $r \neq 0$ we then have

$$\begin{aligned} \text{Var}_{t_0, x_0}[(X_T^{\pm \tilde{u}})^2] &= e^{2rT} \frac{\alpha}{(T - t_0)} \int_{t_0}^T e^{-2rs} ds \\ &= e^{2rT} \frac{\alpha}{(T - t_0)} \frac{e^{-2rt_0} - e^{-2rT}}{2r} \\ &= \alpha \frac{e^{2r(T-t_0)} - 1}{2r(T - t_0)} \\ &> \alpha, \end{aligned}$$

where for the last inequality we used that $T - t_0 \neq 0$ and that for all $x \neq 0$ it holds that $e^x - 1 > x$ by strict concavity of $e^x - 1$.

Again we want to compare the two strategies (for $\delta \neq 0$) under the same aspect as in Remark 4.2. If we compare V_s^1 and V_d^1 point wise we notice that

$$\begin{aligned} V_d^1(t_0, x_0) &= x_0 e^{r(T-t_0)} + 2\sqrt{\alpha} \sqrt{e^{\delta^2(T-t_0)} - 1} \\ &> x_0 e^{r(T-t_0)} + \sqrt{\alpha} \sqrt{e^{\delta^2(T-t_0)} - 1} \\ &= V_s^1(t_0, x_0). \end{aligned}$$

for $\alpha > 0$ and $\delta \neq 0$. This is possible, as the control $u_{2, \alpha}^d$ neither lies in $\mathcal{A}_{t_0, x_0}$, nor satisfies $\text{Var}_{t_0, x_0}[X_T^d] < \alpha$ (X_T^d is actually not well defined), so this does not

contradict the property of $u_{2,\alpha}^s$ being the statically optimal control in Problem 2. In fact we even have that

$$\begin{aligned}
\text{Var}_{t_0, x_0}[X_t^d] &= \text{Var}_{t_0, x_0} \left[2\sqrt{\alpha} e^{-r(T-t)} \frac{\delta}{2} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{\sqrt{e^{\delta^2(T-s)} - 1}} dW_s \right] \\
&= \alpha \delta^2 e^{-2r(T-t)} \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} ds \\
&= \alpha \delta^2 e^{-2r(T-t)} \int_{e^{\delta^2(T-t_0)}}^{e^{\delta^2(T-t)}} \frac{x^2}{x-1} \left(-\frac{1}{\delta^2} \frac{1}{x} \right) dx \\
&= \alpha 2 e^{-2r(T-t)} \int_{e^{\delta^2(T-t)}}^{e^{\delta^2(T-t_0)}} \left(1 + \frac{1}{x-1} \right) dx \\
&= \alpha e^{-2r(T-t)} \left[e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)} + \log \left(\frac{e^{\delta^2(T-t_0)} - 1}{e^{\delta^2(T-t)} - 1} \right) \right] \xrightarrow{t \uparrow T} \infty.
\end{aligned}$$

J.L. Pedersen and G. Peskir refer to this as *dynimc compliance effect* (see [2, Remark 6, p. 15]). However, we omit this discussion here.

Moreover, consider that

$$\begin{aligned}
\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T ((u_{2,\alpha}^d)_t^2 + 1)(X_t^d)^2 dt \right] &\leq \mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T (u_{2,\alpha}^d)_t^2 (X_t^d)^2 dt \right] \\
&= \alpha \frac{\delta^2}{\sigma^2} \int_{t_0}^T \frac{e^{2(\delta^2-r)(T-t)}}{e^{\delta^2(T-t)} - 1} dt \\
&= \infty.
\end{aligned}$$

This is the reason why we defined admissibility without using any integrability assumptions. The problem is that the dynamically optimal controls won't satisfy the needed property $\mathbb{E}_{t_0, x_0} \left[\int_{t_0}^T (u_t^2 + 1)(X_t^u)^2 dt \right]$. However we saw that the proofs already work if we only force the statically optimal controls to satisfy this integrability assumption.

Now we want to compare the two strategies as in Remark 4.2. Consider that

$$\begin{aligned}
\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[V_s^1(t, X_t^s)] &= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^s e^{r(T-t)} + \sqrt{\alpha} \sqrt{e^{\delta^2(T-t)} - 1}] \\
&= \lim_{t \uparrow T} \left(\mathbb{E}_{t_0, x_0}[X_t^s] e^{r(T-t)} + \sqrt{\alpha} \sqrt{e^{\delta^2(T-t)} - 1} \right) \\
&= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0}[X_t^s] \\
&= V_s^1(t_0, x_0)
\end{aligned}$$

$$\begin{aligned}
&< V_d^1(t_0, x_0) \\
&= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [X_t^d] \\
&= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [X_t^d e^{r(T-t)} + 2\sqrt{\alpha} \sqrt{e^{\delta^2(T-t)} - 1}] \\
&= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_d^1(t, X_t^d)].
\end{aligned}$$

So under this aspect the strongly dynamically optimal control in Problem 2 is better than the statically optimal control in Problem 2 given in (4.46).

4.3 Optimal controls in Problem 3

Recall that Problem 3 reads

$$\inf_{\substack{u \in \mathcal{A}_{t_0, x_0} \\ \mathbb{E}_{t_0, x_0}[X_T^u] \geq \beta}} \text{Var}_{t_0, x_0}(X_T^u)$$

for $\beta \in \mathbb{R}$ fixed. Again this will turn out to be a simple application of Theorem 4.1.

Corollary 4.5. *Let X^u solve (2.9) with $X_{t_0}^u = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. for $(t_0, x_0) \in [0, T] \times \mathbb{R}$ given and fixed. Let B solve (2.8) and S solve (2.7). Set $\delta = (\mu - r)/\sigma$ for $\mu \in \mathbb{R}$, $r \in \mathbb{R}$ and $\sigma > 0$.*

(A) *Fix $(t_0, x_0) \in [0, T] \times \mathbb{R}$. A statically optimal control in Problem 3 for $\beta \in \mathbb{R}$ is given by*

$$(4.58) \quad u_{3, \beta}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t_0)-r(T-t)}}{e^{\delta^2(T-t_0)} - 1} \right],$$

if $x_0 e^{r(T-t_0)} < \beta$ and $\delta \neq 0$, and $u_{3, \beta}^s(t, x) = 0$ if $x_0 e^{r(T-t_0)} \geq \beta$ for $(t, x) \in [t_0, T] \times (\mathbb{R} \setminus \{0\})$. The corresponding wealth process is given by

$$(4.59) \quad X_t^s = x_0 e^{r(t-t_0)} + \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{(\delta^2-r)(T-t)}}{e^{\delta^2(T-t_0)} - 1} \\ \times \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t - W_{t_0}) - \frac{\delta^2}{2}(t-t_0)} \right]$$

$$(4.60) \quad = x_0 \frac{B_t}{B_{t_0}} + \left(\beta - x_0 \frac{B_T}{B_{t_0}} \right) \frac{\left(\frac{B_t}{B_T} \right)^{1-\delta^2/r}}{\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - 1} \\ \left[\left(\frac{B_t}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{B_t}{B_{t_0}} \right)^{\delta(1/\sigma + (\delta-\sigma)/(2r))} \left(\frac{S_t}{S_{t_0}} \right)^{-\delta/\sigma} \right]$$

$\mathbb{P}_{t_0, x_0}$ -a.s. if $x_0 e^{r(T-t_0)} < \beta$ and $\delta \neq 0$ (the second equality only holds for $r \neq 0$), and $X_t^s = x_0 e^{r(t-t_0)}$ $\mathbb{P}_{t_0, x_0}$ -a.s. if $x_0 e^{r(T-t_0)} \geq \beta$ for $t \in [t_0, T]$. The static value function $V_s^2(t_0, x_0) := \lim_{t \uparrow T} \text{Var}_{t_0, x_0}[X_t^s] = \text{Var}_{t_0, x_0}[X_T^s]$ is given by

$$(4.61) \quad V_s^2(t_0, x_0) = \frac{\left(\beta - x_0 e^{r(T-t_0)} \right)^2}{e^{\delta^2(T-t_0)} - 1}$$

if $x_0 e^{r(T-t_0)} < \beta$ and $\delta \neq 0$, and $V_s^1(t_0, x_0) = 0$ if $x_0 e^{r(T-t_0)} \geq \beta$.

Moreover, for the case $\delta = 0$ and $x_0 e^{r(T-t_0)} < \beta$ there do not exist any statically optimal controls in Problem 3.

(B) Let $\delta \neq 0$. The unique strongly dynamically optimal control in Problem 3 for $\beta \in \mathbb{R}$ is given by⁹

$$(4.62) \quad u_{3,\beta}^d(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left(\beta - x e^{r(T-t)} \right)^+ \frac{e^{(\delta^2-r)(T-t)}}{e^{\delta^2(T-t)} - 1}$$

for $(t, x) \in [0, T[\times (\mathbb{R} \setminus \{0\})$, where $z^+ := \max\{z, 0\}$ for $z \in \mathbb{R}$. If $x_0 e^{r(T-t_0)} < \beta$, then the corresponding wealth process for $(t_0, x_0) \in [0, T[\times (\mathbb{R} \setminus \{0\})$ fixed is given by

$$(4.63) \quad X_t^d = e^{-r(T-t)} \left[\beta - \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right. \\ \left. \times \exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s - \frac{\delta^2}{2} \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) \right] \\ (4.64) \quad = \frac{B_t}{B_T} \left[\beta - \left(\beta - x_0 \frac{B_T}{B_{t_0}} \right) \left(\frac{\left(\frac{B_T}{B_t} \right)^{\delta^2/r} - 1}{\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - 1} \right)^{1/2 + \sigma/(2\delta) - r/(\sigma\delta)} \right. \\ \left. \times \exp \left(-\frac{\delta}{\sigma} \frac{\left(\frac{B_T}{B_t} \right)^{\delta^2/r}}{\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - 1} \log \left(\frac{S_t}{S_{t_0}} \right) \right. \right. \\ \left. \left. + \frac{\delta^3}{\sigma} \int_{t_0}^t \frac{\left(\frac{B_T}{B_s} \right)^{\delta^2/r}}{\left(\left(\frac{B_T}{B_s} \right)^{\delta^2/r} - 1 \right)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right. \right. \\ \left. \left. - \frac{1}{2} \frac{\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - \left(\frac{B_T}{B_t} \right)^{\delta^2/r}}{\left(\left(\frac{B_T}{B_t} \right)^{\delta^2/r} - 1 \right) \left(\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - 1 \right)} \right) \right]$$

$\mathbb{P}_{t_0, x_0}$ -a.s. (the second equality only holds for $r \neq 0$) with $X_t^d e^{r(T-t)} < \beta$ $\mathbb{P}_{t_0, x_0}$ -a.s. for all $t \in [t_0, T[$ and

$$X_T^d := \lim_{t \uparrow T} X_t^d = \beta \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

If $x_0 e^{r(T-t_0)} \geq \beta$ then $X_t^d = x_0 e^{r(T-t_0)}$ $\mathbb{P}_{t_0, x_0}$ -a.s. with $X_t^d e^{r(T-t)} \geq \beta$ $\mathbb{P}_{t_0, x_0}$ -a.s.

⁹ This formulation is much more precise as the one given in [2, Corollary 7, p. 16]. The dynamically optimal control should not depend on the initial values $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Otherwise one would have to change the definition of dynamic optimality.

We define two value functions $V_d^2(t_0, x_0) := \lim_{t \uparrow T} \text{Var}_{t_0, x_0}(X_t^d)$ and $\tilde{V}_d^2(t_0, x_0) := \text{Var}_{t_0, x_0}(X_T^d)$. If $x_0 e^{r(T-t_0)} \geq \beta$ then

$$V_d^2(t_0, x_0) = \tilde{V}_d^2(t_0, x_0) = 0$$

for $(t_0, x_0) \in [0, T[\times \mathbb{R}$. If $x_0 e^{r(T-t_0)} < \beta$ then

$$V_d^2(t_0, x_0) = \infty \neq 0 = \tilde{V}_d^2(t_0, x_0)$$

for $(t_0, x_0) \in [0, T[\times \mathbb{R}$.¹⁰

For the case $\delta = 0$, $u_{3,\beta}^d \equiv 0$ is strongly dynamically optimal in Problem 3.

Proof. (A): Similar to the proof of Corollary 4.3 we want to apply the method of Kuhn-Tucker. Fix $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Consider the Lagrangian

$$L_{t_0, x_0}(u, c) = \text{Var}_{t_0, x_0}(X_T^u) - c[\mathbb{E}_{t_0, x_0}[X_T^u] - \beta],$$

for $c > 0$ and $u \in \mathcal{A}_{t_0, x_0}$. Assume for each $c > 0$ there exists an admissible control $u_*^c \in \mathcal{A}_{t_0, x_0}$ such that

$$(4.65) \quad L_{t_0, x_0}(u_*^c, c) = \inf_{u \in \mathcal{A}_{t_0, x_0}} L_{t_0, x_0}(u, c).$$

If we can additionally find an $c_* > 0$ such that

$$\mathbb{E}_{t_0, x_0}[X_T^{u_*^{c_*}}] = \beta,$$

we see that for all admissible controls v with $\mathbb{E}_{t_0, x_0}[X_T^v] \geq \beta$ we have that

$$(4.66) \quad \begin{aligned} \text{Var}_{t_0, x_0}(X_T^{u_*^{c_*}}) &= L_{t_0, x_0}(u_*^{c_*}, c_*) \\ &\leq L_{t_0, x_0}(v, c_*) \\ &= \text{Var}_{t_0, x_0}(X_T^v) - c[\mathbb{E}_{t_0, x_0}[X_T^v] - \beta] \\ &\leq \text{Var}_{t_0, x_0}(X_T^v) \end{aligned}$$

Hence $u_{3,\beta}^s := u_*^{c_*}$ is statically optimal in Problem 3. We will see that such an c_* exists if and only if $\delta \neq 0$ and $\beta > x_0 e^{r(T-t_0)}$. Therefore we will have to consider several cases.

¹⁰Note that $\tilde{V}_d^2(t_0, x_0)$ here corresponds to $V_d^2(t_0, x_0)$ in J.L. Pedersen and G. Peskir [2, Corollary 7, p. 16-17]. See Remark 4.6 for further information.

Rewriting the right-hand side of (4.65) leads to

$$(4.67) \quad L_{t_0, x_0}(u_*^c, c) = -c \sup_{u \in \mathcal{A}_{t_0, x_0}} \left[\mathbb{E}_{t_0, x_0}[X_T^u] - \frac{1}{c} \text{Var}_{t_0, x_0}(X_T^u) \right] + c\beta.$$

But this is exactly the problem of finding the statically optimal control in Problem 1 where we replace $c > 0$ with $\frac{1}{c} > 0$. So with the result of Theorem 4.1 we find

$$u_*^c(t, x) = u_{1, 1/c}^s(t, x) \stackrel{(4.1)}{=} \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \frac{c}{2} e^{\delta^2(T-t_0)-r(T-t)} \right]$$

and

$$\mathbb{E}_{t_0, x_0}[X_T^{u_*^c}] = \mathbb{E}_{t_0, x_0}[X_T^{u_{1, 1/c}^s}] \stackrel{(4.38)}{=} x_0 e^{r(T-t_0)} + \frac{c}{2} [e^{\delta^2(T-t_0)} - 1].$$

Case 1: $\delta \neq 0$

Setting above expression equal to β leads to

$$c_* = \frac{2(\beta - x_0 e^{T-t_0})}{e^{\delta^2(T-t_0)} - 1}$$

Consider that $c_* > 0$ if and only if $\beta > x_0 e^{T-t_0}$.

Case 1a: $\delta \neq 0, x_0 e^{r(T-t_0)} \geq \beta, (c_* \leq 0)$

We can simply define $u_{3, \beta}^s \in \mathcal{A}_{t_0, x_0}$ by $u_{3, \beta}^s(t, x) = 0$ for for $(t, x) \in [t_0, T] \times (\mathbb{R} \setminus \{0\})$. We obtain the wealth proces $X_t^s = x_0 e^{r(t-t_0)}$ $\mathbb{P}_{t_0, x_0}$ -a.s. Note that this process is $\mathbb{P}_{t_0, x_0}$ -a.s. deterministic, hence the expectation under $\mathbb{P}_{t_0, x_0}$ is $x_0 e^{r(T-t_0)} \geq \beta$ and the Variance is constantly equal to 0. Hence this control is statically optimal for this case.

Case 1b: $\delta \neq 0, x_0 e^{r(T-t_0)} < \beta, (c_* > 0)$

Set

$$u_{3, \beta}^s(t, x) = u_*^{c_*}(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t_0)-r(T-t)}}{e^{\delta^2(T-t_0)} - 1} \right].$$

By the previous argumentation (see (4.66)), this control is statically optimal in Problem 3. This proves (4.58).

Note that the wealth process is given by Theorem 4.1 with $c = \frac{1}{c_*}$, i.e

$$X_t^s \stackrel{(4.2)}{=} x_0 e^{r(t-t_0)} + \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{(\delta^2-r)(T-t)}}{e^{\delta^2(T-t_0)} - 1} \left[e^{\delta^2(t-t_0)} - e^{-\delta(W_t - W_{t_0}) - \frac{\delta^2}{2}(t-t_0)} \right]$$

$\mathbb{P}_{t_0, x_0}$ -a.s., as claimed in (4.59). (4.60) holds by the same reason. Now we compute the static value function $V_s^2(t_0, x_0) = \text{Var}_{t_0, x_0}(X_T^s)$. Again we can use the result of Theorem 4.1 for $c = \frac{1}{c_*}$ to get

$$V_s^2(t_0, x_0) \stackrel{(4.56)}{=} \frac{c_*^2}{4} \left[e^{\delta^2(T-t_0)} - 1 \right] = \frac{\left(\beta - x_0 e^{r(T-t_0)} \right)^2}{e^{\delta^2(T-t_0)} - 1}.$$

This proves (4.61). Finally with (4.40) we see that

$$\lim_{t \uparrow T} \text{Var}_{t_0, x_0}(X_t^s) = \text{Var}_{t_0, x_0}(X_T^s).$$

Case 2: $\delta = 0$

We know from Step 4 of the proof of Theorem 4.1 (in particular by (4.41)) that for all $u \in \mathcal{A}_{t_0, x_0}$ it holds that

$$(4.68) \quad \mathbb{E}_{t_0, x_0}[X_T^u] = x_0 e^{r(T-t_0)}.$$

Case 2a: $\delta = 0, x_0 e^{r(T-t_0)} \geq \beta$

As in Case 1a we see that $u_{3, \beta}^s = 0$ is statically optimal in Problem 3.

Case 2b: $\delta = 0, x_0 e^{r(T-t_0)} < \beta$

By (4.68) there is no statically optimal control in Problem 3 for this case.

This completes the proof of part (A).

(B): Let $\delta = 0$. Define $u_{3, \beta}^d \equiv 0$. We show that this control is strongly dynamically optimal in Problem 3. Fix any $(t_0, x_0) \in [0, T[\times \mathbb{R}$. If $x_0 e^{r(T-t_0)} < \beta$ (as in Case 2b) then there is nothing to show as $\{u \in \mathcal{A}_{t_0, x_0} \mid \mathbb{E}_{t_0, x_0}[X_T^u] \geq \beta\} = \emptyset$. Assume now that $x_0 e^{r(T-t_0)} \geq \beta$. Then any $u \in \mathcal{A}_{t_0, x_0}$ satisfies $\mathbb{E}_{t_0, x_0}[X_T^u] \geq \beta$. So we can take $w \equiv 0$. But then $w(t_0, x_0)x_0 = 0 = u_{3, \beta}^d(t_0, x_0)x_0$ and $\text{Var}_{t_0, x_0}(X_T^w) = 0 < \text{Var}_{t_0, x_0}(X_T^v)$ by Lemma 3.3 for any $v \in \mathcal{A}_{t_0, x_0}$ different from zero.

Let $\delta \neq 0$. We use the same trick as before to find a weakly dynamically optimal control. Denote by u_{3, β, t_0, x_0}^s the statically optimal control in Problem 3 for $(t_0, x_0) \in$

$[0, T[\times \mathbb{R}$. Define again $u_{3,\beta}^d(t, x) = u_{3,\beta,t,x}^s(t, x)$ for all $(t, x) \in [0, T[\times (\mathbb{R} \setminus \{0\})$. So we have

$$u_{3,\beta}^d(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left(\beta - x e^{r(T-t)} \right)^+ \frac{e^{(\delta^2-r)(T-t)}}{e^{\delta^2(T-t)} - 1},$$

as claimed in (4.62). Note that this control does not depend on the starting point (t_0, x_0) and that the map

$$[0, T[\times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$$

$$(t, x) \mapsto u_{2,\alpha}^d(t, x)x = \frac{\delta}{\sigma} \left(\beta - x e^{r(T-t)} \right)^+ \frac{e^{(\delta^2-r)(T-t)}}{e^{\delta^2(T-t)} - 1}$$

is continuously continuable to a map on $[0, T[\times \mathbb{R}$. Fix any $(t_0, x_0) \in [0, T[\times \mathbb{R}$. Let $v \in \mathcal{A}_{t_0, x_0}$ be an admissible control with $v(t_0, x_0)x_0 \neq u_{3,\beta}^d(t_0, x_0)x_0$ and $\mathbb{E}_{t_0, x_0}[X_T^v] \geq \beta$. Define $w := u_{3,\beta,t_0,x_0}^s$. By the statical optimality of u_{3,β,t_0,x_0}^s it is clear that

$$\text{Var}_{t_0, x_0}[X_T^w] \leq \text{Var}_{t_0, x_0}[X_T^v].$$

Hence if $u_{3,\beta}^d$ is admissible on $[0, T[$, then it is also weakly dynamically optimal. We show that the inequality is strict.

Consider that by Lemma 3.3 we have

$$\text{Var}_{t_0, x_0}[X_T^w] = 0 < \text{Var}_{t_0, x_0}[X_T^v]$$

if $x_0 e^{r(T-t_0)} \geq \beta$, since in this case $u_{3,\beta}^d(t, X_t^d) = 0$ and $X_t^d = x_0 e^{r(t-t_0)}$ for $t_0 \leq t \leq T$. This follows from the fact that if $x_0 e^{r(T-t_0)} \geq \beta$ then $(x_0 e^{r(t-t_0)}) e^{r(T-t)} = x_0 e^{r(T-t_0)} \geq \beta$. Hence if $x_0 e^{r(T-t_0)} \geq \beta$, then $X_t^d e^{r(T-t)} \geq \beta$ $\mathbb{P}_{t_0, x_0}$ -a.s. for all $t_0 \leq t \leq T$.

On the other hand if $x_0 e^{r(T-t_0)} < \beta$, then u_{3,β,t_0,x_0}^s equals the statically optimal control in Problem 1 for $c = 1/c_*$ found in Theorem 4.1 by construction.¹¹ Hence by the proof of Theorem 4.1 we know that

$$\mathbb{E}_{t_0, x_0}[X_T^w] - \frac{1}{c_*} \text{Var}(X_T^w) \stackrel{(4.43)}{>} \mathbb{E}_{t_0, x_0}[X_T^v] - \frac{1}{c_*} \text{Var}(X_T^v).$$

Now as $\mathbb{E}_{t_0, x_0}[X_T^w] = \beta$ it follows that

$$\text{Var}_{t_0, x_0}[X_T^w] = -c_* \left((\mathbb{E}_{t_0, x_0}[X_T^w] - \beta) - \frac{1}{c_*} \text{Var}(X_T^w) \right)$$

¹¹ At this point there was a mistake in the paper. The identity $u_{3,\beta}^s = u_{1,1/c_*}^s$ only holds if $x_0 e^{r(T-t_0)} < \beta$. So proving strongly dynamical optimality would require additional work, namely Lemma 3.3.

$$\begin{aligned}
&< -c_* \left((\mathbb{E}_{t_0, x_0}[X_T^v] - \beta) - \frac{1}{c_*} \text{Var}(X_T^v) \right) \\
&\leq \text{Var}_{t_0, x_0}[X_T^v].
\end{aligned}$$

We compute the wealth process $X^d = X_{3, \beta}^{u^d}$ again in the same way as before. Fix $(t_0, x_0) \in [0, T[\times \mathbb{R}$. For $x_0 e^{r(T-t_0)} \geq \beta$ we already know that $X_t^d = x_0 e^{r(t-t_0)}$ so $V_2^d(t_0, x_0) = \tilde{V}_2^d(t_0, x_0) = 0$.

Assume now that $x_0 e^{r(T-t_0)} < \beta$. Recall that the wealth process solves the SDE

$$dX_t^d = \left(r + (\mu - r) u_{3, \beta}^d(t, X_t) \right) X_t^d dt + \sigma u_{3, \beta}^d(t, X_t) X_t^d dW_t,$$

with $X_{t_0}^d = x_0$ $\mathbb{P}_{t_0, x_0}$ -a.s. We make the ansatz $X_t^d e^{r(T-t)} < \beta$ $\mathbb{P}_{t_0, x_0}$ -a.s. for all $t \in [t_0, T[$. Plugging in (4.62) leads to the equation

$$\begin{aligned}
dX_t^d &= \left(r + (\mu - r) \frac{\delta}{\sigma} \frac{1}{X_t^d} \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{(\delta^2 - r)(T-t)}}{e^{\delta^2(T-t)} - 1} \right) X_t^d dt \\
&\quad + \sigma \frac{\delta}{\sigma} \frac{1}{X_t^d} \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{(\delta^2 - r)(T-t)}}{e^{\delta^2(T-t)} - 1} X_t^d dW_t \\
&= \left(r X_t^d + \delta^2 \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{(\delta^2 - r)(T-t)}}{e^{\delta^2(T-t)} - 1} \right) dt + \delta \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{(\delta^2 - r)(T-t)}}{e^{\delta^2(T-t)} - 1} dW_t.
\end{aligned}$$

Define the process

$$\tilde{Z}_t = \beta - e^{r(T-t)} X_t^d.$$

Applying Itô's formula leads to

$$\begin{aligned}
&d\tilde{Z}_t \\
&= r e^{r(T-t)} X_t^d dt - e^{r(T-t)} dX_t^d + 0[X^d, X^d]_t \\
&= \cancel{r e^{r(T-t)} X_t^d dt} - e^{r(T-t)} \left(\left(\cancel{r X_t^d} + \delta^2 \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{(\delta^2 - r)(T-t)}}{e^{\delta^2(T-t)} - 1} \right) dt \right. \\
&\quad \left. + \delta \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{(\delta^2 - r)(T-t)}}{e^{\delta^2(T-t)} - 1} dW_t \right) \\
&= -\delta^2 \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} dt - \delta \left(\beta - X_t^d e^{r(T-t)} \right) \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} dW_t \\
&= -\delta^2 \tilde{Z}_t \frac{1}{1 - e^{-\delta^2(T-t)}} dt - \delta \tilde{Z}_t \frac{1}{1 - e^{-\delta^2(T-t)}} dW_t.
\end{aligned}$$

By Lemma 2.4 this linear SDE has a unique strong solution

$$\tilde{Z}_t = \tilde{Z}_{t_0} \exp \left(- \int_{t_0}^t \frac{\delta}{1 - e^{-\delta^2(T-s)}} dW_s - \int_{t_0}^t \left[\frac{\delta^2}{1 - e^{-\delta^2(T-s)}} + \frac{1}{2} \frac{\delta^2}{(1 - e^{-\delta^2(T-s)})^2} \right] ds \right)$$

$\mathbb{P}_{t_0, x_0}$ -a.s. on $[t_0, T[$. Note that $\tilde{Z}_{t_0} = \beta - x_0 e^{r(T-t_0)} > 0$. Therefore we indeed have $X_t^d e^{r(T-t)} = \beta - \tilde{Z}_t < \beta$ $\mathbb{P}_{t_0, x_0}$ -a.s. for all $t \in [t_0, T[$. Consider that

$$\begin{aligned} \exp \left(- \int_{t_0}^t \frac{\delta^2}{1 - e^{-\delta^2(T-s)}} ds \right) &= \exp \left(\int_{t_0}^t \frac{-\delta^2 e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} ds \right) \\ &= \exp \left(\left[\log \left(e^{\delta^2(T-s)} - 1 \right) \right]_{s=t_0}^{s=t} \right) \\ &= \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1}. \end{aligned}$$

So therefore

$$\begin{aligned} X_t^d &= e^{-r(T-t)} [\beta - \tilde{Z}_t] \\ &= e^{-r(T-t)} \left[\beta - \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right. \\ &\quad \times \exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s - \int_{t_0}^t \frac{1}{2} \frac{\delta^2}{(1 - e^{-\delta^2(T-s)})^2} ds \right) \Big] \\ &= e^{-r(T-t)} \left[\beta - \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right. \\ &\quad \times \exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s - \frac{\delta^2}{2} \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) \Big] \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}, \end{aligned}$$

as claimed in (4.63). In order to rewrite this expression in terms of $B_t = be^{rt}$ and $S_t = S_0 \exp \left(\sigma W_t + \left(\mu - \frac{1}{2} \sigma^2 \right) t \right)$ for $r \neq 0$, we consider first, that by integration by parts (see (4.44)) for all $f \in C^2([t_0, T]; \mathbb{R})$, $\mathbb{P}_{t_0, x_0}$ -a.s.

$$\int_{t_0}^t f(s) dW_s = f(t) \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) - \frac{1}{\sigma} \int_{t_0}^t f'(s) \log \left(\frac{S_s}{S_{t_0}} \right) ds - \frac{\mu - \frac{1}{2} \sigma^2}{\sigma} \int_{t_0}^t f(s) ds$$

for all $t \in [t_0, T[$. We apply this on

$$f : s \mapsto \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1},$$

hence

$$f'(s) = (-\delta^2) \frac{e^{\delta^2(T-s)} (e^{\delta^2(T-s)} - 1) - e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} = \delta^2 \frac{e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2},$$

for $s \in]t_0, T[$ and obtain

$$\begin{aligned} & \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s \\ &= \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) - \frac{\delta^2}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \\ & \quad - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} ds \\ &= \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) - \frac{\delta^2}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \\ & \quad - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \left[\frac{1}{-\delta^2} \log (e^{\delta^2(T-s)} - 1) \right]_{s=t_0}^{s=t} \\ &= \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} \frac{1}{\sigma} \log \left(\frac{S_t}{S_{t_0}} \right) - \frac{\delta^2}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \\ & \quad + \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \frac{1}{\delta^2} \log \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right) \quad \mathbb{P}_{t_0, x_0}\text{-a.s.} \end{aligned}$$

So it holds $\mathbb{P}_{t_0, x_0}$ -a.s. that

$$\begin{aligned} X_t^d &= \frac{B_t}{B_T} \left[\beta - \left(\beta - x_0 \frac{B_T}{B_{t_0}} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right. \\ & \quad \times \exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s - \frac{\delta^2}{2} \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) \Big] \\ &= \frac{B_t}{B_T} \left[\beta - \left(\beta - x_0 \frac{B_T}{B_{t_0}} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right. \\ & \quad \times \exp \left(-\frac{\delta}{\sigma} \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^3}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right. \\ & \quad \left. \left. - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \frac{1}{\delta} \log \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right) - \frac{\delta^2}{2} \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) \right]. \end{aligned}$$

Consider that with the substitution $x = e^{\delta^2 T - s}$, i.e $s = T - \frac{1}{\delta^2} \log(x)$ and $ds = -\frac{1}{\delta^2} \frac{1}{x}$ we obtain

$$\begin{aligned}
 \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds &= -\frac{1}{\delta^2} \int_{e^{\delta^2(T-t_0)}}^{e^{\delta^2(T-t)}} \frac{x}{(x-1)^2} ds \\
 &= -\frac{1}{\delta^2} \int_{e^{\delta^2(T-t_0)}}^{e^{\delta^2(T-t)}} \frac{1}{x-1} + \frac{1}{(x-1)^2} ds \\
 (4.69) \quad &= -\frac{1}{\delta^2} \left(\log \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right) - \frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right).
 \end{aligned}$$

With this result we get that $\mathbb{P}_{t_0, x_0}$ -a.s.

$$\begin{aligned}
 X_t^d &= \frac{B_t}{B_T} \left[\beta - \left(\beta - x_0 \frac{B_T}{B_{t_0}} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right. \\
 &\quad \times \exp \left(-\frac{\delta}{\sigma} \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^3}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right. \\
 &\quad \left. \left. + \left(\frac{1}{2} - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \frac{1}{\delta} \right) \log \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right) - \frac{1}{2} \frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) \right] \\
 &= \frac{B_t}{B_T} \left[\beta - \left(\beta - x_0 \frac{B_T}{B_{t_0}} \right) \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right)^{\frac{1}{2} + 1 - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \frac{1}{\delta}} \right. \\
 &\quad \times \exp \left(-\frac{\delta}{\sigma} \frac{e^{\delta^2(T-t)}}{e^{\delta^2(T-t)} - 1} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^3}{\sigma} \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right. \\
 &\quad \left. \left. - \frac{1}{2} \frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) \right].
 \end{aligned}$$

Now as

$$1 - \frac{\mu - \frac{1}{2}\sigma^2}{\sigma} \frac{1}{\delta} = \frac{1}{\sigma\delta} \left(\frac{1}{2}\sigma^2 - r \right) = \frac{\sigma}{2\delta} - \frac{r}{\sigma\delta}$$

holds, we end up with

$$X_t^d = \frac{B_t}{B_T} \left[\beta - \left(\beta - x_0 \frac{B_T}{B_{t_0}} \right) \left(\frac{\left(\frac{B_T}{B_t} \right)^{\delta^2/r} - 1}{\left(\frac{B_T}{B_{t_0}} \right)^{\delta^2/r} - 1} \right)^{1/2 + \sigma/(2\delta) - r/(\sigma\delta)} \right]$$

$$\begin{aligned} & \times \exp \left(-\frac{\delta}{\sigma} \frac{\left(\frac{B_T}{B_t}\right)^{\delta^2/r}}{\left(\frac{B_T}{B_{t_0}}\right)^{\delta^2/r} - 1} \log \left(\frac{S_t}{S_{t_0}} \right) + \frac{\delta^3}{\sigma} \int_{t_0}^t \frac{\left(\frac{B_T}{B_s}\right)^{\delta^2/r}}{\left(\left(\frac{B_T}{B_s}\right)^{\delta^2/r} - 1\right)^2} \log \left(\frac{S_s}{S_{t_0}} \right) ds \right. \\ & \left. - \frac{1}{2} \frac{\left(\frac{B_T}{B_{t_0}}\right)^{\delta^2/r} - \left(\frac{B_T}{B_t}\right)^{\delta^2/r}}{\left(\left(\frac{B_T}{B_{t_0}}\right)^{\delta^2/r} - 1\right) \left(\left(\frac{B_T}{B_t}\right)^{\delta^2/r} - 1\right)} \right) \mathbb{P}_{t_0, x_0}\text{-a.s.}, \end{aligned}$$

as claimed in (4.64). Finally we want to show that $X_T^d := \lim_{t \uparrow T} X_t^d = \beta$ $\mathbb{P}_{t_0, x_0}\text{-a.s.}$ Then it will be clear that $\tilde{V}_d^2(t_0, x_0) = \text{Var}_{t_0, x_0}(X_T^d) = 0$. Define

$$M_t = -\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s.$$

Then M_t is a continuous local martingale for $t \in [t_0, T[$ with $\mathbb{P}_{t_0, x_0}\text{-a.s.}$

$$\begin{aligned} [M, M]_t &= \delta^2 \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \\ &\stackrel{(4.69)}{=} - \left(\log \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right) - \frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) \xrightarrow{t \uparrow T} \infty. \end{aligned}$$

By the Dambis-Dubins-Schwarz theorem (see e.g. [3, p.181]) we can find a Brownian motion $\bar{W}$ such that $M_t = \bar{W}_{[M, M]_t}$. Therefore by the law of iterated logarithm

$$M_t - \frac{1}{2}[M, M]_t = \bar{W}_{[M, M]_t} - \frac{1}{2}[M, M]_t \xrightarrow{t \uparrow T} -\infty \quad \mathbb{P}_{t_0, x_0}\text{-a.s.}$$

Hence we have $\mathbb{P}_{t_0, x_0}\text{-a.s.}$

$$X_t^d = e^{-r(T-t)} \left[\beta - \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \exp \left(\bar{W}_{[M, M]_t} - \frac{1}{2}[M, M]_t \right) \right] \xrightarrow{t \uparrow T} \beta.$$

It is clear that $V_d^2(t_0, x_0) = \lim_{t \uparrow T} \text{Var}_{t_0, x_0}(X_t^d) = 0$ for the case $x_0 e^{r(T-t_0)} \geq \beta$. We still need to show that $V_d^2(t_0, x_0) = \infty$ if $x_0 e^{r(T-t_0)} < \beta$. We first use (4.69) to rewrite the closed form expression (4.63) for X^d as

$$\begin{aligned} X_t^d &= e^{-r(T-t)} \left[\beta - \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right. \\ &\quad \left. \times \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right)^{1/2} \exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s - \frac{1}{2} \frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) \right], \end{aligned}$$

$\mathbb{P}_{t_0, x_0}$ -a.s. Hence

$$\begin{aligned} \text{Var}_{t_0, x_0}(X_t^d) &= e^{-2r(T-t)} \left(\beta - x_0 e^{r(T-t_0)} \right)^2 \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right)^2 \\ &\quad \times \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \exp \left(-\frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) \\ &\quad \times \text{Var}_{t_0, x_0} \left(\exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s \right) \right). \end{aligned}$$

Consider that

$$\begin{aligned} &\text{Var}_{t_0, x_0} \left(\exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s \right) \right) \\ &= \mathbb{E}_{t_0, x_0} \left[\exp \left(-2\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s \right) \right] - \mathbb{E}_{t_0, x_0} \left[\exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s \right) \right]^2 \\ &= \exp \left(\frac{1}{2} 4\delta^2 \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) - \left(\exp \left(\frac{1}{2} \delta^2 \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) \right)^2 \\ &= \exp \left(\delta^2 \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) \left(\exp \left(\delta^2 \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right) - 1 \right) \\ &\stackrel{(4.69)}{=} \frac{e^{\delta^2(T-t_0)} - 1}{e^{\delta^2(T-t)} - 1} \exp \left(\frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) \\ &\quad \times \left(\frac{e^{\delta^2(T-t_0)} - 1}{e^{\delta^2(T-t)} - 1} \exp \left(\frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) - 1 \right). \end{aligned}$$

Hence¹²

$$\begin{aligned} \text{Var}_{t_0, x_0}(X_t^d) &= e^{-2r(T-t)} \left(\beta - x_0 e^{r(T-t_0)} \right)^2 \left(\frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right)^2 \\ &\quad \times \left(\frac{e^{\delta^2(T-t_0)} - 1}{e^{\delta^2(T-t)} - 1} \exp \left(\frac{e^{\delta^2(T-t_0)} - e^{\delta^2(T-t)}}{(e^{\delta^2(T-t)} - 1)(e^{\delta^2(T-t_0)} - 1)} \right) - 1 \right) \xrightarrow[t \uparrow T]{} \infty. \end{aligned}$$

This completes the proof of (B) and Corollary 4.5. $\square$

¹²See J.L. Pedersen and G. Peskir [2, Remark 8, p. 20]

Remark 4.6. We first show that the case $\delta = 0$ does not follow from the case $\delta \neq 0$ by taking limits, i.e. that the corresponding part of [2, Corollary 7, p. 16-17] is not correct. This follows from the fact that the limit

$$\begin{aligned} \lim_{\delta \rightarrow 0} \frac{\delta}{e^{\delta^2(T-t_0)} - 1} &= \lim_{\delta \rightarrow 0} \frac{\delta}{\sum_{n=1}^{\infty} \frac{(\delta^2(T-t_0))^n}{n!}} \\ &= \lim_{\delta \rightarrow 0} \frac{1}{\sum_{n=1}^{\infty} \frac{\delta^{2n-1}(T-t_0)^n}{n!}} \end{aligned}$$

does not exist despite for $\delta \neq 0$

$$u_{3,\beta}^s(t, x) = \frac{\delta}{\sigma} \frac{1}{x} \left[x_0 e^{r(t-t_0)} - x + \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t_0)-r(T-t)}}{e^{\delta^2(T-t_0)} - 1} \right].$$

This corollary shows that in the case $x_0 e^{r(T-t_0)} < \beta$ it is dynamically optimal in Problem 3 to hold the wealth strictly below β for $t \in [t_0, T[$ and then converge $\mathbb{P}_{t_0, x_0}$ -a.s. to β . One can think of this strategy as an equivalent to the doubling strategy in discrete time. If an investor is allowed to borrow money as much as one likes, he can reach any wealth with probability one by doubling the investment until he wins. The problem is that he has unbounded risk in between, i.e. that there is no bound for how much he has to borrow until he wins.

Consider that we also have that the expectation of X_t^d converges to β . As

$$\mathbb{E}_{t_0, x_0} \left[\exp \left(-\delta \int_{t_0}^t \frac{e^{\delta^2(T-s)}}{e^{\delta^2(T-s)} - 1} dW_s \right) \right] = \exp \left(\frac{\delta^2}{2} \int_{t_0}^t \frac{e^{2\delta^2(T-s)}}{(e^{\delta^2(T-s)} - 1)^2} ds \right)$$

we have

$$\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [X_t^d] = \lim_{t \uparrow T} e^{-r(T-t)} \left[\beta - \left(\beta - x_0 e^{r(T-t_0)} \right) \frac{e^{\delta^2(T-t)} - 1}{e^{\delta^2(T-t_0)} - 1} \right] = \beta.$$

However, the second moment of X_t^d does not converge to β^2 as shown in the corollary.

Now we compare again (for $\delta \neq 0$) the statically with the dynamically optimal control under the same aspect as in Remark 4.2 and 4.4. Consider that for the interesting case $(t_0, x_0) \in [0, T[\times \mathbb{R}$ with $x_0 e^{r(T-t_0)} < \beta$ we get that

$$\begin{aligned} &\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_s^2(t, X_t^s)] \\ &= \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} \left[\frac{\left(\beta - X_t^s e^{r(T-t)} \right)^2}{e^{\delta^2(T-t)} - 1} \right] \end{aligned}$$

$$\begin{aligned}
&= \lim_{t \uparrow T} \frac{\mathbb{E}_{t_0, x_0} \left[\left((X_t^s e^{r(T-t)} - \beta e^{r(T-t)}) + (\beta e^{r(T-t)} - \beta) \right)^2 \right]}{(e^{\delta^2(T-t)} - 1)^2} \\
&= \lim_{t \uparrow T} \frac{1}{(e^{\delta^2(T-t)} - 1)^2} \left(e^{2r(T-t)} \text{Var}_{t_0, x_0} (X_t^s) \right. \\
&\quad \left. + 2 \left(\mathbb{E}_{t_0, x_0} [X_t^s e^{r(T-t)}] - \beta e^{r(T-t)} \right) (\beta e^{r(T-t)} - \beta) + (\beta e^{r(T-t)} - \beta)^2 \right) \\
&= \lim_{t \uparrow T} \frac{\text{Var}_{t_0, x_0} (X_t^s)}{(e^{\delta^2(T-t)} - 1)^2} \\
&= \lim_{t \uparrow T} \frac{V_s^2(t_0, x_0)}{(e^{\delta^2(T-t)} - 1)^2} \\
&= \infty,
\end{aligned}$$

where we used continuity of $t \mapsto \mathbb{E}_{t_0, x_0} [X_t^s]$ and $t \mapsto \mathbb{E}_{t_0, x_0} [(X_t^s)^2]$ (see (4.39) and (4.40)) to get $\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [(X_t^s - \beta)^2] = \lim_{t \uparrow T} \text{Var}_{t_0, x_0} (X_t^s) = V_s^2(t_0, x_0)$. On the other hand we have

$$\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [\underbrace{\tilde{V}_d^2(t, X_t^s)}_0] = 0 < \infty = \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_s^2(t, X_t^s)].$$

So under this aspect, the weakly dynamically optimal control in Problem 3 given in (4.62) is better than the statically optimal control in Problem 3 given in (4.58). But

$$\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [\underbrace{V_d^2(t, X_t^s)}_\infty] = \infty = \lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_s^2(t, X_t^s)].$$

Hence under this aspect, both controls would be as good (or bad) as each other. In order to be consistent with the previous discussions (see Remark 4.2 and 4.4), it seems to be more intuitive to compare $\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_d^2(t, X_t^s)]$ and $\lim_{t \uparrow T} \mathbb{E}_{t_0, x_0} [V_s^2(t, X_t^s)]$, i.e. V_d^2 with V_s^2 .¹³ Nevertheless J.L. Pedersen and G. Peskir decided to compare only $\tilde{V}_d^2$ with V_s^2 (see [2, Remark 8, p. 19]).

¹³This is the reason why we introduced two different dynamic value functions V_d^2 and $\tilde{V}_d^2$.

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